

Advanced Dimensionality Reduction and Machine Learning Techniques

Pattern Identification in Dense and Scarce Environmental Data
Regimes



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Table of Contents

List of Figures	vii
List of Tables	ix
Nomenclature	xi
1 Motivation	1
I THEORETICAL FOUNDATIONS	3
2 Introduction	5
3 Data-Driven Dimensionality Reduction	7
3.1 Principal Component Analysis	7
3.1.1 Derivation from Maximum Variance Formulation	8
3.1.2 Sparse PCA via Rotation	13
3.1.3 Hilbert PCA	17
3.2 Maximum Covariance Analysis	20
3.2.1 Rotated Hilbert MCA	24
3.2.2 From Covariance to Correlation: The CPCCA Framework	31
4 Gaussian Process Regression	33
4.1 Introduction to Gaussian Processes	33
4.2 Gaussian Processes for Regression	33
4.3 Spatial Modeling with Gaussian Processes	33
4.3.1 Log Gaussian Cox Process	33
4.3.2 Negative Binomial Distribution	34
4.4 Model Implementation and Inference	34

II	PATTERN IDENTIFICATION IN THE DENSE DATA REGIME	35
5	Bridging Theory and Practice with Advanced Dimensionality Reduction in Python	37
5.1	Open-Source Tools for Scalable Climate Data Analysis	37
5.2	Challenges in Large-Scale and Multidimensional Climate Data Analysis	39
5.3	The XEOFS Library: A Solution For Dimensionality Reduction in Climate Science	40
5.4	Conclusion	43
6	Hilbert Maximum Covariance Analysis of the Madden-Julian Oscillation	45
7	Shared Variability between Temporally Lagged SST and Continental Precipitation	47
7.1	Methods & Data	48
7.2	Results and Discussion	53
7.3	Conclusions	65
III	PATTERN IDENTIFICATION IN THE SCARCE DATA REGIME	67
8	A Bayesian Machine Learning Approach to Seasonal Variations in Beach Litter Pollution	69
8.1	Plastic Pollution on Beaches	70
8.2	Materials & Methods	72
8.2.1	Beach Litter Data	72
8.2.2	Log-Gaussian Cox process model	73
8.2.3	LGPC Model Posterior Inference	75
8.2.4	Clustering of Seasonal Patterns	76
8.2.5	Sources of Macroplastics and Seasonal Potential	77
8.3	Results and Discussion	79
8.3.1	Seasonal Variations of Anthropogenic Beach Litter in the North-East Atlantic	80
8.3.2	Identification of Seasonal Hotspots	81
8.3.3	Drivers of Seasonal Hotspots in Anthropogenic Beach Litter	94
8.4	Conclusion	97

TABLE OF CONTENTS

v

9 Conclusion & Future Work

99

References

101

List of Figures

3.1	Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.	18
5.1	Open-source software (OSS) projects in Earth system sciences.	38
5.2	Computational performance of PCA using the XEOFs library (version 3.0.0). . .	42
7.1	Evaluation of varimax-rotated Hilbert MCA applied to SST and continental precipitation.	51
7.2	Seasonal cycle as represented by mode 1 of varimax-rotated Hilbert MCA applied to SST and continental precipitation.	54
7.3	ENSO as represented by mode 2 of varimax-rotated Hilbert MCA applied to SST and continental precipitation.	55
7.4	Oceanic warming as represented by mode 3 of varimax-rotated Hilbert MCA applied to SST and continental precipitation.	57
7.5	TBD	57
7.6	Nonlinear ENSO modulation as represented by mode 4 of varimax-rotated Hilbert MCA applied to SST and precipitation.	59
7.7	Atlantic Meridional Modulation (AMM) as represented by mode 6 of varimax-rotated Hilbert MCA applied to SST and precipitation.	60
7.8	Pacific Decadal Oscillation (PDO) as represented by mode 9 of varimax-rotated Hilbert MCA applied to SST and precipitation.	61
7.9	Indian Ocean Dipole (IOD) as represented by mode 10 of varimax-rotated Hilbert MCA applied to SST and precipitation.	62
7.10	TBD	64
8.1	TBD	81
8.2	TBD	82

8.3	TBD	83
8.4	TBD	84
8.5	TBD	84
8.6	TBD	85
8.7	TBD	86
8.8	TBD	86
8.9	TBD	87
8.10	TBD	88
8.11	TBD	89
8.12	TBD	90
8.13	TBD	91
8.14	TBD	92
8.15	TBD	92
8.16	TBD	93

List of Tables

- 3.1 Computational complexity of PCA algorithms 14
- 5.1 Overview of available methods in the XEOFS library 41
- 8.1 Inference of parameters of the log-Gaussian Cox process model: variances η_i^2 , length scales ρ_i measured in km, with $i \in \{\text{short}, \text{long}\}$, spatial mean μ_μ and over-dispersion ϕ . Shown are the 50th percentile (median) together with the 95 % highest density interval (in parentheses) given by the minimum width Bayesian credible interval (BCI). 82
- 8.2 Median relative difference of macroplastic X between season i and j computed as $\text{med}((X_j - X_i)/X)$ where $X = X_i$ if $\text{med}(X_j - X_i) \geq 0$, else $X = X_j$. Here, $\text{med}(\cdot)$ refers to the median. We then computed estimates for the NEA region as weighted median over all beaches with the individual confidence estimates as weights. For simplicity, all values are shown in percentage, together with their 25th and 75th weighted percentile. 93

Nomenclature

Greek Symbols

π $\simeq 3.14\dots$

Acronyms / Abbreviations

AMO Atlantic Multidecadal Oscillation

BSISO Boreal Summer Intraseasonal Oscillation

CCA Canonical Correlation Analysis

CPCCA Continuum Power CCA

ENSO El Niño–Southern Oscillation

HPC High Performance Computing

IOD Indian Ocean Dipole

MJO Madden-Julian Oscillation

NAO North Atlantic Oscillation

ONI Oceanic Niño Index

OPA Optimal Persistence Analysis

PC Principal Component

PDO Pacific Decadal Oscillation

POP Principal Oscillation Pattern

PSA Pacific-South American

SST Sea Surface Temperature

VC Vector of Coefficients

Motivation

- complexity of our environment: more and more data are available (Earth observation data, satellites, reanalysis, citizen science) to describe very complex systems; research becomes more and more data-driven 3 4 5
- dimensionality reduction methods and non-linear modelling approaches are important to get a better understanding 6 7
- software tools are necessary to lower the barrier to entry for non-experts 8

Part I

Theoretical Foundations

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2

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Introduction

- complexity of our environment: more and more data are available (Earth observation data, satellites, reanalysis, citizen science) to describe very complex systems; research becomes more and more data-driven 3 4 5
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Data-Driven Dimensionality Reduction

nicrie: Motivation for Dimensionality Reduction: Explain why dimensionality reduction is necessary in climate science. e.g. ERA5 contains more than 9 PB of data. Discuss the need to simplify complex datasets by identifying the most significant patterns and features, which can help in understanding underlying processes and improving predictability.

nicrie: provide a brief outline of this chapter; link to what is coming in the next chapters

3.1 Principal Component Analysis

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of a dataset while preserving most of the variance. This is achieved by identifying the principal axes, which are the directions in which the data varies the most. These directions correspond to the eigenvectors of the covariance matrix of the data, with the associated eigenvalues representing the magnitude of variance along these directions. PCA is particularly useful for dimensionality reduction, data compression, feature extraction and visualization of high-dimensional datasets.

The origins of PCA can be traced back to Pearson [196] in 1901 and Hotelling [111] in 1936. It gained significant traction in meteorology through the work of Lorenz [161] in 1956, who introduced the technique under the term *empirical orthogonal function* (EOF) analysis¹. The method has since been extensively applied in meteorology and climate science to identify dominant modes of variability in the atmosphere and oceans, as discussed in numerous textbooks [e.g. 52, 96, 124, 204, 244, 281].

In this section, we derive PCA from the maximum variance formulation², as outlined by Bishop [14, pp. 561-563]. This approach involves projecting the data onto a lower-dimensional linear space, known as the principal subspace, to maximize the variance of the projected data. We

¹Although EOF analysis remains the preferred term in meteorology and climate science, we will use the more commonly recognized term, PCA.

²Hotelling [111] originally proposed this formulation. An alternative derivation, known as the minimum error formulation by Pearson [196], seeks a linear projection that minimizes the average projection error, defined as the mean squared distance between the original data points and their projections. This alternative approach shows that PCA is an optimal solution for approximating data in the squared error sense.

will then explore the limitations of PCA, particularly in the context of climate science, and discuss common strategies to address these limitations, such as rotated PCA and Hilbert PCA.

nicrie: make link to Karhunen-Loève expansion

3.1.1 Derivation from Maximum Variance Formulation

Consider a data set represented as a complex-valued, two-dimensional $N \times D$ matrix $\mathbf{Z}$, where N denotes the number of observations (or samples) and D the number of variables (or features). We will refer to the n^{th} observation as the D -dimensional vector $\mathbf{z}_n$. For simplicity, we assume the data is centered, meaning the mean of each variable is zero, i.e. $\bar{\mathbf{z}} = \frac{1}{N} \sum_{n=1}^N \mathbf{z}_n = \mathbf{0}$. The goal is to project the data onto an M -dimensional space ($M < D$) of M latent variables while maximizing the variance of the projected data.

To start, consider a one-dimensional projection ($M = 1$). We define the direction of this space using a D -dimensional unit³ vector $\mathbf{v}_1$, such that $\mathbf{v}_1^H \mathbf{v}_1 = 1$, where $\mathbf{v}_1^H$ denotes the conjugate transpose of $\mathbf{v}_1$. The data points $\mathbf{z}_n$ are then projected onto this direction, $\mathbf{v}_1^H \mathbf{z}_n$, and the variance of the projected data is

$$\frac{1}{N-1} \sum_{n=1}^{N-1} (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}})^2 = \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1, \quad (3.1)$$

where $\mathbf{S}_{ZZ}$ is the covariance matrix of $\mathbf{Z}$ given by

$$\mathbf{S}_{ZZ} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{z}_n \mathbf{z}_n^H = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z}. \quad (3.2)$$

The goal is to find the vector $\mathbf{v}_1$ that maximizes this variance, formulated as:

$$\underset{\mathbf{v}_1}{\operatorname{argmax}} \quad \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 \quad \text{subject to} \quad \mathbf{v}_1^H \mathbf{v}_1 = 1. \quad (3.3)$$

To solve this optimization problem, we maximize the projected variance $\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. To enforce the normalization constraint $\mathbf{v}_1^H \mathbf{v}_1 = 1$, a Lagrange multiplier λ_1 is introduced, leading to the unconstrained maximization of:

³This particular normalization is for convenience, and any other would do just fine, since we are only interested in the direction of $\mathbf{v}_1$, not its magnitude.

$$\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^H \mathbf{v}_1). \quad (3.4) \quad 1$$

By setting the derivative with respect to $\mathbf{v}_1$ to zero, we find a stationary point when: 2

$$\mathbf{S}_{ZZ} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1. \quad (3.5) \quad 3$$

This indicates that $\mathbf{v}_1$ must be an eigenvector of $\mathbf{S}_{ZZ}$, with λ_1 being the corresponding real-valued⁴ eigenvalue. The variance $\mathbf{v}_1^T \mathbf{S}_{ZZ} \mathbf{v}_1$ reaches its maximum when $\mathbf{v}_1$ aligns with the eigenvector associated with the largest eigenvalue⁵ λ_1 . The eigenvector is referred to as the first **vector of coefficients**⁶ (VC) of the data set. 4
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To find additional VCs, we iteratively select new directions that maximize the projected variance while being orthogonal to the previously chosen directions. For an M-dimensional projection space, the optimal linear projection that maximizes the variance of the projected data is defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the covariance matrix $\mathbf{S}_{ZZ}$, corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This result can be derived using proof by induction, leading to the concise expression: 8
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$$\mathbf{S}_{ZZ} \mathbf{V} = \mathbf{V} \mathbf{\Lambda}, \quad (3.6) \quad 14$$

where $\mathbf{V}$ is unitary, i.e. $\mathbf{V}^H \mathbf{V} = \mathbf{I}_M$, and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_M)$ is the diagonal matrix of eigenvalues. 15
16

Thus, the eigenvectors of $\mathbf{S}_{ZZ}$ define a new unitary coordinate system that aligns with the primary patterns of variability in the data. In other words, PCA identifies the directions (principal axes) along which the data exhibits the most significant variations. However, these principal components are constrained to be orthogonal to each other. In situations where the underlying patterns in the data are not naturally aligned with these orthogonal directions, a 17
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⁴This follows from the property of the covariance matrix being hermitian.

⁵This can easily be verified by left-multiplying Equation (3.5) with $\mathbf{v}_1^H$ and making use of the normalization condition.

⁶Naming conventions can vary significantly across different scientific domains. For example, in machine learning, the terms principal components and principal component scores refer to the eigenvectors (here $\mathbf{V}$) and their corresponding data projections (here $\mathbf{P}$) [e.g. 14]. In statistics, these are often called vectors of coefficients/loadings and principal components [e.g. 124], while in climate science and meteorology, they are known as empirical orthogonal functions and principal components/expansion coefficients [e.g. 244]. In this thesis, I aim to maintain consistency with the terminology commonly used in statistics.

rotation technique such as varimax (discussed in Section 3.1.2) can be applied to achieve a more interpretable solution.

According to the spectral theorem for normal matrices [e.g. 110, p. 133], the hermitian covariance matrix $\mathbf{S}_{ZZ}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding real-valued eigenvalues λ_i :

$$\mathbf{S}_{ZZ} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^H. \quad (3.7)$$

Each eigenvalue λ_i represents the amount of variance captured by the corresponding component, typically expressed as a fraction γ_i of the total variance:

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (3.8)$$

Projecting the data $\mathbf{Z}$ onto the M -dimensional subspace thus provides a lower-dimensional representation, called the principal components (PCs),

$$\mathbf{P} = \mathbf{ZV}, \quad (3.9)$$

which also enables an approximate reconstruction of the original data from the projected data:

$$\mathbf{Z} \approx \mathbf{PV}^H. \quad (3.10)$$

This approximation is optimal in the least squares sense and becomes exact only when $M = \text{rank}(\mathbf{X})$.

However, calculating the eigenvector decomposition is not always computationally efficient. A more efficient method involves using Singular Value Decomposition (SVD) [e.g. 110, p. 150], which decomposes the data matrix as

$$\mathbf{Z} = \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^H, \quad (3.11)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are unitary matrices, $\mathbf{U}'^H \mathbf{U}' = \mathbf{V}'^H \mathbf{V}' = \mathbf{I}_M$, and $\mathbf{\Sigma}$ is a diagonal matrix containing the singular values σ_i of $\mathbf{Z}$. The covariance matrix $\mathbf{S}_{ZZ}$ can then be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z} = \frac{1}{N-1} \mathbf{V}' \mathbf{\Sigma}^H \mathbf{U}'^H \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^H = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H, \quad (3.12)$$

where, by comparison with Equation (3.6) and (3.9), $\mathbf{V} = \mathbf{V}'$, $\mathbf{\Lambda} = \frac{1}{N-1} \mathbf{\Sigma}^T \mathbf{\Sigma}$ and $\mathbf{P} = \mathbf{U}' \mathbf{\Sigma}$. This establishes a direct relationship between the eigenvalues of the covariance matrix and the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (3.13)$$

Comparing (3.10) with (3.11) shows that the PC scores $\mathbf{P}$ can be normalized

$$\mathbf{U} \equiv \mathbf{U}' = \mathbf{P} \mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \mathbf{P} \mathbf{\Lambda}^{-1/2}, \quad (3.14)$$

$$\text{or alternatively } \mathbf{P} = (N-1)^{1/2} \mathbf{U} \mathbf{\Lambda}^{1/2}. \quad (3.14)$$

Thus, $\mathbf{U}^H \mathbf{U} = \mathbf{I}_M$ ensures that the PC scores $\mathbf{U}$ are unitary and uncorrelated.

A common scaling method for the vector of coefficients $\mathbf{V}$ involves scaling by the square root of the corresponding eigenvalues, referred to as **loadings**:

$$\tilde{\mathbf{V}} = \mathbf{V} \mathbf{\Lambda}^{1/2}. \quad (3.15)$$

Although not strictly unitary, these loadings form a generalized unitary matrix, as shown by

$$\tilde{\mathbf{V}}^H \tilde{\mathbf{V}} = \mathbf{\Lambda}. \quad (3.16)$$

Using this scaling, the Equation 3.10 can be expressed as

$$\mathbf{X} = (N-1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^T. \quad (3.17)$$

Weighting of Data Variables In some cases, it may be necessary to weight the data variables differently. This can be achieved by scaling the data matrix $\mathbf{Z}$ by a diagonal matrix $\mathbf{W}$, where the j^{th} diagonal element w_j corresponds to the weight of the j^{th} variable. A very common situation is that the observed variables exhibit stark differences in their variance, which can sometimes lead to mask structure in the data as the PCA solution is solely driven by the different scales of variance (so the first principal axis will closely align to the variable with the highest variance) in which case the obvious choice is to use the inverse of the standard deviation of each variable, which ensures that all variables are equally weighted in which case the covariance matrix becomes the correlation matrix. Another common situation in climate science is that the physical fields are represented on regular longitude-latitude grids. This representation leads to a higher density of grid points near the poles, which typically results in an exaggerated importance of components spatially focused near the poles. To address this, the data matrix can be weighted by the square root of the cosine of the latitude [184], $w_j = \sqrt{\cos(\text{lat}_j)}$, where lat_j is the latitude of the j^{th} grid point. This weighting ensures that the grid points are equally weighted in the PCA solution, regardless of their geographical location.

Computational Considerations The computational cost of performing eigendecomposition on the covariance matrix scales with $O(D^3)$ [e.g. 83, Section 8.3], making it impractical for modern datasets, such as those in climate science, where typical data sizes are on the order of $D = O(10^6)$ or even larger. For example, ERA5 [43], a widely used reference for state-of-the-art reanalysis climate data, distributes a single climate variable on a regular 0.25° grid, resulting in $D = 1\,036\,800$ grid cells.

In practice, more efficient algorithms, such as SVD or power methods [83], have been traditionally employed. Recently, randomized low-rank approximation techniques [54, 92] have further reduced computational complexity to $O(ND \log(M) + (N + D)M^2)$. These advancements offer feasible alternatives for handling large, out-of-memory datasets (see Table 3.1).

Shortcomings Despite its widespread use and versatility, PCA has several limitations and potential drawbacks that should be considered:

1. **Linearity Assumption:** PCA fundamentally relies on linear transformations to reduce the dimensionality of data. Specifically, PCA seeks to identify PCs that are linear combinations of the original variables (Equation (3.9)), which may not hold in complex systems such as climate data, where relationships between variables often exhibit nonlinear patterns [e.g. 94].
2. **Interpretability of VCs:** The orthogonality of the vectors of coefficients is a mathematical construct that may not always correspond to the physical reality of the data, potentially leading to misinterpretations [213].

3. **Variance Focus:** Focusing on directions of maximum variance may not always capture the most relevant features for a specific application, potentially missing critical but less variable information [68].
4. **Static Nature:** As a static method, PCA does not account for temporal dynamics or changes over time, which is particularly limiting in climate science, where time evolution is a key aspect.
5. **Sensitivity to Outliers:** PCA can be sensitive to outliers, as extreme values can disproportionately influence the VCs, leading to misleading results [26].
6. **Sensitivity to Scaling:** The results are sensitive to the scaling of the input data, meaning that variables with larger ranges can dominate the VCs unless the data is appropriately standardized.
7. **Loss of Information:** While dimensionality reduction is often the goal, PCA inherently discards some information, which could lead to the loss of important details crucial for accurate analysis and interpretation in specific applications.

Over the past few decades, numerous modifications and extensions have been proposed to address these limitations, especially in the context of climate science. For instance, to address the linearity assumption, various PCA variants based on non-parametric kernel [21, 232, 276] or neural network [132] approaches have been developed. For improved interpretability, rotated [213] and sparse [125] PCA methods have been proposed. Robust PCA [26] offers a more resilient approach to handling sensitivity to outlier. Furthermore, extended [19, 275] and complex [10, 108] PCA methods have been introduced to account for temporal dynamics. In addition, eigenmethods like Optimal Persistence Analysis (OPA) [49, 50] and Principal Oscillation Pattern (POP) analysis [101] provide alternative approaches independent to variance maximization to understanding dominant patterns in physical climate fields.

In the following subsections, we will explore some of these shortcomings and proposed solutions in greater detail. Specifically, we will discuss how additional rotation of the PCA solution can enhance physical interpretability (Section 3.1.2) and how complexifying the data can provide insights into time evolution patterns (Section 3.1.3).

3.1.2 Sparse PCA via Rotation

PCA is widely used for dimensionality reduction, as it maximizes the explained variance and provides orthogonality, making it optimal in a least-squares sense [14]. It effectively identifies principal axes that capture the covariance among variables, allowing for the examination of interrelationships between stations, grid points or variables. However, PCA is not ideal for identifying the cause of variation or deriving physical modes of variability [124, 150, 213].

Table 3.1 — Computational costs of different algorithms used to compute either full or approximate M-rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [14] and Halko *et al.* [92].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

One limitation is that real-world processes do not necessarily adhere to the orthogonality imposed by PCA's eigenvectors. Additionally, the dense nature of these VCs – where all elements are non-zero – can make interpretation challenging. In the context of climate and environmental science, where variables are often represented across spatial grids, these limitations can lead to recognizable but artificial patterns known as Buell patterns [20, 149, 213]. These patterns typically start with a positive component and evolve into increasingly complex structures, such as bipole or tripole patterns (Figure 3.1). Such issues highlight the disconnect between standard PCA and underlying physical processes [213], prompting the development of alternative methods that produce sparse VCs, characterized by many zero or near-zero elements. This sparsity – which necessarily implies the loss of orthogonality for at least one of the VCs and PC scores – can enhance interpretability by localizing the patterns and reducing the density of the VCs.

In climate science, the most common approach to achieving sparse PCA is through rotation techniques [109, 212, 213], a concept borrowed from factor analysis [cf. 124, pp. 269]. In this "factor analysis viewpoint" of PCA [124, pp. 150-161], a linear transformation is applied to a subset of the VCs to derive a new set of rotated VCs that are "simpler" in a heuristically defined manner, ideally aligning better with the underlying physical processes.

As noted in Equation (3.17), once the orthogonality constraint is relaxed, PCA solutions exhibit inherent indeterminacy, allowing any non-singular transformation $\mathbf{R}$ to satisfy the PCA formulation [e.g. 213, p. 305]:

$$\mathbf{Z} = (N - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^H = (N - 1)^{1/2} \mathbf{U} \mathbf{R} \mathbf{R}^{-1} \tilde{\mathbf{V}}^H. \quad (3.18)$$

Rotated PCA exploits this indeterminacy by choosing $\mathbf{R}$ to optimize a simplicity criterion f , resulting in rotated loadings⁷ $\tilde{\mathbf{V}}_R = \tilde{\mathbf{V}} \mathbf{R}$ that maximize f :

⁷In principle, it is possible to rotate the unitary matrix of coefficients $\mathbf{V}$ instead of the loadings matrix $\tilde{\mathbf{V}}$ [124, 172]. However, this approach is seldom used in practice. The primary reason is that rotating the unitary coefficient

$$\operatorname{argmax}_{\mathbf{R}} f(\tilde{\mathbf{V}}_{\mathbf{R}}). \quad (3.19) \quad 1$$

There are two main categories of rotation: *orthogonal*⁸ and *oblique*. In orthogonal rotation, $\mathbf{R}$ is a unitary matrix, preserving the uncorrelated nature of the PCs since $\mathbf{U}_{\mathbf{R}}^{\mathbf{H}} \mathbf{U}_{\mathbf{R}} = \mathbf{R}^{\mathbf{H}} \mathbf{U}^{\mathbf{H}} \mathbf{U} \mathbf{R} = \mathbf{I}_M$. In oblique rotation, $\mathbf{R}$ is a general non-singular matrix, resulting in correlated PCs, as $\mathbf{U}_{\mathbf{R}}^{\mathbf{H}} \mathbf{U}_{\mathbf{R}} = \mathbf{R}^{\mathbf{H}} \mathbf{R}$. Regardless of the method, the rotated loadings are not orthogonal, 2 3 4 5

$$\tilde{\mathbf{V}}_{\mathbf{R}}^{\mathbf{H}} \tilde{\mathbf{V}}_{\mathbf{R}} = \mathbf{R}^{\mathbf{H}} \tilde{\mathbf{V}}^{\mathbf{H}} \tilde{\mathbf{V}} \mathbf{R} \stackrel{(3.16)}{=} \mathbf{R}^{\mathbf{H}} \mathbf{\Lambda} \mathbf{R} \neq \mathbf{I}_M, \quad (3.20) \quad 6$$

which is crucial for achieving a more interpretable solution. Various simplicity criteria f have been proposed [see e.g. 213], with most focusing on making the rotated loadings more localized and physically interpretable. In practice, the specific choice of simplicity criterion often has little impact on the results [213]. 7 8 9 10

Varimax Orthogonal Rotation The most widely used orthogonal rotation technique is the varimax criterion [127], which aims to simplify the VCs by maximizing the variance of the squared loadings, 11 12 13

$$f(\tilde{\mathbf{V}}_{\mathbf{R}}) = \sum_{m=1}^M \left(p \sum_{j=1}^P b_{jm}^2 - \left(\sum_{j=1}^P b_{jm} \right)^2 \right) \quad (3.21) \quad 14$$

where b_{jm} are the squared norm of the rotated loadings $|\tilde{v}_{R,jm}|^2/h_j^2$ normalized by the communalities $h_j^2 = \sum_{m=1}^M |\tilde{v}_{R,jm}|^2$ of variable j . The varimax rotation pushes the elements of $\tilde{\mathbf{V}}_{\mathbf{R}}$ towards either 0 or 1, creating a sparse solution with primarily large or small values, which is particularly useful for identifying localized physical processes, such as atmospheric or oceanic teleconnections. 15 16 17 18 19

matrix preserves the orthogonality of the VCs but results in the loss of orthogonality among the PCs, which does not address issues like the Buell patterns. Moreover, this method does not consider the relative importance of the spatial patterns, as indicated by the eigenvalues, making the results of the rotation process highly sensitive to the selection of the number of components to rotate.

⁸In the broader context of complex data, this would be referred to as a unitary rotation. However, since factor analysis has primarily been applied to real-valued data, the terms orthogonal and oblique rotations have become the standard. To avoid confusion, I use these historical terms while acknowledging that they also encompass the more general case of complex data.

Promax Oblique Rotation The promax criterion [102], a generalization of the varimax criterion, is one of the most common oblique rotation techniques. It is based on the premise that an orthogonal simple structure solution, such as varimax, is often close to an optimal oblique solution. The promax method constructs a hypothetical oblique target matrix $\mathbf{T}$ by performing an oblique Procrustes transformation [118] on the varimax-rotated loadings matrix,

$$\mathbf{T} = \tilde{\mathbf{V}}_R \mathbf{R} + \mathbf{E}, \quad (3.22)$$

where $\mathbf{R}$ is the transformation matrix and $\mathbf{E}$ is a matrix of residuals. The normal equation for solving the complex Procrustes problem is given by:

$$\mathbf{R} = (\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R)^{-1} \mathbf{T}^H \tilde{\mathbf{V}}_R. \quad (3.23)$$

To define the target matrix $\mathbf{T}$, the promax criterion increases the discrimination between high and low loadings by raising the loadings to a power q while retaining the original sign,

$$t_{jm} = \frac{|a_{jm}|^{q+1}}{a_{jm}}, \quad (3.24)$$

where q controls the degree of obliqueness and a_{jm} are the elements of the max-normalized, varimax-rotated loadings matrix $\mathbf{A}$, defined as

$$\mathbf{A} = \tilde{\mathbf{A}} \mathbf{K}^{-1}, \quad (3.25)$$

$$\tilde{\mathbf{A}} = \mathbf{H}^{-1} \tilde{\mathbf{V}}_R, \quad (3.26)$$

with $\mathbf{K}$ being a diagonal matrix containing the maximum absolute values of each column of $\tilde{\mathbf{A}}$, $a_m = \max_m |A_{pm}|$, and $\mathbf{H}$ a diagonal matrix with the square root of the communalities h_j for the j^{th} variable.

Application and Criticism Both orthogonal and oblique rotation techniques, particularly varimax and promax, have been extensively used since the 1980s to identify stationary modes of variability in climate data [e.g. 12, 75, 109, 119, 148, 158, 173]. Notable climate oscillations, such as the North Atlantic Oscillation (NAO), have been defined based on varimax-rotated

patterns of loadings [12]. These techniques also help identify "ghost patterns", artificial patterns identified by PCA that are not physically meaningful [119, 156].

Rotated modes offer several advantages, such as greater robustness against temporal and spatial subsampling [32] compared to unrotated modes. They also effectively mitigate the statistical artifacts associated with Buell patterns, resulting in more realistic representations of the underlying autocorrelation structure and thus increased interpretability [156, 213, 214]. Moreover, rotation techniques have been successfully integrated into other PCA extensions dealing with the non-linear and non-stationary nature of climate oscillations [21, 90, 108].

However, rotation techniques are not without criticism, mainly due to the ad-hoc nature of the various simplicity criteria [cf. 124, pp. 270-274]. This has led to the exploration of alternative approaches to enforce sparsity. Notably, Jolliffe *et al.* [125] introduced the Simplified Component Technique - LASSO (SCoTLASS), a sparse PCA variant based on the least absolute shrinkage and selection operator (LASSO), originally from regularized regression [258]. This method imposes a constraint on the ℓ_1 norm of the vector of loadings, resulting in some loading coefficients being exactly zero.

Despite its advantages, SCoTLASS has drawbacks, particularly when the number of variables D is much larger than the number of observations N . In such cases, SCoTLASS can only select at most N non-zero elements in the loading vectors, which is not ideal in climate science where often $D \gg N$. Additionally, since SCoTLASS solutions cannot be obtained via SVD and require a numerical approach using projected gradient methods [260], the computational demand is typically higher than for (rotated) PCA. While this method has inspired many other sparse PCA techniques based on additional constraints (usually the 1-norm or 2-norm) [see e.g. 63, 291, 292], these formulations are often complex and require algorithmic innovations to avoid issues like local optima and ensure convergence. They may also face challenges with memory limitations, slow computation speeds and limited effectiveness in modelling the data [25, 89].

It is likely for these reasons that rotated PCA remains one of the most commonly used techniques for achieving sparse PCA solutions in climate science. It is computationally efficient, straightforward to implement and provides a more interpretable understanding of the data compared to standard PCA.

nicrie: note: so far only minimal caption, details are missing

3.1.3 Hilbert PCA

While PCA efficiently identifies the principal axes along which the covariance among all variables is distributed, and rotated PCA provides a more sparse solution related to underlying physical processes, both methods are inherently static. They do not for temporal dynamics or changes over time, as they only consider the current state of the physical field. This limitation is

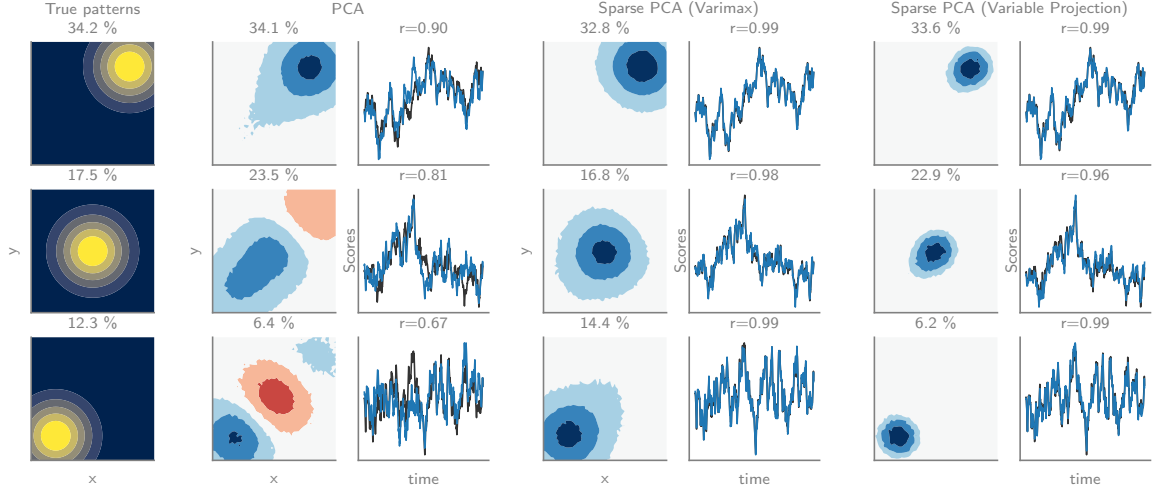


Figure 3.1 — Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

significant in climate science, where time evolution is crucial. For instance, standard PCA does not differentiate between the order of data points over time, resulting in the same eigenvectors and eigenvalues regardless of temporal permutations.

This static nature poses a challenge for capturing oscillatory patterns, such as QBO, MJO, ENSO and BSISO, which involve spatial patterns propagating over time. These phenomena cannot be adequately captured by standard PCA.

Hilbert Transform as a Momentum Operator

To address this limitation, one approach is to model both the current state of the climate and its tendency or ‘momentum’, analogous to the Hamiltonian principle in mechanics, where a system’s future is described by first-order differential equations for location (state) and momentum (tendency) [cf. 244, p. 353]. In classical mechanics, propagating features are represented using canonical coordinates: each state variable is augmented by a momentum variable, allowing oscillations to be depicted as rotations in a two-dimensional phase space. From a data perspective, this augmentation can be achieved via the Hilbert transform ($\mathcal{H}$). For a real time series $x(t)$, the Hilbert transform $x'(t)$ is defined as

$$x'(t) \equiv \mathcal{H}\{x\}(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{\tau - t} d\tau, \quad (3.27)$$

where p.v. denotes the Cauchy principal value ensuring convergence despite the singularity at $\tau = t$. The Hilbert transform emphasizes the local properties of $x(t)$ by convolution with $\frac{1}{\pi t}$. In the frequency domain, it is related to the Fourier transform $\mathcal{F}$ by

$$\mathcal{F}\{x'\}(\omega) = -i \operatorname{sgn}(\omega) \mathcal{F}\{x\}(\omega) \quad (3.28)$$

where the imaginary unit i indicates a phase shift of $\pi/2$ for the frequency components ω of x , placing x' in quadrature to x . This provides an intuitive interpretation of the Hilbert transform, as it acts as a linear filter that alters the phase of the signal's components by 90 degrees without changing their amplitude.

By using the Hilbert transform, a real-valued time series $x(t)$ can be complexified into $z(t) = x(t) + i\mathcal{H}\{x\}(t)$, known as the *analytic signal*, which allows for the computation of the instantaneous amplitude and phase of $x(t)$:

$$A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x\}(t)^2}, \quad \phi(t) = \arg(z(t)) = \arctan\left(\frac{\mathcal{H}\{x\}(t)}{x(t)}\right). \quad (3.29)$$

Similar results can be obtained for the discrete case [188, pp. 942-956], and this concept can be extended to multivariate data [e.g. 96, pp. 105-106], defining the analytic signal of a data matrix X as

$$Z = X + i\mathcal{H}\{X\}, \quad (3.30)$$

where $\mathcal{H}\{X\}$ is the feature-wise Hilbert transform of X .

Rasmusson *et al.* [208] introduced the concept of Hilbert PCA using this data augmentation technique⁹. This method, refined by Barnett [10] and Horel [108], involves applying PCA to the augmented data matrix Z . By incorporating phase information, Hilbert PCA can capture the direction and speed of wave-like patterns, unlike traditional PCA, which only identifies standing oscillations. The resulting eigenvectors and principal component scores are complex, with the real and imaginary components linked via the Hilbert transform. The variance captured by each component is represented by the real eigenvalues of the complex covariance matrix of Z (see Section 3.1).

⁹Although Rasmusson *et al.* [208] originally termed it *Hilbert Singular Decomposition*, the terms *Hilbert EOF/PCA* and *Complex EOF* analysis are more commonly used in the literature [23, 96, 124, 244].

The main advantage of Hilbert PCA is the ability to identify and reconstruct propagating waves, which is especially valuable in studying climate phenomena characterized by such patterns, including MJO, ENSO, QBO and BSISO. For instance, Cai & Baines [24] utilized Hilbert PCA to analyze SST in the Southern Hemisphere, linking the findings to ENSO, while Bürger [23] employed it to develop an index for BSISO. However, to reliably identify oscillatory patterns, sufficiently long time series are required to resolve at least one full oscillation [171]. Additionally, interpreting of the Hilbert transform as representing the instantaneous frequency is valid holds for narrowband signals; the phase of the Hilbert transform is not well-defined for broadband signals [16, 72]. Moreover, real-time monitoring is challenging with Hilbert PCA because new real-valued data cannot be directly projected onto the existing patterns without knowing or estimating the exact momentum (phase) of the new data, which requires future information (Equation (3.27)). To mitigate this, an approximative real-time (causal) Hilbert transform can be employed [22, 23], offering a practical remedy to the issue of real-time applicability. Finally, Hilbert PCA shares PCA's limitations regarding physical interpretability due to the orthogonality of the VCs. To address this, rotated Hilbert PCA has been proposed to enhance interpretability [15, 109].

3.2 Maximum Covariance Analysis

In contrast to finding latent variables that maximize variance within a single dataset, Maximum Covariance Analysis (MCA) aims to identify the latent variables that are common in two sets of data and maximize covariance. For example, in the case of the MJO [164], which is the dominant mode of intraseasonal variability in the tropical atmosphere, characterized by an eastward-moving cell of overturning wind circulation and associated regions of enhanced and suppressed precipitation, MCA allows us to focus on shared variability between different variables while ignoring dominant variations associated with only one variable, such as diurnal cycles linked to surface processes absent at higher atmosphere levels. MCA gained popularity in the field of meteorology¹⁰ through the works of Bretherton *et al.* [17] and Wallace *et al.* [271], since when it has been extensively applied [e.g. 13, 153, 180, 209, 261, 271].

MCA identifies patterns of maximum covariance between datasets by finding pairs of linear subspaces¹¹ with high covariance [e.g. 244, p. 321]. Unlike PCA, which operates on a single

¹⁰According to Jolliffe [124, p. 225], the technique was first used in psychology in the 1950s under the name *inter-battery factor analysis*. As a variant of Partial Least Squares (PLS), it was later applied in biomedical sciences by Sampson *et al.* [227] and was referred to as *PLS-SVD* by Wegelin [277]. In meteorology, Bretherton *et al.* [17] introduced the term *SVD analysis*, which remains in use today [e.g. 180]. Occasionally, the method is also called *Canonical Covariance Analysis*, similar to Canonical Correlation Analysis [e.g. 35, 96].

¹¹Cherry [36] provided an interesting, complementary view: MCA can be seen as two PCAs followed by rotation to align the results of the two analyses as closely as possible. Based on this view, it becomes immediately clear that MCA reduces to PCA when both sets are identical.

set of variables, MCA considers two¹² sets of variables and seeks components that exhibit high covariance across datasets, sometimes called cross-covariance. This approach can be formulated as an optimization problem, aiming to find pairs of (complex-valued) coupled patterns $\mathbf{v}_{s,1}$ for datasets $\mathbf{Z}_s$, $s = \{x, y\}$, such that the projection maximizes the covariance between the datasets:

$$\underset{\mathbf{v}_{x,1}, \mathbf{v}_{y,1}}{\operatorname{argmax}} \mathbf{v}_{x,1}^H \mathbf{C}_{xy} \mathbf{v}_{y,1} \quad \text{subject to} \quad \|\mathbf{v}_{x,1}\|_2 = \|\mathbf{v}_{y,1}\|_2 = 1, \quad (3.31)$$

where the cross-covariance matrix between the two datasets is given by

$$\mathbf{C}_{xy} = \frac{1}{N-1} \mathbf{Z}_x^H \mathbf{Z}_y. \quad (3.32)$$

The solution to this problem gives rise to a generalized eigenvalue equations similar to that of PCA, which can also be solved by using the SVD of the cross-covariance matrix,

$$\mathbf{C}_{xy} = \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^H, \quad (3.33)$$

where the k^{th} column of the left and right singular vectors, $\mathbf{V}_x$ and $\mathbf{V}_y$, contains the pattern vectors $\mathbf{v}_{x,k}$ and $\mathbf{v}_{y,k}$, and $\mathbf{\Sigma}$ a diagonal matrix containing the singular values σ_k in descending order, with $k = 1 \dots M$ and $M = \operatorname{rank}(\mathbf{C}_{xy})$. The first left and right singular vectors correspond to the maximum covariance, given by the largest singular value σ_1 .

The associated time series of these pattern vectors, typically called expansion coefficients, $\mathbf{P}_s$ for each dataset are obtained by projecting the data onto the pattern vectors:

$$\mathbf{P}_s = \mathbf{Z}_s \mathbf{V}_s, \quad (3.34)$$

with the expansion coefficients $\mathbf{P}_s$, $s \in \{x, y\}$. Note that in MCA an exact reconstruction of the original data is in general not possible, in contrast to PCA (see Equation (3.10)), as the cross-covariance matrix has in general lower rank than the original data matrices, i.e. $\operatorname{rank}(\mathbf{C}_{xy}) \leq \min(M_x, M_y)$, where M_x and M_y are the number of variables in each dataset. In other words, the singular vectors $\mathbf{V}_x$ and $\mathbf{V}_y$ span a subspace of the original data space, capturing the shared variability between the two datasets, and thus—in general—do not provide

¹²In theory, this can be extended to more than two sets of data [129].

a complete basis for either $\mathbf{Z}_x$ or $\mathbf{Z}_y$. Equation (3.34) can be used though to reconstruct the shared variability into the original data space of each dataset:

$$\hat{\mathbf{Z}}_s \approx \mathbf{P}_s \mathbf{V}_s^H. \quad (3.35)$$

It can be shown that the expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are jointly orthogonal:

$$\frac{1}{N-1} \mathbf{P}_x^H \mathbf{P}_y \stackrel{(3.34)}{=} \frac{1}{N-1} \mathbf{V}_x^H \mathbf{Z}_x^H \mathbf{Z}_y \mathbf{V}_y \stackrel{(3.32)}{=} \mathbf{V}_x^H \mathbf{C}_{xy} \mathbf{V}_y \stackrel{(3.33)}{=} \boldsymbol{\Sigma}. \quad (3.36)$$

However, the expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are not orthogonal within themselves, meaning their covariance matrices $\mathbf{P}_s^H \mathbf{P}_s$ are generally not diagonal.

Equation (3.36) can also be interpreted as the total covariance between $\mathbf{X}$ and $\mathbf{Y}$, given by the trace of $\boldsymbol{\Sigma}$. Therefore, one may use the singular values to assess the relative importance of each mode. However, this total covariance is *not* equal to the sum of covariances between $\mathbf{X}$ and $\mathbf{Y}$ before decomposition. The reason is that covariances can be positive or negative, and summing them directly may result in cancellations. Moreover, the singular values σ_k are always non-negative, representing the magnitude of covariance captured by each mode, without accounting for sign. That is, the total covariance is *not invariant* before and after decomposition.

A more appropriate measure to assess the relative importance of each mode in capturing the shared variability between the two datasets is the *squared* covariance. The reason for this can be seen by computing the squared Frobenius norm of the cross-covariance matrix,

$$\|\mathbf{C}_{xy}\|_F^2 \equiv \left(\sum_i^N \sum_j^P |c_{ij}|^2 \right) = \sum_i^M \sigma_i^2, \quad (3.37)$$

where M is the rank of the cross-covariance matrix. The result shows that the total squared covariance of the cross-covariance matrix $\mathbf{C}_{xy}$ is an invariant and the same as the sum of the squared singular values. A formal proof for this statement is provided in Appendix ??.

Therefore, the relative importance of mode k can be quantified by the squared covariance fraction $f_{xy,k}$:

$$f_{xy,k} = \frac{\sigma_k^2}{\sum_{i=1}^M \sigma_i^2}. \quad (3.38) \quad 1$$

Alternatively, this expression can be formulated as a residual-based measure [248]: 2

$$f_{xy,k} = 1 - \frac{\|(\mathbf{Z}_x - \mathbf{Z}_{x,k})^H (\mathbf{Z}_y - \mathbf{Z}_{y,k})\|_F^2}{\|\mathbf{Z}_x^H \mathbf{Z}_y\|_F^2} \quad (3.39) \quad 3$$

where $\mathbf{Z}_{x,k}$ and $\mathbf{Z}_{y,k}$ represent the shared variability of the k^{th} mode reconstructed in the original data space using Equation (3.35). This fraction provides a measure of how much of the shared variability between the two datasets is captured by the k^{th} mode. 4
5
6

To assess the relative importance of the patterns in relation to each dataset, it is necessary to consider the variance within the individual datasets explained by each mode relative to the total variance of the individual datasets. This can be expressed using the squared Frobenius norm of the individual datasets: 7
8
9
10

$$f_{ss,k} = \frac{\mathbf{v}_{s,k}^H \mathbf{C}_{ss} \mathbf{v}_{s,k}}{\text{tr}(\mathbf{Z}_s)} = \frac{\mathbf{p}_{s,k}^H \mathbf{p}_{s,k}}{\|\mathbf{Z}_s\|_F^2}. \quad (3.40) \quad 11$$

Note that this projection-based measure of explained variance is mathematically the same in MCA as the following residual-based measure: 12
13

$$f_{ss,k} = 1 - \frac{\|\mathbf{Z}_s - \mathbf{Z}_{s,k}^2\|}{\|\mathbf{Z}_s\|_F^2} \quad (3.41) \quad 14$$

While the squared covariance fraction $f_{xy,k}$ indicates the amount of shared variability between the two datasets, it does not provide information about the strength of the association or coupling between them. To measure this, we use the correlation coefficient ρ_k for the expansion coefficients of mode k between the two datasets: 15
16
17
18

$$\rho_k = \frac{\text{cov}(\mathbf{p}_{1,k}, \mathbf{p}_{2,k})}{(\text{var}(\mathbf{p}_{1,k}) \text{var}(\mathbf{p}_{2,k}))^{1/2}} = \frac{\sigma_k}{\|\mathbf{p}_{1,k}\|^2 \|\mathbf{p}_{2,k}\|^2}. \quad (3.42) \quad 19$$

1 The homogeneous covariance patterns [17] for the pattern vectors are:

$$2 \quad \mathbf{h}_{M,x,k} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{x,k}) = \mathbf{S}_{xx} \mathbf{v}_{x,k}, \quad (3.43a)$$

$$3 \quad \mathbf{h}_{M,y,k} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{y,k}) = \mathbf{S}_{yy} \mathbf{v}_{y,k}, \quad (3.43b)$$

4 These patterns provide an indicator of which variables or features are important for the co-
5 varying components in both fields. The corresponding heterogeneous covariance patterns are
6 defined as:

$$7 \quad \mathbf{h}_{T,x,k} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{y,k}) = \sigma_i \mathbf{v}_{y,k}, \quad (3.44a)$$

$$8 \quad \mathbf{h}_{T,y,k} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{x,k}) = \sigma_i \mathbf{v}_{x,k}. \quad (3.44b)$$

9 These patterns indicate how well the variables in one field can be predicted from the k^{th}
10 expansion coefficient of the other field. One may also use standardized data to obtain the
11 respective homogeneous and heterogeneous correlation patterns.

12 3.2.1 Rotated Hilbert MCA

13 The method's orthogonal patterns ensure efficient dimensionality reduction of the covariance
14 but limit physical interpretability, similar to PCA. To enhance interpretability, rotation of
15 the MCA solution can be employed. In the following, we outline the rotation procedure as
16 proposed by Cheng & Dunkerton [31].

17 A 'loaded' composite matrix $\mathbf{L}$ of the left and right singular vectors truncated to the first R
18 modes, $\tilde{\mathbf{V}}_s$, is constructed and a simplicity criterion (e.g. varimax) is applied,

$$19 \quad \mathbf{L} = \begin{bmatrix} \tilde{\mathbf{V}}_x \\ \tilde{\mathbf{V}}_y \end{bmatrix} \Sigma^{1/2}, \quad (3.45)$$

20 where the term 'loaded' refers to the fact that the truncated singular vectors are weighted by
21 the square root of the singular values. The rotated pattern vectors $\tilde{\mathbf{V}}_{R,s}$ are then obtained by
22 applying the rotation matrix $\mathbf{R}$ to the 'loaded' $\tilde{\mathbf{V}}_s$:

$$23 \quad \tilde{\mathbf{V}}_{R,s} = \tilde{\mathbf{V}}_s \Sigma^{1/2} \mathbf{R} \mathbf{W}_s^{-1} \quad (3.46)$$

where $\mathbf{W}_s^{-1}$ is a diagonal matrix with the elements that normalize the rotated pattern vectors to unit length. Similarly, the associated rotated expansion coefficients $\tilde{\mathbf{P}}_{R,s}$ are then obtained by rotating the first R non-rotated expansion coefficients $\tilde{\mathbf{P}}_s$:

$$\tilde{\mathbf{P}}_{R,x} = \tilde{\mathbf{P}}_x \Sigma^{-1/2} \mathbf{R} \mathbf{W}_s \quad (3.47)$$

For an orthogonal rotation, the rotated expansion coefficients remain uncorrelated:

$$\begin{aligned} \tilde{\mathbf{P}}_{R,x}^H \tilde{\mathbf{P}}_{R,y} &= \left(\tilde{\mathbf{P}}_x \Sigma^{-1/2} \mathbf{R} \mathbf{W}_x \right)^H \left(\tilde{\mathbf{P}}_y \Sigma^{-1/2} \mathbf{R} \mathbf{W}_y \right) \\ &= \mathbf{W}_x \mathbf{R}^H \Sigma^{-1/2} \tilde{\mathbf{P}}_x^H \tilde{\mathbf{P}}_y \Sigma^{-1/2} \mathbf{R} \mathbf{W}_y \\ &= (N - 1) \mathbf{W}_x \mathbf{W}_y. \end{aligned} \quad (3.48)$$

while the unitary property of the rotated pattern vectors is removed:

$$\begin{aligned} \tilde{\mathbf{V}}_{R,x}^H \tilde{\mathbf{V}}_{R,y} &= \left(\tilde{\mathbf{V}}_x \Sigma^{1/2} \mathbf{R} \mathbf{W}_x \right)^H \left(\tilde{\mathbf{V}}_y \Sigma^{1/2} \mathbf{R} \mathbf{W}_y^{-1} \right) \\ &= \mathbf{W}_x \mathbf{R}^H \Sigma^{1/2} \tilde{\mathbf{V}}_x^H \tilde{\mathbf{V}}_y \Sigma^{1/2} \mathbf{R} \mathbf{W}_y \\ &= \mathbf{W}_x \mathbf{R}^H \Sigma \mathbf{R} \mathbf{W}_y. \end{aligned} \quad (3.49)$$

To assess the relative importance of the rotated modes, Cheng & Dunkerton [31] proposed to use the fraction of covariance explained, defined as

$$f_{CD,k} = \frac{\sigma_k}{\sum_{i=1}^M \sigma_i}. \quad (3.50)$$

Cheng & Dunkerton [31] argued that they chose this measure to be consistent with PCA and rotated PCA, although they acknowledge that it is the total squared covariance which is an invariant in MCA.

In this thesis, we demonstrate that while the total covariance between two datasets is preserved under rotation, the total *squared* covariance is not. To the best of our knowledge, this distinction has not been previously reported. This finding has practical implications for interpreting rotated MCA modes.

1 Total Squared Covariance Is Not Preserved Under Rotation

2 Our goal is to verify whether the total squared covariance of the first R rotated modes of
3 datasets $\mathbf{Z}_x$ and $\mathbf{Z}_y$ equals that of the original modes obtained from MCA. We'll show that,
4 except in trivial cases, it does not.

5 First, recall that the total squared covariance between $\mathbf{Z}_x$ and $\mathbf{Z}_y$ is given by the sum of the
6 squared singular values σ_k^2 of their cross-covariance matrix $\mathbf{C}_{xy}$ (Equation (3.37)):

$$\|\mathbf{C}_{xy}\|_F^2 = \sum_{k=1}^M \sigma_k^2. \quad (3.51)$$

8 Next, according to Equation (3.48), the covariance of the rotated expansion coefficients $\tilde{\mathbf{P}}_{R,x}$
9 and $\tilde{\mathbf{P}}_{R,y}$ is given by:

$$\frac{1}{N-1} \tilde{\mathbf{P}}_{R,x}^H \tilde{\mathbf{P}}_{R,y} \stackrel{(3.48)}{=} \mathbf{W}_x \mathbf{W}_y. \quad (3.52)$$

11 To relate $\mathbf{W}_x$ and $\mathbf{W}_y$ to the original singular values σ_k , we use the normalization condition
12 for the k^{th} rotated pattern vector $\tilde{\mathbf{V}}_{R,s,k}$, for $s \in \{x, y\}$:

$$|\tilde{\mathbf{V}}_{R,s,k}|^2 = 1. \quad (3.53)$$

14 Using the expression for the rotated pattern vectors, we have:

$$\begin{aligned} 1 &= \|\tilde{\mathbf{V}}_{R,s,k}\|^2 \\ &= \tilde{\mathbf{V}}_{R,s,k}^H \tilde{\mathbf{V}}_{R,s,k} \\ &\stackrel{(3.46)}{=} \left(\tilde{\mathbf{V}}_s^H \Sigma^{1/2} \mathbf{R}_{:,k} (\mathbf{W}_s^{-1})_{kk} \right)^H \left(\tilde{\mathbf{V}}_s^H \Sigma^{1/2} \mathbf{R}_{:,k} (\mathbf{W}_s^{-1})_{kk} \right) \\ &= (\mathbf{W}_s^{-1})_{kk}^2 \left(\mathbf{R}_{:,k}^H \Sigma \mathbf{R}_{:,k} \right), \end{aligned}$$

19 where $\mathbf{R}_{:,k}$ is the k^{th} column of the rotation matrix $\mathbf{R}$.

20 Solving for $(\mathbf{W}_s)_{kk}$, we find:

$$(\mathbf{W}_s)_{kk}^2 = \left(\mathbf{R}_{:,k}^H \Sigma \mathbf{R}_{:,k} \right). \quad (3.54) \quad 1$$

This equation holds for both $\mathbf{W}_x$ and $\mathbf{W}_y$ implying that $\mathbf{W}_x = \mathbf{W}_y \equiv \mathbf{W}$. Therefore, the covariance between the rotated expansion coefficients simplifies to $\mathbf{W}^2$. 2
3

The total squared covariance of the first R rotated modes is then: 4

$$\begin{aligned} \text{tr } \mathbf{W}^4 &= \sum_{k=1}^r \left((\mathbf{R}^H \Sigma \mathbf{R})_{kk} \right)^2 & 5 \\ &\leq \text{tr} \left((\mathbf{R}^H \Sigma \mathbf{R})^2 \right) & 6 \\ &= \text{tr} \left(\Sigma^2 \right) & 7 \\ &= \sum_{k=1}^r \sigma_k^2. & (3.55) \quad 8 \end{aligned}$$

The inequality arises because the sum of the squares of the diagonal elements is less than or equal to the trace of the squared matrix. This shows that the total squared covariance after rotation is generally less than before rotation, unless the rotation matrix $\mathbf{R}$ is the identity matrix. Thus, total squared covariance is not preserved under rotation. 9
10
11
12

Total Covariance Is Preserved Under Rotation 13

In contrast, the total covariance (without squaring) remains unchanged under rotation. Here's how we can show this: 14
15

$$\begin{aligned}
& \text{tr } \mathbf{W}^2 = \sum_{k=1}^R (\mathbf{R}^H \Sigma \mathbf{R})_{kk} \\
& = \sum_{k=1}^R \left(\sum_{i=1}^R \mathbf{r}_{ik}^H \Sigma_{ii} \mathbf{r}_{ik} \right) \\
& = \sum_{k=1}^R \left(\sum_{i=1}^R \sigma_i r_{ik}^2 \right) \\
& = \sum_{i=1}^R \sigma_i \left(\sum_k r_{ik}^2 \right) \\
& = \sum_{i=1}^R \sigma_i.
\end{aligned} \tag{3.56}$$

where we used the orthonormality of the rotation matrix $\mathbf{R}$, which ensures that for the i^{th} row $\mathbf{r}_i$ of $\mathbf{R}$, the sum of the squares of its elements equals ones ($\sum_k r_{ik}^2 = 1$). This shows that the total covariance is preserved under rotation.

The implication is that Equation (3.50) can be used to assess the relative importance of each rotated mode. However, it is important to interpret this with caution. While total covariance remains invariant before and after rotation, it does not hold as an invariant for MCA itself. Therefore, Equation (3.50) should be viewed as a general measure of the relative importance of each mode rather than the amount of explained covariance between Z_x and Z_y .

Unfortunately, because total squared covariance is not preserved under rotation, we encounter a limitation. In MCA, total squared covariance is an invariant, but this invariance does not hold after rotation. Conversely, total covariance is preserved under rotation but is not an invariant within MCA. For practical applications of this method (discussed in Section ??), we choose to use the SCF as a measure of relative importance, acknowledging that it does not sum to one after rotation.

Finally, before applying rotated MCA to real-world data, we must make two key decisions: (1) determining the optimal number of modes to rotate, and (2) choosing between orthogonal or oblique rotation.

Determining the Optimal Number of Rotated Modes

Determining the optimal number of modes to retain and rotate is crucial for capturing relevant patterns without introducing noise from higher-order modes [213]. To address this challenge, we introduce the *mode stability index*, denoted as κ , which provides an objective criterion for selecting the appropriate number of modes to rotate.

The mode stability index quantifies the consistency of mode patterns across different rotation solutions. The fundamental idea is that if the rotation solution is stable, the leading modes' patterns should remain consistent even as the number of rotated modes varies within a reasonable range. This consistency indicates that the essential structures have been captured and that including additional modes does not significantly alter the results.

To compute the mode stability index, we proceed as follows:

1. **Generate multiple rotation solutions:** We perform rotations on the first R modes for a range of R values. The values of $R \in [R_{\min}, R_{\max}]$ should cover a broad range to ensure that the most stable solution is identified. The range $[1, R_{\min}]$ defines the set of modes to be assessed for stability.
2. **Extract pattern vectors:** For each rotated solution, we extract the amplitude pattern vectors, $|\mathbf{v}_k^R|$ of the leading modes, where $k \in [1, R]$ indicates the mode number.
3. **Measure pairwise similarity between patterns:** We compare the amplitude patterns of mode k across different rotation solutions using the cosine similarity. The cosine similarity between two patterns $\mathbf{v}_k^R$ and $\mathbf{v}_{k'}^{R'}$ is given by:

$$c_{k,k'}^{R,R'} = \frac{|\mathbf{v}_k^R|^T |\mathbf{v}_{k'}^{R'}|}{\|\mathbf{v}_k^R\| \cdot \|\mathbf{v}_{k'}^{R'}\|}. \quad (3.57)$$

The cosine similarity ranges from -1 to 1 , where 1 indicates identical patterns, 0 indicates orthogonal patterns, and -1 indicates patterns that are exact opposites.

4. **Determine mode similarity:** For a given mode k in rotation solution R , we identify the most similar mode k' in rotation solution R' by maximizing the cosine similarity:

$$C_k^{R,R'} = \max_{k' \in [1, R_{\min}]} (c_{k,k'}^{R,R'}). \quad (3.58)$$

To consider the fact that the order of modes may vary slightly across different rotation solutions, we maximize the cosine similarity over all leading modes, i.e. the range $[1, R_{\min}]$.

5. **Define an indicator for consistency:** We introduce an indicator function $\delta_k^{R,R'}$ to signify whether the patterns are sufficiently similar:

$$\delta_k^{R,R'} = \begin{cases} 1, & \text{if } C_k^{R,R'} > \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (3.59)$$

Here, τ is a threshold value (e.g. 0.9) chosen to represent a high degree of similarity.

6. **Compute the mode stability index:** The mode stability index κ for the rotation solution R is calculated by averaging the indicator function over all pairs of rotation solutions and modes:

$$\kappa^R = \frac{1}{R_{\min}(R_{\max} - R_{\min})} \sum_{R'=R_{\min}}^{R_{\max}} \sum_{k=1}^{R_{\min}} \delta_k^{R,R'}. \quad (3.60)$$

The mode stability index κ represents the average fraction of modes that remain consistent across different rotation solutions. A higher value of κ (closer to 1) indicates greater stability, meaning the mode patterns are robust to changes in the number of rotated modes. Conversely, a lower κ suggests that the mode patterns are sensitive to the number of modes rotated, indicating potential instability or the influence of noise. By plotting κ against the number of modes rotated, we can identify a point where κ reaches a maximum or plateaus.

Selecting Between Orthogonal and Oblique Rotation

Choosing between orthogonal and oblique rotation involves balancing interpretability with complexity. Orthogonal rotations keep the expansion coefficients uncorrelated, while producing more localized spatial patterns, which may simplify their interpretation. However, the lack of correlation between coefficients may not accurately reflect the interconnected nature of geophysical processes. In contrast, oblique rotations allow the expansion coefficients to be correlated, potentially capturing more realistic relationships, while still producing simpler spatial structures. This added realism comes at the cost of increased complexity in interpretation.

In this thesis, we compared the outcomes of both orthogonal and oblique rotations to evaluate the degree of correlation between the expansion coefficients. When the correlation between leading modes was weak (for example, $< \pm 0.2$), we opted for orthogonal rotation to maintain simplicity and clarity in the results.

Hilbert MCA

Beyond rotation for better interpretability, Hilbert MCA can be applied to capture the temporal evolution of coupled patterns, akin to Hilbert PCA (see Section 3.1.3). This is especially useful for analyzing propagating patterns like the MJO, which involve covarying patterns in different climate variables, such as wind speed in the troposphere and lower stratosphere, and precipitation. By augmenting data with the Hilbert transform, we obtain complexified fields and solve the maximization problem (Equation (3.31)) for these augmented data, capturing the temporal evolution of coupled patterns. So far, it seems that Hilbert MCA has seen limited application in climate science [e.g. 58].

3.2.2 From Covariance to Correlation: The CPCCA Framework

While the MCA framework effectively identifies shared variability between two datasets by maximizing squared covariance, it does not necessarily optimize correlation. The distinction between covariance and correlation becomes particularly significant when the datasets operate on vastly different scales. Moreover, there are scenarios where maximizing correlation is more important than achieving high covariance.

Canonical Correlation Analysis (CCA) [111] addresses this by identifying shared patterns that maximize correlation:

$$\operatorname{argmax}_{\mathbf{v}_{x,1}, \mathbf{v}_{y,1}} \mathbf{v}_{x,1}^H \mathbf{C}_{xy} \mathbf{v}_{y,1} \quad \text{subject to} \quad \left\| \mathbf{v}_{x,1}^H \mathbf{Z}_x \right\|_2 = \left\| \mathbf{v}_{y,1}^H \mathbf{Z}_y \right\|_2 = 1, \quad (3.61)$$

Unlike MCA, where pattern vectors $\mathbf{V}_s$ are orthogonal, resulting in correlated expansion coefficients, CCA produces pattern vectors that are not necessarily orthogonal but yield uncorrelated expansion coefficients, known as canonical variates.

However, a common drawback of CCA is that its leading modes may exhibit strong correlations while accounting for only a small portion of the overall variance in each dataset. To mitigate this, CCA is often performed in a reduced PC space, focusing on the most significant modes of variability and reducing computational demands [244, pp. 320-321]. Despite these advantages, this approach can overlook important correlations involving lower-variance PCs, as noted by Jolliffe [124, p. 223].

In practice, a balance is desired between maximizing covariance (as in MCA) and maximizing correlation (as in CCA). To achieve this, Swenson [248] introduced the Continuum Power Canonical Correlation Analysis (CPCCA) framework [248]. CPCCA partially whitens¹³ the datasets using a fractional whitening parameter, α_s , allowing for a continuous adjustment between maximizing correlation (CCA) and maximizing covariance (MCA). This provides a flexible framework to identify shared patterns between datasets.

In this thesis, we extended the CPCCA framework similarly to our approach with MCA by applying it to Hilbert complexified data and incorporating an additional rotation step for enhanced interpretability, resulting in rotated Hilbert CPCCA. We utilized both MCA and CPCCA to identify and interpret ocean-atmospheric teleconnections in the climate system (Section ??), and the Madden-Julian Oscillation (Section ??).

nicrie: add PCA preprocessing as a common step in CPCCA

¹³Whitening, also known as sphering, refers to the process of transforming a dataset so that its features or variables are uncorrelated and each has unit variance. This often involves adjusting the data to have a zero mean and an identity covariance matrix. See DelSole & Tippet [51] for more details on its relevance in climate science.

Gaussian Process Regression

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

4.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

4.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

4.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

4.3.1 Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

4.3.2 Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

4.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. Inference and Prediction: Explain how to perform inference and make predictions using the fitted model.

Part II

Pattern Identification

in the Dense Data Regime

1

2

3

4

Bridging Theory and Practice with Advanced Dimensionality Reduction in Python

Over the pas few decades, Earth system sciences have experienced an exponential increase in the volume and complexity of data [193], driven by the proliferation of Earth observations satellites [84], high-resolution climate models [99] and citizen science contributions [71]. As a result, researchers must now navigate the immense challenge of analyzing vast datasets to extract meaningful insights about Earth’s climate system [230]. Dimensionality reduction techniques have become indispensable for addressing this challenge by simplifying complex data and revealing key underlying patterns.

In the previous chapter, we explored the theoretical foundations of several dimensionality reduction techniques commonly applied in Earth system sciences. We demonstrated how common PCA extensions can enhance MCA, yielding more physically interpretable patterns that reflect the shared dynamics between two datasets. This chapter we shift focus to the practical challenges that arise when applying these techniques to the increasingly large Earth system datasets and the solutions provided by open-source software. In particular, Section 5.3 will introduce the XEOFS Python library, which is designed to facilitate scalable dimensionality reduction with a particular focus on climate science. This package provides user-friendly access to both traditional and advanced methods, enabling researching to conduct sophisticated analyses without needing extensive expertise in data science. This section is based upon the work presented in:

- **Rieger, N.** & Levang, S. J. xeofs: Comprehensive EOF Analysis in Python with Xarray. *Journal of Open Source Software* 9, 6060. doi:10.21105/joss.06060 (2024)

5.1 Open-Source Tools for Scalable Climate Data Analysis

Climate science is inherently data-intensive, relying on the analysis of large, complex datasets to understand the Earth’s climate system. However, traditional analysis methods often struggle to handle the scale and complexity of modern climate data, making the application of these techniques increasingly difficult [65]. This creates a growing need for tools that lower the barrier to entry for non-experts, allowing them to manage and analyze vast datasets effectively.

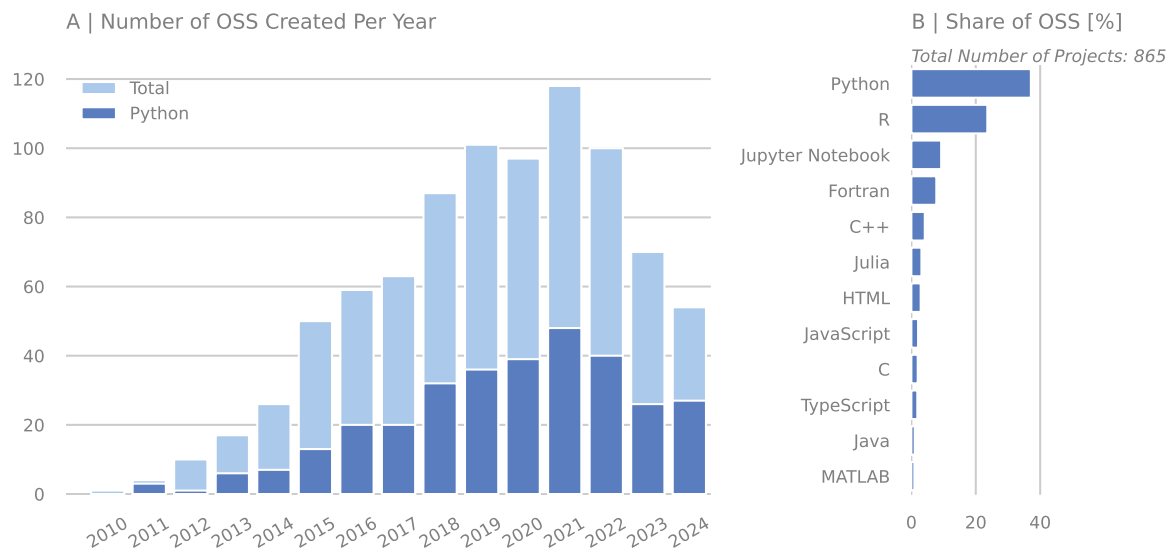


Figure 5.1 — Open-source software (OSS) projects in Earth system sciences. Panel A illustrates the number of new OSS projects created annually, highlighting the growing importance of Python-based projects. Panel B shows the relative share of programming languages used in all OSS projects since 2010, excluding languages with a share of $\leq 0.5\%$. The OSS project data is sourced from *Open Sustainable Technology* (OST) [7]. For this analysis, Earth system science projects include the OST categories *Atmosphere*, *Climate Change*, *Cryosphere*, *Hydrosphere* and *Biosphere*. Data for 2024 includes only the first half of the year, up to August.

One of the central challenges in climate science today is the "data deluge" – the overwhelming volume of climate data and the difficulty of extracting useful insights from it. As Schnase *et al.* [231] noted, 'data analytics, the capacity to perform useful scientific analyses over large quantities of data in reasonable amounts of time' is a critical challenge for climate science. Without effective tools to transform large datasets into actionable knowledge, the mission of climate science falters.

To ensure that data analysis remains accessible to the broader scientific community, free, robust, and scalable software tools are essential [56]. These tools must allow researchers to work with large datasets, even without specialized programming skills, and simplify the processing of data produced by climate models.

It therefore comes as no surprise that the open-source software (OSS) movement has gained significant traction in the scientific community (Figure 5.1), with researchers increasingly recognizing its benefits for fostering collaboration, enhancing transparency, and ensuring reproducibility in research [121, 134, 175]. Python, in particular, has become a preferred language for scientific computing (Figure 5.1 B), presumably due to its ease of use, extensive library support and active developer community. Notable examples of successful Python-based OSS projects include NUMPY [97, 186], SCIPY [186, 269] and MATPLOTLIB [116] libraries, which have become essential tools for scientific computing and visualization contributing

to the democratization of scientific tools, driving innovation and facilitating new scientific discoveries [241].

The *Pangeo* project [194] has been especially influential in promoting open-source tools for big data geoscience. Two key components of the Pangeo ecosystem are the `XARRAY` [112] and `DASK` [47] Python libraries. `XARRAY` enhances the user experience by introducing labeled data structures that simplify the manipulation of multidimensional arrays, while `DASK` enables parallel and distributed computing. Together, these tools have revolutionized the way climate researchers handle large datasets, allowing them to focus on scientific questions rather than technical hurdles.

5.2 Challenges in Large-Scale and Multidimensional Climate Data Analysis

Analyzing climate data often involves multivariate statistical techniques to identify patterns within large, multidimensional datasets. In Section ??, we only explored a small subset of available methods for pattern identification in climate science [see e.g. 96, 114]. Most existing techniques assume that data can be structured as a two-dimensional (2D) matrix. However, climate data, whether from Earth observations or simulations, often exhibit two key characteristics:

1. **Multi-dimensionality:** Climate data span several dimensions, including spatial (longitude, latitude, height), temporal (time steps, forecast lead times), and categorical (variables, sensors).
2. **Large Volume:** These datasets are often so large that they exceed the memory capacity of a single computer and even of high performance computing (HPC) infrastructures.

These characteristics have led to the widespread adoption of tools like `XARRAY` and `DASK` which facilitate the handling of large, multidimensional datasets in the Python ecosystem.

However, applying dimensionality reduction techniques such as PCA and MCA (Section ??) presents several challenges. Climate datasets typically require significant pre-processing, including variable weighting (Section 3.1), removing invalid grid points (e.g., land cells in sea surface temperature data), and reshaping multidimensional arrays into 2D matrices. After analysis, researchers must reverse this process – transforming the 2D results back into their original multidimensional form, reinserting missing data (such as land grid cells), and removing weights. This workflow can be cumbersome, error-prone, and time-consuming.

Earlier tools, such as the `EOFS` and `xMCA` libraries, addressed PCA and MCA applications in climate science. However, they lack full compatibility with `XARRAY` and `DASK` limiting their scalability and effectiveness when dealing with large datasets. For example, these libraries do not support `XARRAY`'s `Dataset` object, and neither leverages randomized linear algebra

techniques [62, 92], which can significantly improve computational performance for large datasets.

In addition, researchers often need more than dimensionality reduction. Assessing the statistical significance of the extracted patterns, for example, requires bootstrapping techniques [57], adding further complexity to the analysis process. Implementing these advanced methods often requires substantial coding expertise, which can be a barrier for many scientists [284].

Moreover, several classical methods developed over the years for pattern extraction in climate science have not yet been implemented in Python or integrated with XARRAY and DASK. This lack of availability limits researchers' ability to apply these well-established techniques to their datasets.

5.3 The xeoofs Library: A Solution For Dimensionality Reduction in Climate Science

The XEOFS library [216] was developed to address these challenges by providing a Python package for applying dimensionality reduction techniques in climate science. Built with compatibility in mind, XEOFS integrates seamlessly with XARRAY and DASK, making it suitable for large, multidimensional datasets. The library offers a user-friendly interface to both traditional and advanced methods commonly used in climate data analysis (Table 5.1). Its development adheres to key principles of open-source, scientific programming [8, 207], including clean coding practices, rigorous test design [117], version control [200], robustness [252], usability [160], and thorough documentation¹ [147]. The library also follows best practices in open-source software development. For Python specifically, XEOFS aligns with the guidelines set forth by the Pangeo project [194] and the *pyOpenSci* Packaging Guide [273].

Key features of the XEOFS library include:

Multi-dimensional & Labeled Data Designed specifically for XARRAY objects, XEOFS retains data labels and dimensional structures throughout the analysis, which helps to maintain clarity and to reduce errors.

Out-of-Memory Computations Leveraging DASK, XEOFS can process large datasets that exceed the memory capacity of a single machine or HPC systems.

Computational Efficiency By employing randomized linear algebra, XEOFS dramatically improves the efficiency of dimensionality reduction techniques, particularly for large datasets. Benchmark comparisons show that XEOFS is approximately an order of magnitude faster than EOFs for medium to large datasets (Figure 5.2).

Additionally, XEOFS offers several auxiliary features:

¹The XEOFS online documentation is available at <https://xeofs.readthedocs.io>

Table 5.1 — Overview of available methods in the XEOFS library [216] (as of September 2024) for single-set (one dataset), cross-set (two datasets), and multi-set (more than two datasets) analysis. PCA: Principal Component Analysis; POP: Principal Oscillation Pattern; OPA: Optimal Persistence Analysis; CCA: Canonical Correlation Analysis; RDA: Redundancy Analysis; MCA: Maximum Covariance Analysis; CPCCA: Continuum Power CCA. Rotation techniques include orthogonal varimax [127] and oblique promax [102]. Combinations like rotated Hilbert PCA [108] are also supported. Multi-set CCA is based on the MCCA class of the CCA-ZOO library [27] and Sparse PCA uses the RISTRETTO library [62].

	METHOD	REFERENCE
SINGLE-SET ANALYSIS	PCA	[111, 161, 196]
	Complex PCA	[137, 244]
	Hilbert PCA	[10, 109, 208]
	Rotated PCA	[109, 212]
	Extended PCA	[19, 275]
	GWPCA	[98]
	Sparse PCA	[63]
	POP	[101]
	OPA	[49, 50]
CROSS-SET ANALYSIS	CCA	[17, 111]
	RDA	[244]
	MCA	[17, 244]
	CPCCA	[248]
	Complex CCA	
	Complex RDA	
	Complex MCA	
	Hilbert CCA	
	Hilbert RDA	
	Hilbert MCA	[58]
	Rotated MCA	[31]
	Rotated CPCCA	
MULTI-SET ANALYSIS	CCA	[129]
	Regularized CCA	[268]

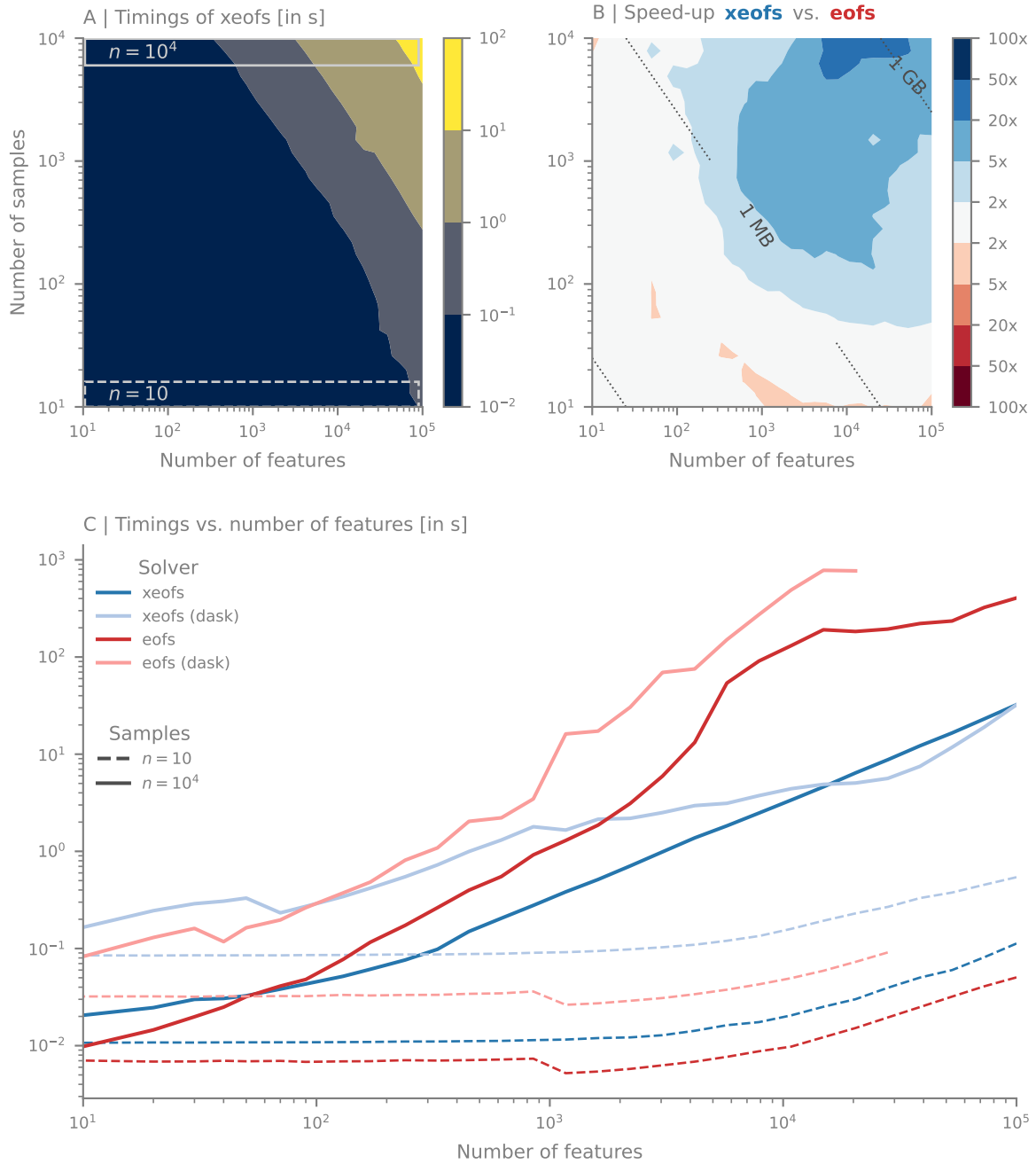


Figure 5.2 — Computational performance of PCA using the XEOFS library (version 3.0.0). Panel A displays computation times (in seconds) for varying dataset sizes, with different numbers of observations (samples) and variables (features). Panel B illustrates the speedup achieved by XEOFS compared to the EDFS library (version 1.4.1). Panel C presents the performance of delayed (lazy) Dask computation versus eager computation in both small data ($n = 10$ samples) and large data ($n = 10,000$ samples) scenarios. The results demonstrate a significant efficiency gain for medium to large datasets (greater than 1 MB) where XEOFS outperforms EDFS by approximately an order of magnitude. Additionally, XEOFS's Dask implementation for chunked out-of-memory data offers further improvements for larger datasets. Interestingly, this efficiency gain is observed primarily for datasets with a balanced number of features and samples, while performance does not significantly improve for extremely "tall-and-skinny" or "short-and-wide" matrices. Notably, the Dask implementation in EDFS performs worst on larger datasets. For small datasets (less than 1 MB), the Python overhead is more pronounced, with EDFS being slightly more efficient, although the difference is generally less than a factor of two. The benchmark was conducted on a system with Dual Intel Xeon Gold 6248R 24C 3.0 GHz processors and 192 GB of RAM, running Rocky Linux 8.

Handling Missing Values The library can manage typical cases of missing data, such as masked ocean or land cells.

Weighted Analysis XEOFS supports weighted analyses, allowing researchers to quickly apply common weighting schemes (e.g. standard deviation or latitude-based cosine weighting, see Section 3.1). It also supports custom weighting schemes.

Significance Testing A built-in bootstrapping interface enables users to test the significance of extracted modes, reducing the need for additional coding.

Model Serialization Trained models can be saved and loaded in HDF5, netCDF4 or cloud-optimized Zarr [174] formats ensuring compatibility with standard climate data formats [100].

The XEOFS library is modular and object-oriented, allowing for future extensions and integration of new dimensionality reduction methods. It provides an intuitive application programming interface (API) similar to the popular machine learning SCIKIT-LEARN library [197] that simplifies the application of these techniques, even for researchers with limited programming experience.

The library is available on the Python Package Index (PyPI) and can be installed via pip and conda. Its source code is hosted on GitHub, where users can contribute to its development and report any issues. Licensed under the MIT license, XEOFS is free to use and modify for both academic and commercial purposes.

5.4 Conclusion

The development of the XEOFS library represents a contribution toward addressing the challenges of scalable and efficient dimensionality reduction in climate data analysis. However, there are several opportunities for further refinement and expansion to better serve the needs of the climate science research community.

Future updates to XEOFS could introduce additional state-of-the-art methods that have recently gained attention. Nonlinear dimensionality reduction techniques, such as those proposed by Bueso *et al.* [21] as well as spectral methods as proposed by Guilloteau *et al.* [90], would enable researchers to capture more complex relationships within climate datasets. Incorporating approaches like nonlinear PCA, kernel-based methods [232], and advanced sparse algorithms to improve interpretability [89, 237] would provide users with a broader toolkit to explore the intricate dynamics often present in climate data.

Another area for improvement is the expansion of bootstrapping techniques for evaluating the statistical significance of extracted patterns, particularly in MCA and rotated PCA. Bootstrapping helps researchers assess the robustness of their findings, distinguishing meaningful

1 patterns from noise. By integrating more advanced bootstrapping algorithms, such as those
2 discussed by Timmerman *et al.* [259], XEOFS could simplify the workflow for significance
3 testing, making these analyses more accessible without requiring additional coding expertise.
4 The next chapter will focus on applying dimensionality reduction techniques to real-world
5 climate datasets, highlighting the practical use of XEOFS. These applications will demonstrate
6 how the package can help extract meaningful insights from complex datasets. by exploring
7 interactions between the ocean and atmosphere.

Hilbert Maximum Covariance Analysis of the Madden-Julian Oscillation

nicrie: analysis done; to be added if time allows

Shared Variability between Temporally Lagged SST and Continental Precipitation

The Earth's climate system is highly complex, with various components interacting across different spatial and temporal scales. A key aspect of these interactions is atmospheric teleconnections—large-scale patterns of pressure and circulation anomalies that link climate and weather conditions across distant regions, often thousands of kilometers apart. These teleconnections enable climate variations in one area to influence weather and climate in far-reaching locations. For example, the El Niño-Southern Oscillation (ENSO) significantly affects global weather patterns, altering rainfall and temperature distributions far from the Pacific Ocean where it originates [221, 229]. Similarly, the North Atlantic Oscillation (NAO) impacts weather in Europe and North America.

A critical feature of atmospheric teleconnections is their lagged nature; the influence of a climatic anomaly in one region on another does not occur instantaneously but with a temporal delay. For instance, the atmospheric response to ENSO typically manifests several months after the peak of an El Niño event, with the maximum response in the tropical North Atlantic occurring approximately five months later [61, 130, 228]. Understanding these delayed relationships is essential for enhancing the predictability of climate variability, thereby improving climate forecasts from sub-seasonal to decadal timescales [243, 289]. Additionally, incorporating lagged teleconnections into climate models leads to more accurate simulations and projections by accounting for previously unrecognized interactions.

While teleconnections have been extensively studied across various timescales—from intraseasonal [e.g. 243, 289] to decadal [e.g. 155]—many studies have employed hypothesis-driven approaches. There is a growing need for data-driven, exploratory analyses that can identify dominant patterns of shared variability without prior assumptions. This chapter addresses this need by examining the shared variability between sea surface temperatures (SST) and continental precipitation across seasonal to decadal timescales. We utilize rotated Hilbert Maximum Covariance Analysis (MCA; see Section 3.2) to uncover and interpret the dominant patterns of covariability, implicitly capturing lagged teleconnections through the Hilbert transform and enhancing interpretability through mode rotation.

Preliminary results from this research were published in

- **Rieger, N. et al.** Lagged Teleconnections of Climate Variables Identified via Complex Rotated Maximum Covariance Analysis. *Journal of Climate* **34**, 9861–9878. doi:[10.1175/JCLI-D-21-0244.1](https://doi.org/10.1175/JCLI-D-21-0244.1) (2021),

demonstrating the effectiveness of Hilbert MCA in identifying lagged teleconnections between SST and precipitation. This thesis chapter extends those findings in several significant ways. First, it expands the spatial and temporal scope of the analysis by incorporating newly available data. Second, it introduces a more objective method for determining the number of rotated modes, thereby increasing the robustness of the results. Finally, it explores the implications of focusing on correlation instead of covariance as a metric for shared variability, utilizing and extending the Continuous Power Canonical Correlation Analysis (CPCCA) framework introduced by Swenson [248].

The structure of this chapter is as follows: We begin by describing the data sources and methodological framework (Section 7.1), with a particular focus on the data preprocessing and analysis decisions. Next, we present the results, highlighting the first ten rotated modes of shared variability between SST and continental precipitation (Section 7.2). These findings are then discussed in relation to existing climate indices and known teleconnection patterns, highlighting both consistencies and novel insights. Finally, we conclude with a summary of the key findings and their implications for future climate research and forecasting (Section 7.3).

7.1 Methods & Data

We used the fifth-generation ERA5 reanalysis dataset provided by the European Centre for Medium-Range Weather Forecasts (ECMWF) [44] to investigate the shared variability between sea surface temperature (SST) and continental precipitation on seasonal to decadal time scales. ERA5 offers global coverage on a regular longitude-latitude grid with a spatial resolution of 0.25° and monthly temporal resolution. Spanning from 1940 to the present, it allows us to capture long-term climate variability, trends, and potential teleconnections between SST and precipitation. ERA5 has proven to be a reliable source for climate studies due to its high spatial resolution and strong agreement with observations [103, 240, 287]. Precipitation is particularly well represented in the extratropics of the Northern Hemisphere, where most continental landmass resides, making it a suitable proxy for observational data. However, the representation of precipitation in the tropics is less accurate [141], which we need to consider when discussing dominant patterns of variability in tropical precipitation. Nevertheless, the recent extension of ERA5 back to 1940—which we used here—has significantly improved the representation of precipitation during the 1950s and 1960s [240].

Our geographical focus is between 60°S and 70°N because the majority of landmasses experiencing mostly liquid precipitation lie within these latitudes. Excluding Antarctica helped

us avoid biases due to its extreme dryness and sparse observational data. Additionally, SST variability is less reliable in polar regions because of partial sea ice coverage throughout the year. The Southern Hemisphere, in general, has historically fewer observations, affecting data reliability [240].

We concentrated on continental precipitation by masking out oceanic precipitation using the land-ocean mask at a 1:110 million scale provided by Natural Earth¹. To emphasize teleconnections operating on annual to decadal time scales, we applied a centered 7-month moving average to both the SST and precipitation time series. This approach removed high-frequency variability, such as monsoonal circulations, while retaining low-frequency phenomena like ENSO and PDO. Importantly, we retained the seasonal cycle to test whether we can recover dominant modes of variability using rotated Hilbert MCA (see Section 3.2) despite its presence in the data.

Our study aims to identify teleconnections—connections between SST and continental precipitation in regions that are not necessarily geographically close. However, local connections between SST and precipitation are common; for example, coastal precipitation and river plumes are related to changes in SST [e.g. 151]. Since ERA5’s high spatial resolution (approximately 30 km at the equator) can resolve these processes, it’s likely that an MCA of both datasets might also detect these local-scale phenomena. To mitigate the influence of small-scale processes and focus on larger-scale teleconnections, we coarsened the data by applying spatial averaging from a 0.25° to a 1.00° longitude-latitude grid, effectively smoothing out localized patterns. This processing step left us with 36,588 and 16,920 non-masked grid cells (features) for SST and continental precipitation, respectively. Additionally, to counter the effect of grid size distortion toward the poles [184], we applied latitude weighting to the data as described in Section 3.1.

To detect lagged covariance between SST and precipitation, we applied the Hilbert transform to each time series individually, converting them into complex-valued datasets. To minimize boundary effects and spectral leakage caused by the Hilbert transform (see Section 3.1), we first extended each time series in both directions using an exponentially decaying function with a three-month e -folding time. This extension smoothed the data, preventing abrupt jumps and aligning the decay with the dominant seasonal cycle in our dataset. As a result, each time series became three times longer than the original. After applying the Hilbert transform to the extended series, we trimmed the extra sections, retaining only the original time series within the complex data. This approach helps prevent the Hilbert transform from artificially inflating the data’s variance, which could lead to spurious correlations in the Hilbert MCA analysis [215].

Handling the large cross-covariance matrix—in the order of $10^4 \times 10^4$ —poses significant computational challenges. To address this, we performed Hilbert MCA in the principal component (PC)

¹naturalearthdata.com

space [11]. Initially, we transformed each dataset into its principal axes and retained enough principal components to explain 99 % of the variance in each dataset. This reduction resulted in 74 components for SST and 199 components for continental precipitation, shrinking the cross-covariance matrix to a more manageable size of 74×199 . Reducing the dimensionality not only made the computations feasible but also stabilized the solution by regularizing the initially ill-conditioned problem.

With the dimensionality reduced, we conducted MCA on the Hilbert-complexified, PC-transformed data to identify shared variability patterns between SST and continental precipitation. In the PC space, the total shared variability is represented by 74 modes. Analysis of the squared covariance fraction (SCF) for the initial non-rotated modes (Figure 7.1 A) revealed that nearly all squared covariance is captured within the first six modes, with the SCF dropping below 0.01 % for subsequent modes. This rapid decline can also be understood by the fact that *squared* covariance is measured.

To enhance interpretability, we applied an additional rotation to the MCA solution (Section 3.2.1). To ensure a stable rotation solution, We introduced the mode stability index κ (Equation (3.60)) to determine the optimal number of modes to rotate. We generated rotation solutions for $R \in [10, 40]$, assuming that the most relevant modes would be among the first 10. Figure 7.1 D indicates that the highest stability is achieved when rotating 22 modes, where the stability index plateaus. Adding more modes did not significantly change the results. Therefore, we focused our analysis on the solution obtained by rotating the first 22 modes.

Finally, we assessed whether oblique rotation was necessary due to potential correlations between modes. We rotated the first 22 modes using the promax rotation scheme [102] with a weak rotation power of $p = 2$, as recommended by Richman [213]. The resulting absolute Pearson correlation coefficients between the first 10 promax-rotated, indicating only weak correlations between leading modes. Consequently, we opted for orthogonal rotation (promax rotation power $p = 1$). The results of the oblique rotation are presented in the appendix (Figure ??-??).

Indices of known Climate Oscillations For comparative purposes, we obtained climate indices of monthly means from the National Oceanic and Atmospheric Administration (NOAA; <https://psl.noaa.gov/data/climateindices/list/>), including the Oceanic Niño Index (ONI) provided by the NOAA Climate Prediction Center, the PDO as defined by Mantua & Hare [166], and the Atlantic Meridional Mode (AMM) as defined by Chiang & Vimont [39]. We calculated the ENSO Modoki Index (EMI) following Ashok *et al.* [5] using the formula $EMI = R_A - \frac{1}{2}R_B - \frac{1}{2}R_C$, where R denotes the area-averaged ERA5 SST anomalies over regions A (165°E - 140°W ; 10°S - 10°N), B (110°W - 70°W ; 15°S - 5°N), and C (125°E - 145°E ; 10°S - 20°N). Additionally, we define an Oceanic Warming Index (OWI) as the area average of ERA5 SST anomalies

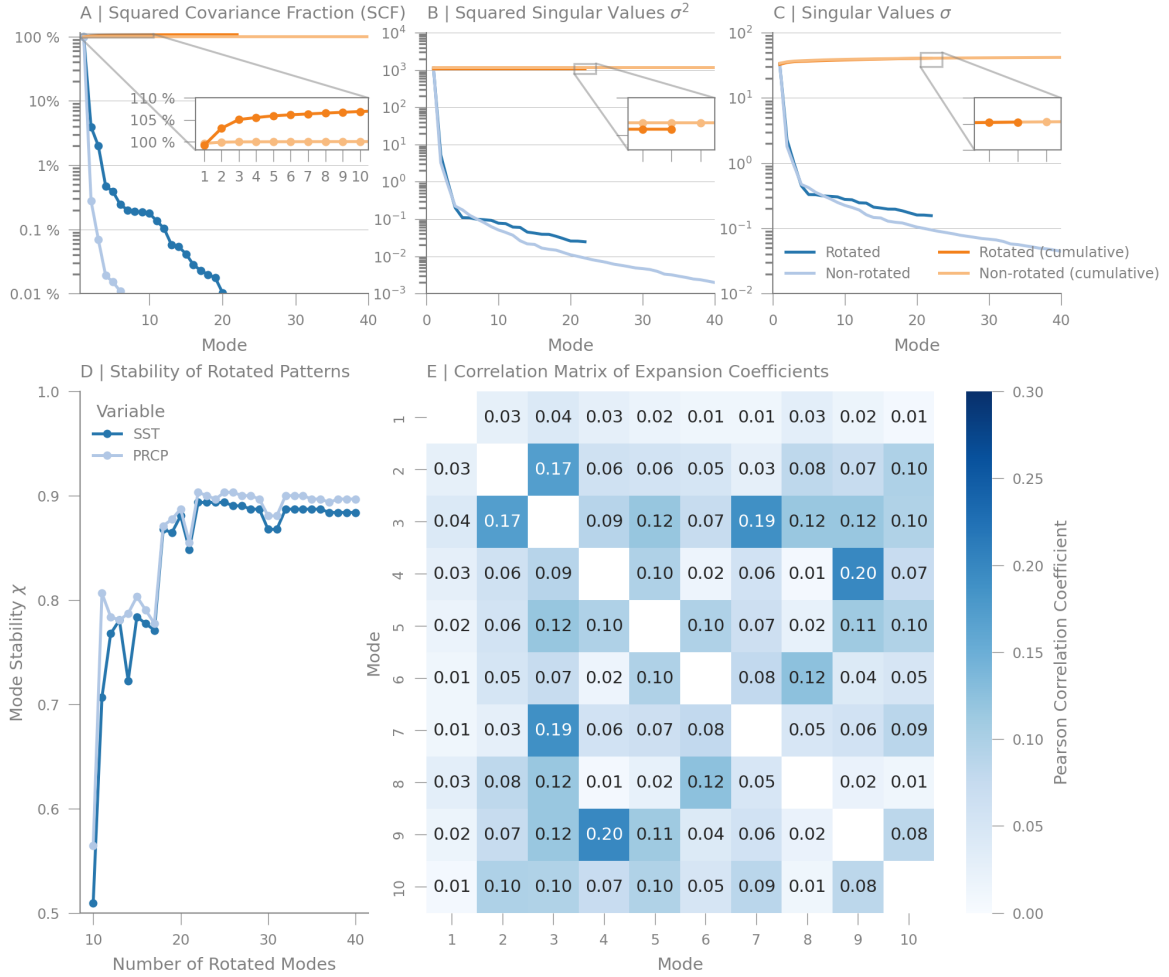


Figure 7.1 — Evaluation of varimax-rotated Hilbert MCA applied to SST and precipitation. (A) Squared covariance fraction (SCF) of the rotated modes, indicating that the SCF does not sum to 100 % after rotation. The SCF is computed according to Equation (3.39). (B) Squared covariance before and after rotation; the inset highlights that squared covariance is not conserved under rotation. Squared covariance is computed according to Equations (3.36) (before rotation) and (3.48) (after rotation). (C) Covariance (non-squared) before and after rotation; the inset demonstrates that covariance is preserved under rotation. (D) Mode stability index κ showing highest stability when 22 modes are rotated. (E) Pairwise Pearson correlation coefficients among the varimax-rotated expansion coefficients of SST, illustrating generally low correlations between different modes; the diagonal elements are masked for clarity.

between 60°S and 70°N, using the same latitude weighting as in our MCA data preparation. All indices are smoothed using a centered 7-month moving average to align with our data preprocessing and are restricted to the period from 1940 to 2022. We detrend all indices except the OWI using linear regression and further center and standardize them for better comparison with our obtained expansion coefficients.

Power Spectral Density We computed the power spectral density (PSD) for the first 10 expansion coefficients of SST and continental precipitation, as well as for the climate indices. For the complex-valued expansion coefficients, we extracted the real part after phase-shifting the coefficients so that, for each mode, the phase at the location with the maximum absolute value of the SST heterogeneous patterns is zero. This step is necessary because the complex-valued singular vectors from our Hilbert MCA are defined only up to a phase shift. By aligning the phase this way, the real part represents the maximum fraction of the lagged variability and captures the temporal evolution of the strongest signal in the data. Prior to PSD computation, all time series were linearly detrended. The PSD was computed using Welch's Overlapped Segment Averaging (WOSA) method [e.g., 199, Section 8.8], with 256 data points per segment and a Hanning window with 50 % overlap, resulting in a total of six segments. All PSDs were normalized to represent the fraction of energy density per frequency band.

Approximate confidence intervals for PSDs are often computed using the χ^2 distribution; however, this approach is valid only for a particular frequency and isn't directly applicable across multiple frequencies or the entire frequency range [cf. 206, p. 468]. Since we don't know in advance which frequencies are of interest, we employed the circular moving block bootstrap [203] to estimate confidence intervals for the PSDs. We generated 10 000 bootstrap samples by resampling the original time series with replacement, using a block length of 9 years. This choice balances the need for sufficient independent bootstrap samples with the requirement for long enough blocks to resolve the frequencies of interest. Note that this choice limits our ability to resolve periodicities longer than 9 years because we effectively scramble the autocorrelation function beyond this time lag. To preserve the annual cycle, we only considered January as a valid starting point for blocks. We computed the PSD for each bootstrap sample, and the 2.5th and 97.5th percentiles of the bootstrap PSDs are used as the lower and upper bounds of the confidence intervals.

Since climate data often exhibit a 'red noise' spectrum, we compared the PSDs of the expansion coefficients against a theoretical 'red noise' spectrum [e.g., 281, Section 10.3]. We estimated the 'red noise' spectrum using a first-order autoregressive process (AR(1)) and compute the theoretical PSD using the formula given by Gilman *et al.* [80]. The autoregressive coefficient ρ was estimated from the data as the lag-1 autocorrelation coefficient. We then tested the null hypothesis by comparing the spectral density of the data to that of the AR(1) process using the F-statistic at a significance level of $\alpha = 0.01$.

7.2 Results and Discussion

The rotated modes exhibit a more gradual decrease in cumulative squared covariance fraction (SCF) compared to the unrotated solution (Figure 7.1 A), with each of the first 12 modes explaining more than 0.1 % of the total squared covariance. A significant drop in SCF is noticeable after the 10th mode, where higher modes explain considerably less and decrease exponentially, indicating they may represent noise. Therefore, we focus our discussion on selected modes within the first 10. It is important to note that neither the cumulative SCF nor the sum of squared singular values are conserved under rotation (Figures 7.1 A, B). Only in the special case of orthogonal varimax rotation is the sum of singular values conserved (Figure 7.1 C).

Mode 1 – The Seasonal Cycle Mode 1 accounts for more than 99 % of the squared covariance between SST and continental precipitation (SCF1 = 99.1 %) and exhibits a clear one-year periodicity, unequivocally representing the annual cycle (Figures 7.2, 7.10 A). This mode shows a very strong link between SST and precipitation, evidenced by a Pearson correlation coefficient of $\text{rec}_1 = 0.99$ between their expansion coefficients. As expected, the annual cycle manifests globally in both variables, with strong expressions in the extratropics for SST and within the tropics for precipitation. Exceptions to this pattern occur in transition zones, such as north of the Arabian Sea, where precipitation shifts between summer and winter-dominated regimes over a narrow region, or in areas with multiple rainy seasons like central East Africa.

The phase functions (Figures 7.2 C, D) correctly separate the seasonal cycle according to the timing of the annual peak, identifying, for example, the anticorrelation (phase shift of approximately π) between the northern and southern oceans. In SST, we observe that the eastern equatorial Pacific and the equatorial Indian Ocean exhibit a positive phase shift of $+\pi/4$ (about six weeks) relative to the dominant oceanic seasonal cycle. On the continents, the phase function illustrates the division between January-February (light blue) and July-August (dark red) dominated rainfall systems, identifying corresponding transition zones such as over the South American rainforest and central North America. It also highlights interesting dynamical regions that stand out from their surroundings, namely the west and east coasts of North America, the Mediterranean region, and, to a lesser extent, the East Asian monsoon region in China.

We also note that this first mode explains significantly more variance in SST (80.4 %) than in continental precipitation (62.1 %). This is expected because the seasonal cycle is a more dominant feature in SST, whereas precipitation is generally a noisier variable with a more complex spatiotemporal structure. Specifically, precipitation is more directly influenced by atmospheric variability, which exhibits high internal variability. Additionally, the ocean's thermal inertia allows it to absorb or release large amounts of heat with only minor temperature

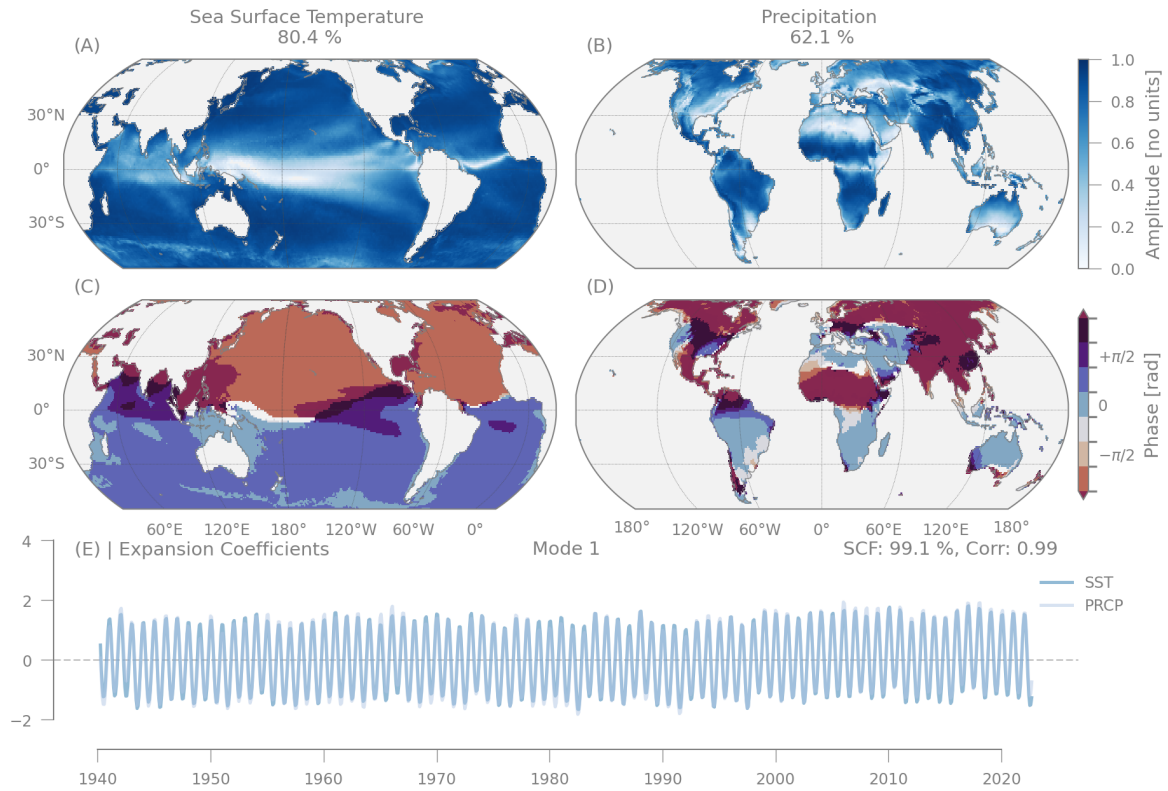


Figure 7.2 — Seasonal cycle as represented by mode 1 of varimax-rotated Hilbert MCA applied to SST and precipitation. Absolute heterogeneous correlation patterns for SST (A) and continental precipitation (B). Spatial phase patterns for SST (C) and continental precipitation (D). Associated expansion coefficients (E) for SST (blue) and continental precipitation (light blue). The squared covariance fraction (SCF) indicates the mode’s relative importance in summarizing the overall covariability between SST and precipitation. The Pearson correlation coefficient (CORR) between the expansion coefficients reflects the strength of their linkage, while the explained variance for each dataset (noted below the variable name) shows the mode’s significance within each dataset. Amplitude and spatial phase are plotted only where p-values, adjusted via the Benjamini-Hochberg correction for multiple testing, are below 0.05. Additionally, the spatial phase is masked in regions where the spatially max-normalized amplitude is below 0.125 to eliminate noisy phase values and enhance visualization. For clarity, the spatial phase is discretized into bins of 45 degrees ($\pi/4$ radians).

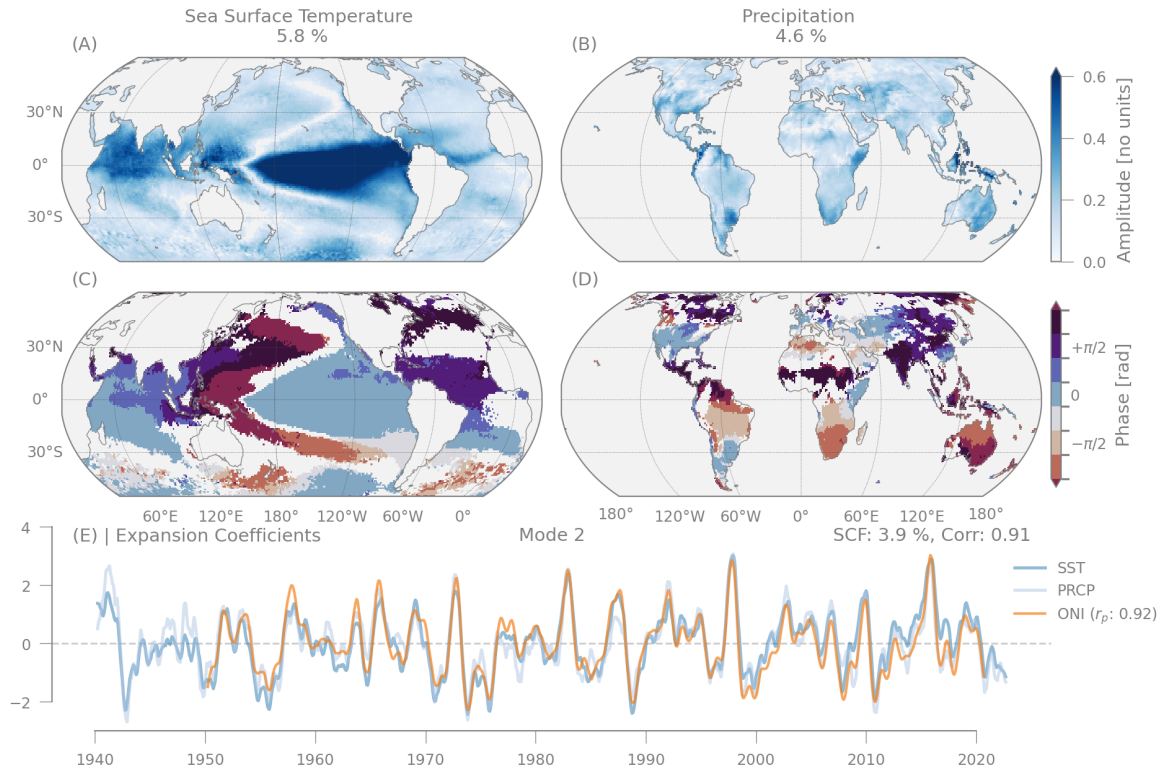


Figure 7.3 — ENSO as represented by mode 2 of varimax-rotated Hilbert MCA applied to SST and continental precipitation. The orange line represents the Oceanic Niño Index (ONI) for comparison, along with its Pearson correlation coefficient with the SST expansion coefficients. For more details, refer to the description in Figure 7.2.

changes, making SSTs generally less variable than atmospheric variables. Finally, while SSTs change smoothly and slowly over vast spatial scales, continental precipitation is highly localized, so interannual variations in global circulation patterns can produce drastic changes in local precipitation patterns.

Mode 2 – ENSO Mode 2 ($SCF_2 = 3.9\%$, $rec_2 = 0.91$) is clearly associated with the ENSO phenomenon (Figure 7.3). The expansion coefficients of SST for this mode show a strong correlation with the ONI, with a Pearson correlation coefficient of $r_{2,ONI} = 0.92$, indicating a strong association with ENSO. Many of the resulting teleconnections illustrated in Figures 7.3 A-B have been previously reported and analyzed. For instance, during El Niño events, higher SSTs in the central and eastern Pacific and lower SSTs in the western Pacific positively correlate with increased rainfall along the west coast and southeastern coast of South America [253], the east and west coasts of North America [222], the East Asian monsoon region [278], and the Horn of Africa [122]. Simultaneously, precipitation decreases in northern South America [253], Oceania [46], South Africa [76], and the Indian monsoon region [34]. During La Niña events,

these correlations are reversed. Recently, similar teleconnections have been identified using event coincidence analysis [280]. For a relatively recent summary of established ENSO-related rainfall patterns during El Niño and La Niña, see Lenssen *et al.* [152].

Furthermore, our results demonstrate the lagged dynamics between the core ENSO region in the western and central Pacific and other regions such as the Indian Ocean [133], the South China Sea [130], and the tropical North Atlantic [2, 38, 61, 228], consistent with previous studies. ENSO-related SST teleconnections in remote oceans often occur with delays: the Indian Ocean typically peaks 3 months, and the South China Sea and tropical North Atlantic about 5 months after the SST peak in the Pacific ENSO region [61, 130, 228]. The mechanism behind these lagged responses involves characteristic atmospheric circulation changes during El Niño events, leading to alterations in cloud cover and evaporation over remote oceans, resulting in increased net heat flux and SSTs [130, 140].

However, due to the broadband frequency spectrum of the SST expansion coefficients (Figure 7.10 B), with most energy between periodicities of roughly 2 and 5 years, the phase cannot be directly translated into a precise time shift. Assuming these periodicity limits, we can compute an approximate time shift. For example, the tropical Atlantic exhibits a phase shift of $+\pi/2$, translating into time shifts between 6 and 15 months, which is close to but slightly larger than the observed average time lag of 5 months. Nevertheless, the tropical North Atlantic clearly shows more positive phase values compared to the Pacific El Niño region, indicating it is positively time-shifted relative to the Pacific.

Recently, Scaife *et al.* [229] provided evidence for an even longer time lag—a one-year lagged extratropical response to ENSO—most pronounced in the North Atlantic. While they used sea surface pressure as the observational variable to assess the ENSO teleconnection, our SST-based analysis also indicates a weak signal of shared variability with ENSO in the North Atlantic, with a phase shift of approximately $+3\pi/4$. Using the periodicity limits from our analysis, this lag corresponds to time shifts between 9 and 22.5 months, aligning well with the reported one-year lagged response by Scaife *et al.* [229].

Considering all the lagged covariance collectively, ENSO accounts for approximately 5.8 % of SST variability and about 4.6 % of continental precipitation variability since 1940.

Mode 3 – Global Warming Mode 3 ($SCF_3 = 2.0\%$, $rec_3 = 0.80$) represents changes in SST and the associated precipitation patterns due to global warming, as evidenced by the strong correlation with the OWI, $r_{OWI} = 0.78$ (Figure 7.4). The warming SST patterns are evident across all major ocean basins, with a clear spatial focus along the equator, except in the eastern equatorial Pacific (Figure 7.4 A). Additionally, amplitudes are slightly more pronounced in the Northern Hemisphere due to the asymmetric response of the northern and southern trade winds to global warming [285]. We also observe pronounced warming in the western parts of

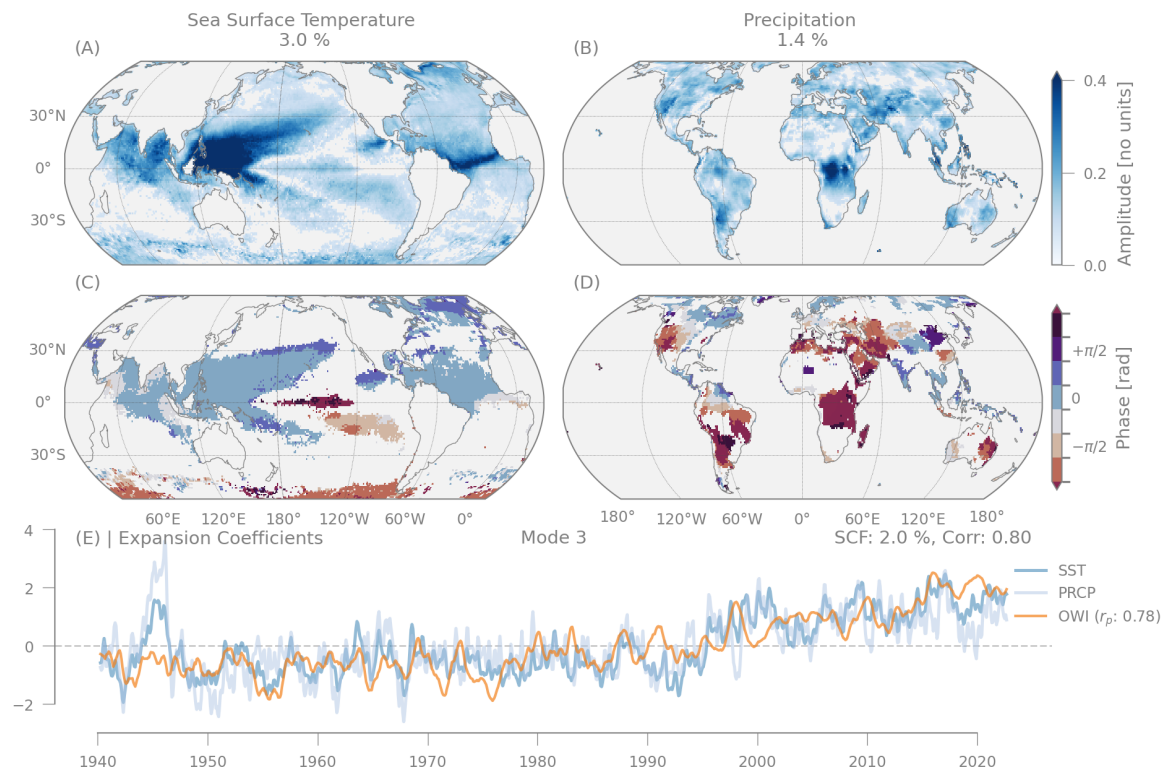


Figure 7.4 — Oceanic warming as represented by mode 3 of varimax-rotated Hilbert MCA applied to SST and continental precipitation. The orange line represents the Oceanic Warming Index (OWI) for comparison, along with its Pearson correlation coefficient with the SST expansion coefficients. For more details, refer to the description in Figure 7.2.

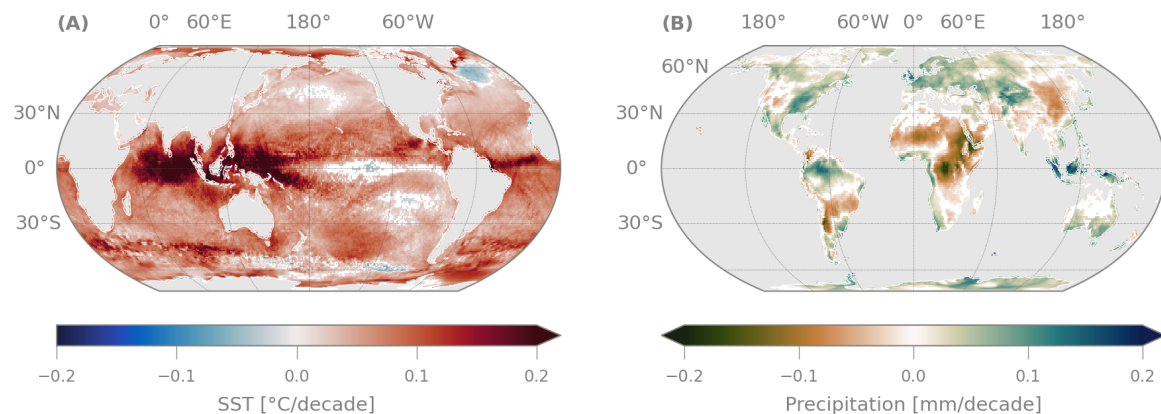


Figure 7.5 — Caption Description

both the Pacific and Atlantic basins, regions associated with enhanced ocean heat transport [95, 285].

Similarly, we observe global trends in precipitation patterns, with the highest amplitudes in Central Africa, parts of Oceania, South and North America, and Central Asia (Figure 7.4 B). From the phase functions, we deduce an increase in precipitation (phase shift of zero) around tropical Oceania, while a decrease in precipitation (phase shift of $\pm\pi$) is suggested for Central Africa, central South Africa, western North America, and the Arabian Peninsula and surrounding regions. These results are in strong spatial agreement with trends obtained from reanalysis data for both SST and continental precipitation (Figure 7.5), and largely concur with studies based on observational data [88] and, more recently, on CMIP5 climate simulations [81]. Collectively, these findings support the interpretation of this mode as representing the shared variability due to global warming.

However, some minor discrepancies exist between the obtained phase functions and the computed trends from reanalysis data. Notably, reanalysis data indicates strong positive precipitation trends of about 1 mm/decade in the eastern part of North America, which is not evident in our phase function. Additionally, the phase functions suggest an increase in precipitation in the southern parts of the Indian subcontinent, whereas reanalysis data shows a more complex pattern, with increases in the southeast and decreases in the western part.

Finally, it is important to emphasize that the expansion coefficients for both SST and precipitation only allow for meaningful interpretation of phases of zero (correlation) and $\pm\pi$ (anticorrelation), due to their broadband power frequency spectra (Figure 7.10 C). For phase shifts different from these values, no clear conclusions can be drawn—for example, in western Australia, where the phase shift is approximately $-\pi/4$.

Mode 4 – Nonlinear ENSO Modulation Mode 4 ($SCF_4 = 0.5\%$, $rec_4 = 0.82$) describes an oscillating SST anomaly mainly confined to the central Pacific (Figure 7.6), resembling the pattern popularized by Ashok *et al.* [5] as ‘El Niño Modoki’, represented by the EMI ($r_{EMI} = 0.56$). This pattern consists of higher SSTs in the central Pacific and lower SSTs in the eastern Pacific, which correlate with reduced precipitation in the East Asian monsoon region [69], Australia [251], parts of South America [253], and South Africa [210], and vice versa. Some of these teleconnections have recently been identified through event coincidence analysis [280], although our analysis suggests additional significant links to the East Asian monsoon region. Later research by Takahashi *et al.* [249], however, indicated that El Niño Modoki and the classical El Niño are essentially the same phenomenon, and that the perceived differences highlight the dynamic range of the system, specifically the nonlinearity of ENSO. Following this interpretation, the teleconnections identified in our analysis should not be considered

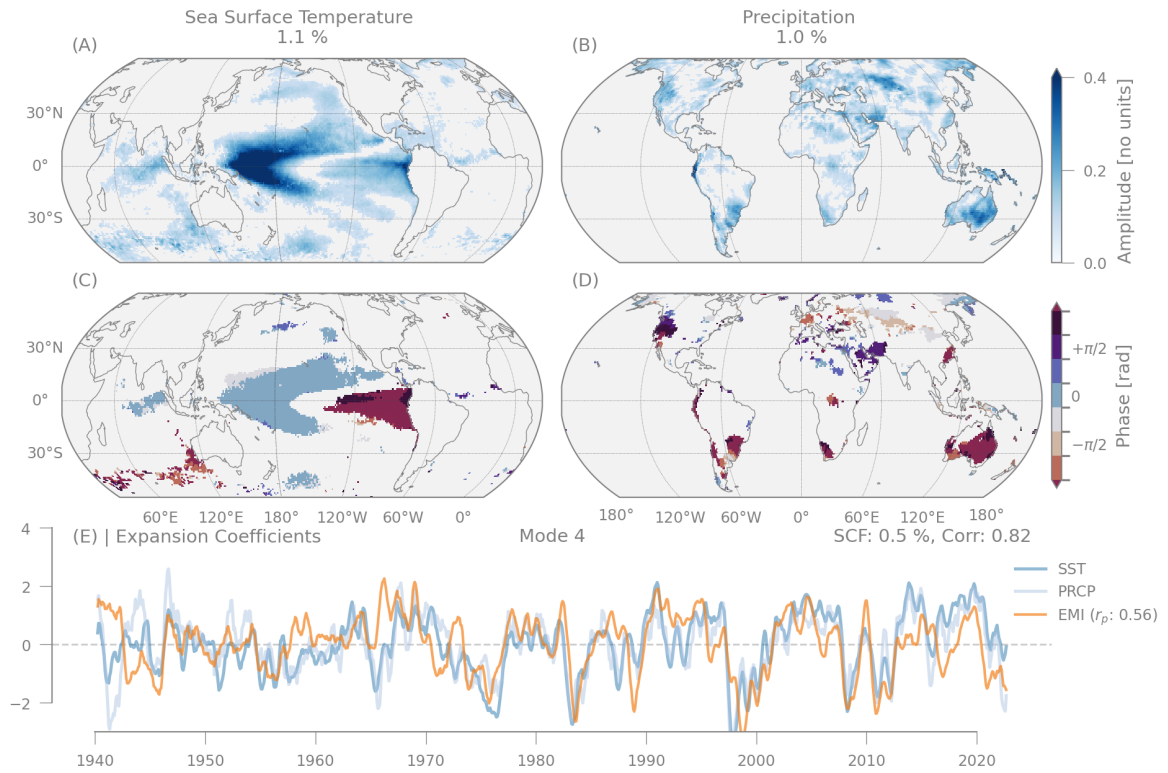


Figure 7.6 — Nonlinear ENSO modulation as represented by mode 4 of varimax-rotated Hilbert MCA applied to SST and precipitation. The orange line represents the El Niño Modoki (EMI) index for comparison, along with its Pearson correlation coefficient with the SST expansion coefficients. For more details, refer to the description in Figure 7.2.

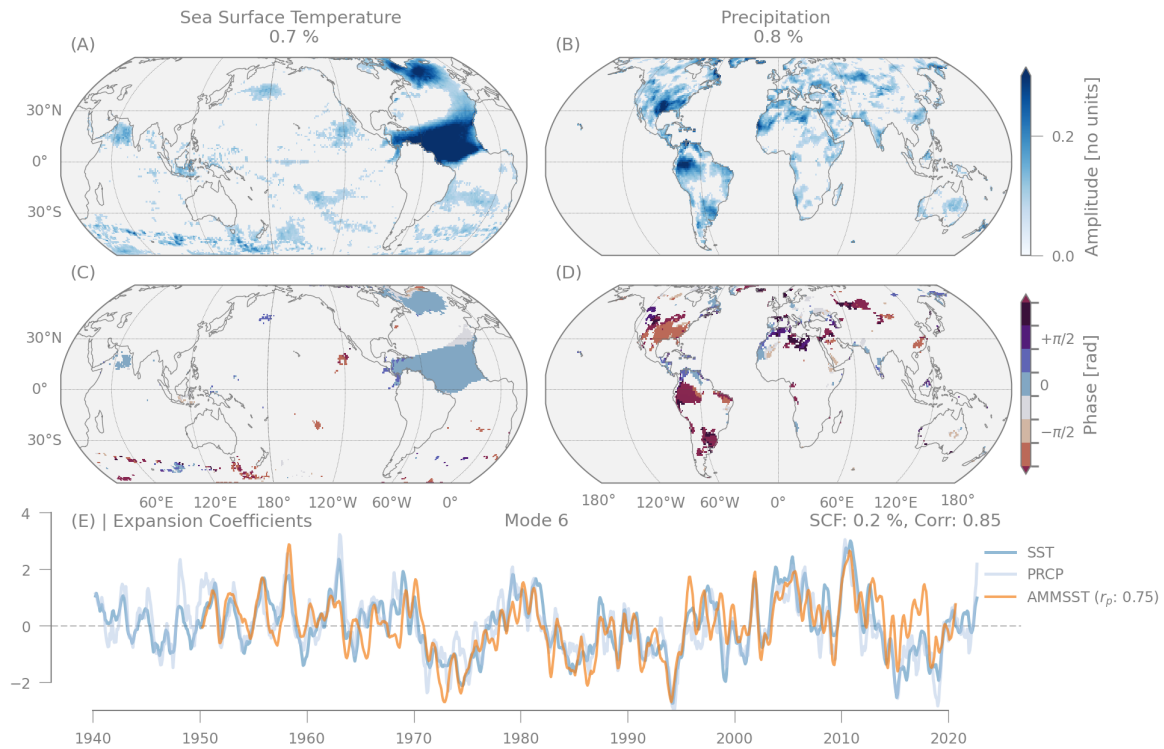


Figure 7.7 — Atlantic Meridional Modulation (AMM) as represented by mode 6 of varimax-rotated Hilbert MCA applied to SST and precipitation. The orange line represents the AMM index for comparison, along with its Pearson correlation coefficient with the SST expansion coefficients. For more details, refer to the description in Figure 7.2.

1 a separate mode but rather a modulation of ENSO Mode 2 due to the system's nonlinear
2 behavior.

3 The power spectral density of this ENSO modulation shows a rather broadband energy
4 distribution mostly indistinguishable from the theoretical 'red noise' spectrum, with the
5 exception from some periodicities around 18 months (Figure 7.10 D). In particular, the energy
6 distribution is markedly different from the periodicities typically associated with ENSO (2 to 7
7 years). Overall, the phase does not provide a clear interpretation apart from zero and $\pm\pi$ phase
8 shifts.

9 **Mode 6 – AMM** Mode 6 ($SCF_6 = 0.2\%$, $rec_6 = 0.85$) is characterized by an SST pattern
10 concentrated in the tropical and subarctic North Atlantic (Figure 7.7), explaining about 0.7%
11 of the overall SST variability. This pattern represents the AMM ($r_{AMM} = 0.75$), the dominant
12 coupled ocean-atmosphere phenomenon in the tropical Atlantic [236]. The AMM's impacts
13 on precipitation—explaining about 0.8% of total precipitation variability—in the African Sahel
14 zone, Central America, and northern South America are well documented [139, 168]. Only
15 recently, Vittal *et al.* [270] found a link between the AMM and Indian summer monsoon rainfall,

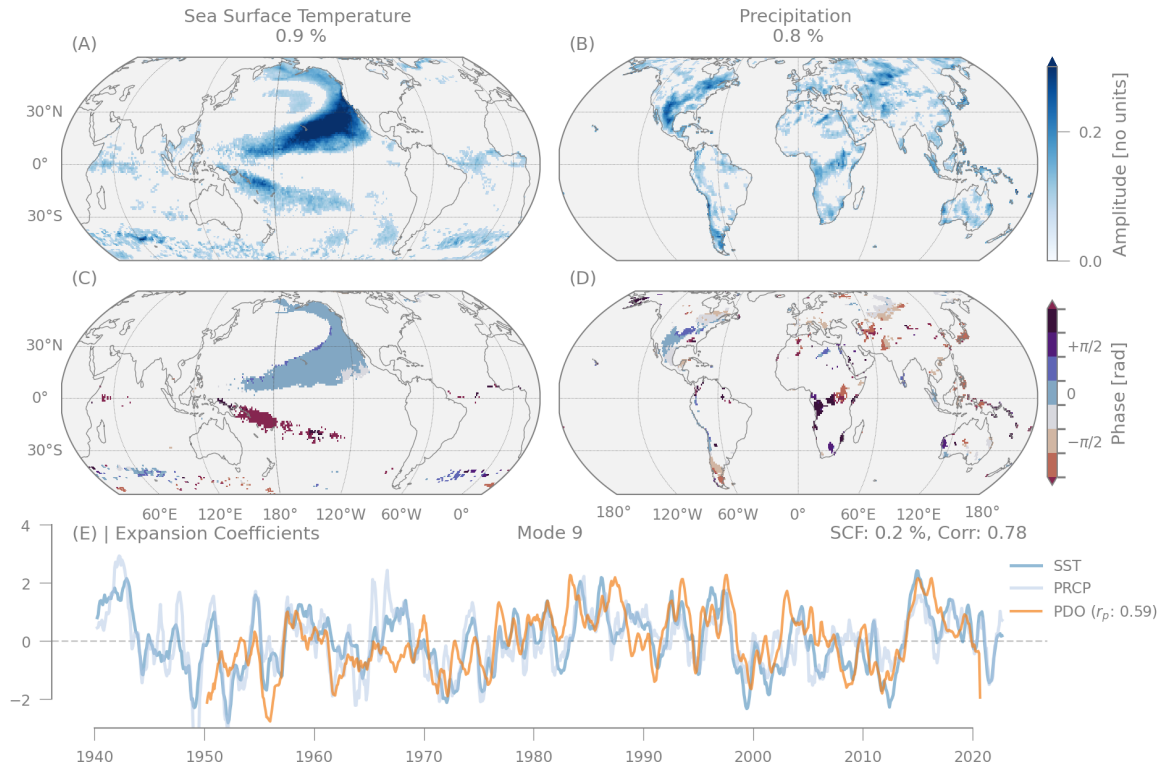


Figure 7.8 — Pacific Decadal Oscillation (PDO) as represented by mode 9 of varimax-rotated Hilbert MCA applied to SST and precipitation. The orange line represents the PDO index for comparison, along with its Pearson correlation coefficient with the SST expansion coefficients. For more details, refer to the description in Figure 7.2.

which they used to improve precipitation forecasts. Additionally, our results suggest further teleconnections between the AMM and precipitation variability over North America, the Mediterranean, and the East Asian monsoon region, offering potential for enhancing local rainfall predictions in these areas. Of particular interest is the southeastern United States, where a phase shift of $-3\pi/4$ —corresponding to a time lag of about 9 months assuming a periodicity of 2 years (Figure 7.10 E)—is observed.

Mode 9 – PDO Mode 9 ($SCF_9 = 0.2\%$, $rec, 9 = 0.78$) exhibits a slow-oscillating SST pattern in the northeastern Pacific, correlating with localized precipitation patterns distributed across all continents, predominantly in North America (Figures 7.8 A, B). This spatial SST pattern is known as the PDO [167] ($r_{PDO} = 0.59$), a well-established climate index. Various processes originating in the tropics and extratropics have been proposed as the physical source of the PDO [183], with ENSO and PDO likely responding to the same forcing function [202]. Our analysis provides a means to disentangle ENSO- and PDO-related precipitation patterns, which are often similar in western North America [115]. Typically, the PDO exhibits strong

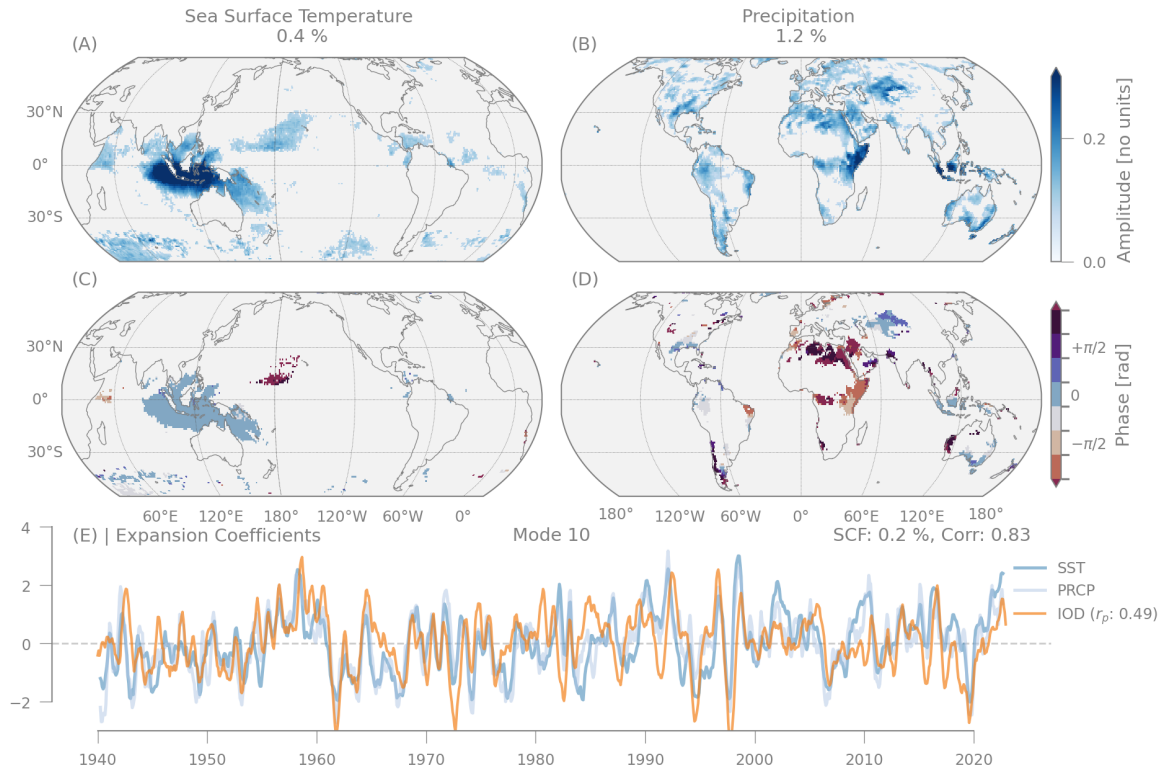


Figure 7.9 — Indian Ocean Dipole (IOD) as represented by mode 10 of varimax-rotated Hilbert MCA applied to SST and precipitation. The orange line represents the IOD index for comparison, along with its Pearson correlation coefficient with the SST expansion coefficients. For more details, refer to the description in Figure 7.2.

energy around periodicities of 30 years; however, due to our preprocessing (using block sizes of 9 years), we cannot resolve such long periodicities in our analysis. Instead, we find higher-frequency, multi-year periodicities around 4 years (Figure 7.10 I), likely due to the significant feedbacks between ENSO and PDO modes [176, 250]. This finding is interesting because this spectral peak is not visible in the official PDO index, indicating slight differences in the composition of the PDO mode in our analysis compared to the traditional PDO index.

Mode 10 - IOD Mode 10 ($SCF_{10} = 0.2\%$, $r_{ec,10} = 0.83$) resembles the Indian Ocean Dipole (IOD) ($r_{IOD} = 0.49$), a coupled ocean-atmosphere phenomenon in the tropical Indian Ocean (Saji et al. 1999) that explains about 0.4 % of the total SST variability. The SST pattern is characterized by a dipole structure with higher SST in the western and lower SST in the eastern Indian Ocean. The associated precipitation pattern is mainly concentrated in the surrounding tropical land masses, namely central east Africa and Oceania, explaining 1.2 % of the total precipitation variability (Figure 7.9 A,B). The phase function shows a clear phase shift of $+\pi/2$, i.e. anti-correlation between precipitation patterns around the eastern and western Indian

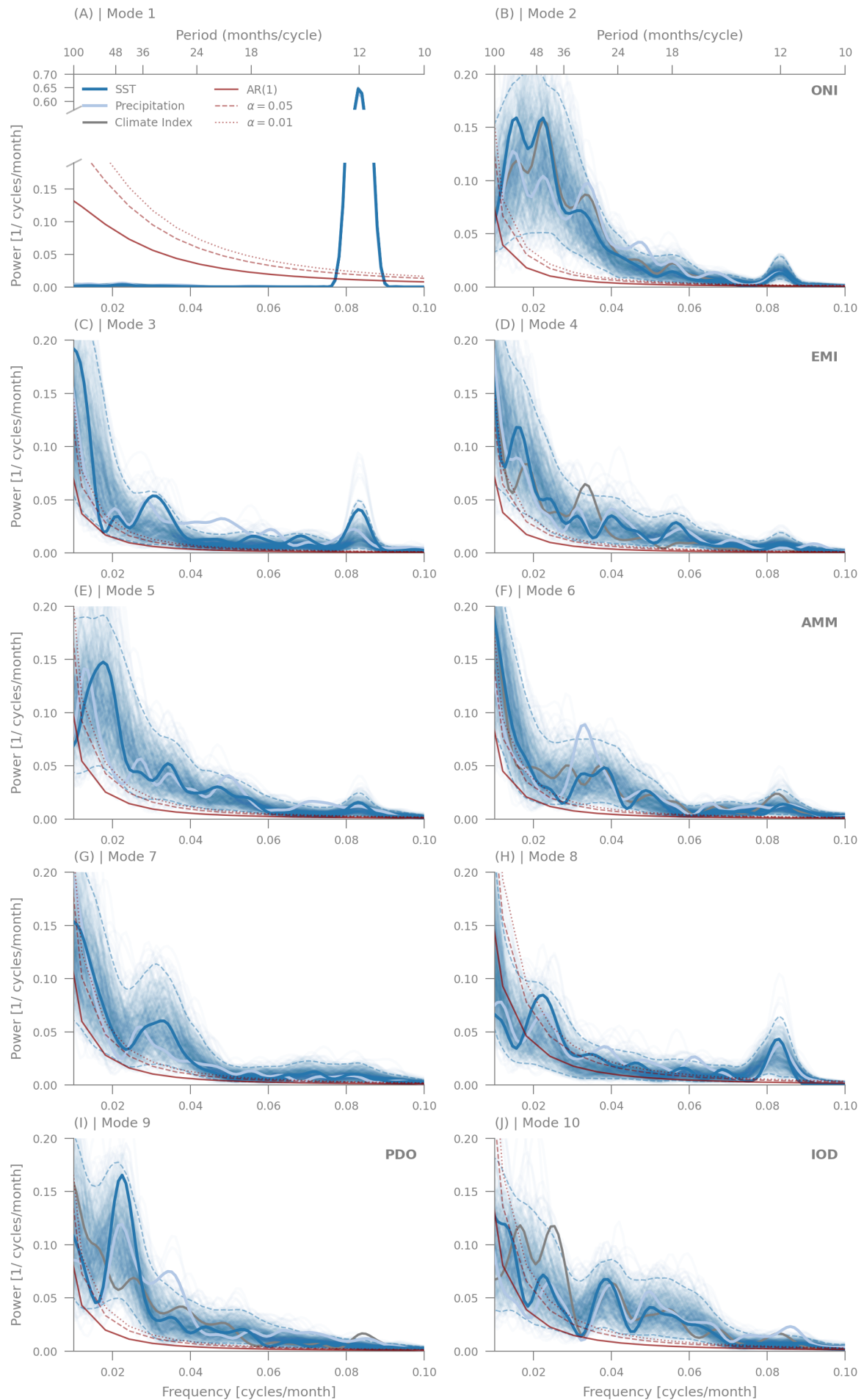
ocean, mirroring the dipole in the tropical Indian SSTs. Note that our results have only a weakly expressed dipole, with a much stronger center of attention in the eastern Indian ocean and much lower amplitude in the western Indian ocean (Figure 7.9 C, D). Although the PSD of this mode shows significant energy around 18-24 months compared to the ‘red noise’ model, the spectral density remains relatively flat across a wide range of frequencies, reflecting the aperiodic and irregular nature of the IOD (Figure 7.10 J).

Other Modes of Shared Variability In our previous discussion, we focused on the dominant modes of shared variability identified by Rieger *et al.* [215] using the MCA framework. With the expanded dataset used here, covering a broader spatial and temporal scope, we identified additional modes—specifically modes 5, 7, and 8—that bring new insights worth noting. Here, we provide a brief overview of these modes, pointing to some interesting preliminary observations while leaving detailed analysis for future work.

Mode 5 (Figure ??) closely mirrors aspects of the atmospheric Pacific-South American (PSA) mode, which influences multiscale spatiotemporal patterns in the upper South Pacific Ocean. This mode significantly drives oceanic variability in the South Pacific subtropics, producing a potentially predictable climate signal connected to tropical regions [162]. This mode shows increased rainfall over southeastern South America, consistent with patterns observed during ENSO events [177], though it does not reflect the typical rainfall deficits in northeastern Brazil associated to the PSA mode (Figure ?? B, D). Mode 7 (Figure ??) reveals a distinct variability pattern in the eastern Indian Ocean, with precipitation teleconnections extending to central Africa and the southeastern United States. This suggests interconnected weather systems across these regions, highlighting the global reach of this mode. Mode 8 (Figure ??) is characterized by SST variability primarily north of the Arctic Circle. This mode shows a positive trend combined with slow-frequency fluctuations, indicating a possible link to global warming. Its unique spatial focus may be related to Arctic amplification, where the Arctic region warms faster than the rest of the planet, representing a different, potentially nonlinear aspect of climate variability.

Beyond Covariance – Applying the CPCCA Framework As outlined in Section 3.2.2, the CPCCA framework offers a balanced approach between maximizing covariance and correlation, complementing the MCA method. By applying CPCCA to our SST and precipitation datasets, we identified additional modes of shared variability that emphasize correlation over covariance. These CPCCA modes exhibit higher correlations but account for less variance within each dataset compared to their MCA counterparts (see Figure ??–?? for varimax-rotated modes and Figure ??–?? for promax-rotated modes).

A particularly noteworthy finding is CPCCA mode 5 (Figure ??), which reveals a unique teleconnection pattern linking SST anomalies in the western tropical Pacific to precipitation in



Northern Europe—a pattern not identified by MCA. This discovery aligns with recent research by Straaten *et al.* [245], who identified this SST pattern as a predictor for European summer temperatures on sub-seasonal timescales. Our results suggest that this source of predictability may also extend to seasonal precipitation, presenting an exciting direction for future research. Overall, our analysis demonstrates that the complexified and rotated CPCCA modes are similar to MCA modes but show higher correlations, reflecting a stronger emphasis on correlation over covariance. Consequently, CPCCA modes tend to explain less variance within each dataset compared to MCA modes. Additionally, some mode mixing was observed, such as the combination of MCA Modes 4 and 9 into a single CPCCA mode (Figure ??), indicating sensitivity to the chosen fractional parameter value α_s .

7.3 Conclusions

In this chapter, we investigated the shared variability between SST and continental precipitation to better understand atmospheric teleconnections—patterns that connect climate variations across distant regions. By applying varimax-rotated Hilbert MCA to an extended dataset of SST and precipitation, we decomposed the complex, high-dimensional covariability into a few dominant modes of shared variability.

Our analysis successfully identified key modes corresponding to well-established climate phenomena such as the seasonal cycle, ENSO, PDO, AMM, and IOD. These modes align well with known climate indices and teleconnection patterns, reinforcing the validity of our approach. Notably, the use of the Hilbert transform and varimax rotation allowed us to capture both the seasonal cycle and lagged teleconnections within a single framework, without the need to remove the seasonal cycle explicitly. This comprehensive approach provides deeper insights into how SST and continental precipitation are interconnected across the globe.

In addition to these established modes, we uncovered additional patterns of shared variability—specifically modes 5, 7, and 8—that offer new avenues for research. For instance, Mode 5 resembles the Pacific-South American (PSA) pattern, influencing climate variability in the South Pacific subtropics and potentially offering predictability linked to tropical regions. Mode 7 highlights a distinct variability in the eastern Indian Ocean with teleconnections to central Africa and the southeastern United States. Mode 8 is characterized by SST changes north of the Arctic Circle, suggesting a link to Arctic amplification and offering insights into the regional impacts of global warming.

To explore the shared variability from a different perspective, we extended our analysis by incorporating the CPCCA framework. By combining CPCCA with Hilbert complexification and varimax rotation, we focused on maximizing correlation rather than covariance. This approach revealed modes with higher correlations but lower explained variance, providing a

1 complementary view of the interactions between SST and precipitation. Notably, CPCCA Mode
2 5 identified a unique teleconnection pattern linking SST anomalies in the western tropical
3 Pacific to precipitation in northern Europe—a pattern not captured by MCA but recently
4 highlighted in other studies as a source of predictability for European summer climate.

5 Looking forward, applying non-linear methods could enhance our understanding of shared
6 variability, especially for capturing the inherent nonlinearity of phenomena like ENSO. Tech-
7 niques such as non-parametric kernel methods [1] or neural network approximations [6, 113]
8 offer promising avenues. The recent availability of tools like the Python library `cca-zoo` [27]
9 makes these advanced methods more accessible. However, challenges remain, particularly in
10 interpreting the underlying spatial patterns when using kernel methods due to the pre-image
11 problem. Recent suggestions using linear approximations [21] or composite patterns [94] to
12 overcome this, could make non-linear methods more practical for climate research.

13 In summary, our study demonstrates that combining varimax-rotated Hilbert extensions with
14 MCA and CPCCA provides a robust framework for identifying and interpreting dominant
15 modes of shared variability between SST and continental precipitation. These methods enhance
16 our understanding of atmospheric teleconnections and offer valuable insights that could
17 improve climate predictions on seasonal to decadal timescales.

Part III

Pattern Identification in the Scarce Data Regime

1

2

3

4

A Bayesian Machine Learning Approach to Seasonal Variations in Beach Litter Pollution

While the era of Earth observation has resulted in massive datasets propelling advances in fields like weather and climate science [29, 205], some environmental challenges remain difficult to tackle due to a lack of sufficient data. A prime example is the pervasive issue of global plastic pollution. Despite recent progress in our scientific understanding of this problem¹, much remains unknown, largely because of extremely limited and scattered observational data. This scarcity presents a significant obstacle to improving our comprehension of the inherent spatio-temporal dynamics of plastics in the environment. Even the most comprehensive large-scale observational datasets rarely exceed a few dozen observations at a given location, making the inference of spatio-temporal trends extremely challenging. Therefore, this problem of pattern identification sits at the lower end of the data density spectrum presented in Chapter 2.

In this chapter, we tackle the challenge of limited and scattered observational data on plastic pollution by using Gaussian process regression within a Bayesian framework to model its spatio-temporal patterns. This approach allows us to address the scarcity and irregularity of data while enabling statistically robust inferences. We focus specifically on beach litter for two main reasons: (1) monitoring beaches is labor-intensive, leading to particularly scarce data, and (2) beaches tend to accumulate high concentrations of marine litter, making them critical areas of concern. The results presented here have also been published in

- **Rieger, N. et al.** Seasonal Hotspots of Beach Litter in the North-East Atlantic Linked to Aquaculture and River Runoff. *Communications Earth & Environment* 5, 1–12. doi:10.1038/s43247-024-01913-7 (2024).

This chapter is organized as follows: we begin with a comprehensive overview of the current state of plastic pollution on beaches (Section 8.1). Next, we describe the data sets and methods used in our analysis (Section 8.2). We then present and discuss our findings (Section 8.3) before concluding with final thoughts and implications (Section 8.4).

¹The political dimension also hinders progress. As of December 2024, negotiators in Busan, South Korea, failed after over two years of talks to finalize a United Nations treaty aimed at significantly reducing plastic pollution [242]. The main obstacle was differing priorities: some countries seek to limit the production of non-essential plastics, while others, particularly petrochemical producers, prefer to focus on waste management.

8.1 Plastic Pollution on Beaches

Beach litter, predominantly composed of plastics [138, 179], has emerged as a significant environmental challenge, impacting societies globally with its adverse effects on marine ecosystems [30], wildlife [3, 9, 256], and human health [220, 266]. Since the 1950s, plastic production has surged almost exponentially—from 2 million tonnes to 400 million tonnes annually—with less than 9 % being recycled and an estimated 23 % ending up as trash in the environment² [78, 79, 185]. Current estimates indicate that marine plastic inputs range between 0.5 million tonnes to 0.7 million tonnes per year [126, 290], resulting in a standing stock of approximately 3.0 million tonnes to 3.4 million tonnes of marine plastic litter [126]. Plastic litter can now be found everywhere on Earth, from the most isolated islands [120] and remote polar regions [272] to the depths of marine trenches [40].

Beaches, despite covering less than 0.02 % of the global marine environment³, likely accumulate up to 2 % of global marine plastics [126]. This leads to relatively high concentrations of pollution in these areas compared to other parts of the marine environment. Recent studies suggest a balanced contribution of plastic pollution from land-based activities and marine sources, particularly from fishing-related activities, each accounting for about half of the total [126, 154]. Concurrently, aquaculture production has witnessed significant growth [66, 195], now accounting for 50 % of global fish production [66], and increasingly contributes to marine pollution [18, 105, 136, 159].

Although much progress has been made in understanding how and where plastics accumulate in our marine environment [e.g. 28, 165, 179, 246], many open questions remain regarding their spatio-temporal variations. One of the main reasons for this is the lack of sufficient and consistent observational data [91]. Most of the data available today—from lakes, rivers, beaches, and the open ocean—are based on in situ measurements. These are often costly in terms of time and resources, resulting in a very sparse set of observations. Additionally, due to a lack of standardized survey protocols, not all globally available data are easily comparable.

Beach litter observations remain particularly scarce as they mostly rely on cumbersome in situ surveys, which are difficult to automate. While remote sensing of beach litter using unmanned aerial vehicles such as drones combined with machine learning classification has been demonstrated as a more efficient method to survey beach litter (see, for instance,

²Almost half (49 %) of plastic waste is now sent to landfills, and the remaining 19 % is incinerated [185]. Over all lifecycle stages, plastics emit about 1.79 billion tonnes of CO₂eq, which accounts for 3.3 % of global greenhouse gas emissions and fuels climate change.

³This estimate was calculated by approximating the area of ice-free shorelines: The Earth's total surface area is approximately 510 million km², with 361 million km² (70.8 %) covered by water. According to Luijendijk *et al.* [163], the total length of ice-free shorelines is about 1.11 million km. Assuming an average beach width of 50 meters, the area of beaches is calculated as 1.11 million km × 0.05 km = 55 500 km², which is roughly 0.015 % of the global marine environment. This estimate likely overstates the area of sandy beaches, as they constitute only about one-third of all shorelines [163].

Gonçalves *et al.* [85] and references therein for advances in this direction), this approach is still under development. For the open ocean, it was recently shown that it is possible to measure plastic pollution—which often forms large coherent structures or filaments—using satellites [45], marking a major step forward. Unfortunately, applying the same approach to beach litter is challenging due to the need for much higher resolution imagery.

Seasonal variations in beach litter present a complex picture, with various studies reporting differing peak pollution periods across seasons [4, 48, 73, 218, 224, 233, 274, 282]. A consistent large-scale seasonal pattern remains elusive [181, 233], possibly because of insufficient survey data [91, 234]. Improved understanding of beach litter seasonality is vital for identifying anthropogenic sources and setting effective, evidence-based management policies. Notably, seasonality in plastic emissions has been observed from riverine discharges [146, 264, 265] and capture fisheries [135, 195], while the seasonal dynamics of marine aquaculture (mariculture) contributions are less documented. The anticipated rise in global plastic waste and seafood demand [70, 145], driven by population growth and socioeconomic factors, underscores the necessity to explore these seasonal dynamics comprehensively.

In this study, we aim to (1) investigate the seasonal variations of anthropogenic beach litter, (2) identify regions where these variations are particularly pronounced—referred to as ‘seasonal hotspots’, and (3) understand the underlying drivers. Focusing on the North-East Atlantic (NEA), we utilize the OSPAR dataset [190], one of the most comprehensive in-situ beach litter datasets globally. Despite its extensive coverage, the infrequent surveys of many beaches present challenges in detecting statistically significant seasonality [87, 233, 234]. Therefore, we employ a spatial log-Gaussian Cox process [53, 178] to model macroplastic distributions, leveraging its Bayesian machine learning framework to capture complex spatial patterns while allowing robust statistical inference. This approach enables us to more accurately model probability distributions of beach litter by incorporating information from nearby beaches and exploiting spatial autocorrelation patterns [87].

This study seeks to enrich the discourse on sustainable, evidence-based management strategies for beach litter [157, 267], aligning with the EU’s Marine Strategy Framework Directive (MSFD) and directly supporting the marine litter objectives outlined in the OSPAR North-East Atlantic Environment Strategy 2030 [192]. By providing insights into the seasonal dynamics and sources of beach litter, our research underscores the need for targeted actions to prevent and significantly reduce marine litter inputs, in line with OSPAR’s Strategic Objective 4. While our focus is regional, these efforts also contribute to broader Sustainable Development Goals (SDGs), mainly related to marine life conservation (SDG 14). Furthermore, as macroplastics are a major source of microplastics entering the marine environment [246], better management and mitigation of macroplastic emissions can support efforts regarding good health (SDG 3) and clean water (SDG 6).

8.2 Materials & Methods

Throughout this work, we define the seasons as follows: winter (December to February), spring (March to May), summer (June to August), and autumn (September to November). Following previous OSPAR data assessments [138], macroplastics are defined as artificial polymer materials larger than 2.5 cm. In this manuscript we use the term beach litter interchangeably with macroplastics.

8.2.1 Beach Litter Data

We utilized the OSPAR dataset [138, 190], comprising 4,305 surveys. Our regional focus is on the North-East Atlantic (NEA), specifically the Atlantic coastlines of Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden, and the United Kingdom. This area closely aligns with OSPAR maritime regions II (Greater North Sea), III (Celtic Seas), and IV (Bay of Biscay and Iberian Coast) but differs slightly from the official OSPAR definition of the NEA.

Specifically, we excluded overseas regions such as the Azores, the Canary Islands, the Faroe Islands, and Greenland, as well as areas above 65° north latitude and data from the Mediterranean Sea, to maintain a consistent geographic scope. Our focus on this specific subregion is motivated by data availability and considerations of spatial scale and model complexity. Regions like Greenland and Iceland were excluded due to limited seasonal coverage, with surveys predominantly conducted during summer and autumn, making it challenging to establish seasonal differences throughout the year. Additionally, including remote areas would introduce larger spatial scales, requiring modifications to our model that could increase complexity without significant benefit from the limited additional data.

Entries where external cleanups may have been performed before the OSPAR survey, as noted in the 'Survey: Remarks' category, were excluded. For surveys deviating from the standard 100-meter length, we extrapolated beach litter counts when the actual survey length was provided; otherwise, these surveys were excluded. We then aggregated all OSPAR categories of artificial polymer materials larger than 2.5 cm (macroplastics) into a common category [138].

According to the OSPAR survey guidelines [279], beaches are surveyed quarterly, specifically in January, April, July, and October. We identified 87 instances of multiple surveys per season for the same beach and year; to mitigate sample frequency bias in beach litter detection [225], we selected the maximum litter item count from these surveys for each season, beach, and year. The refined dataset comprised 3,866 surveys from 168 beaches, documenting over 2 million items.

We estimated the number of items originating from land-based, fishing-related and mariculture-related activities, including only categories that are relatively unambiguous. For mariculture-related items, we used OSPAR categories explicitly linked to mariculture [239] (OSPAR IDs 28, 29, 30 and 114). Fishing-related activities were based on the SEA category [138], excluding items associated with mariculture and ambiguous items like strings [239]. Land-based items were defined by the residual categories, excluding all generic categories (OSPAR IDs 46, 47 and 48). As a result, 39.9% of items within the preprocessed OSPAR dataset could be assigned to either land, fishing or mariculture.

8.2.2 Log-Gaussian Cox process model

We utilized a log-Gaussian Cox process (LGCP) [178] in this study to map the spatial distribution and intensity of macroplastic beach litter. In the LGCP framework, the logarithm of the intensity function is modeled as a Gaussian process ($\mathcal{GP}$) [14], making it particularly suitable for environmental and ecological data exhibiting spatial dependence.

The observed over-dispersion in the beach litter data (Supplementary Figure 11), where the variance exceeds the mean, necessitated adapting the LGCP to use a Negative Binomial distribution instead of the traditional Poisson distribution. The Negative Binomial distribution effectively models count data with over-dispersion. The over-dispersion parameter ϕ modulates the variance independently of the mean, capturing the inherent variability in litter quantities.

Let Y denote the macroplastic counts, assumed to follow a Negative Binomial distribution with mean μ and over-dispersion parameter ϕ :

$$Y \sim \mathcal{NB}(\mu, \phi). \quad (8.1)$$

The mean μ is linked to spatial covariates (longitude and latitude of each beach) through a Gaussian process after applying a Yeo-Johnson transformation [288] $\mathcal{T}$:

$$\mathcal{T}(\mu) \sim \mathcal{GP}(\mu_\mu, K) \quad (8.2)$$

The Yeo-Johnson transformation, accommodating both positive and zero-valued data, ensures positivity and adaptability to the data's skewness:

$$\mathcal{T}(x) = \begin{cases} ((x+1)^\lambda - 1)/\lambda & \lambda \neq 0, \\ \ln x + 1 & \lambda = 0. \end{cases} \quad (8.3)$$

The parameter λ was estimated via maximum likelihood estimation using all beaches and seasons, resulting in $\lambda = 4.27 \times 10^{-3} \approx 0$.

The covariance function K of the $\mathcal{GP}$ models the spatial correlation. A combination of two Matérn 3/2 kernels [283] was chosen to capture different scales of spatial dependence:

$$k(b, b') = k_1^{[3/2]}(b, b') + k_2^{[3/2]}(b, b'), \quad (8.4)$$

where

$$k_i^{[3/2]}(b, b') = \eta_i^2 \left(1 + \frac{\sqrt{3}d}{\rho_i} \right) \exp \left(-\frac{\sqrt{3}d}{\rho_i} \right), \quad i = 1, 2. \quad (8.5)$$

Here, η_i and ρ_i represent the variance and spatial scale parameters of the kernels, and d denotes the chordal distance [123] between locations b and b' . The parameters ρ_i define the typical length scales over which spatial autocorrelation occurs, while η_i modulate the amplitude of these correlations.

Prior distributions were specified for all model hyperparameters to reflect underlying assumptions and ensure robust model inference. For the mean function μ_μ of the GP, a weakly informative prior $\mu_\mu \sim \mathcal{N}(\alpha, \beta)$ was selected. The parameters α and β were defined such that the 95 % highest density interval (HDI) of the prior encompasses the 2.5th and 97.5th percentiles of the average beach litter densities, $[\mathcal{T}(q_{0.025}), \mathcal{T}(q_{0.975})]$, determined from the data. This range includes previously estimated median pollution levels of 149 [93] and 252 [138] items per 100 meters of beachline.

For the length scales ρ_i , positive-valued Inverse Gamma distributions $\rho_i \sim \mathcal{IG}(\gamma_i, \delta_i)$ were used. The hyperparameters γ_i and δ_i were defined such that the 95 % HDIs extend from 10 to 100 km for short length scales ($i = 1$) and from 100 to 5,000 km for medium-to-long length scales ($i = 2$), effectively capturing different spatial autocorrelation ranges.

Weakly informative Gamma priors $\eta_i \sim \mathcal{G}(\epsilon_i, \kappa_i)$ were set for the variance parameters. The parameters ϵ_i and κ_i were defined such that the 95 % HDIs cover the interval $[1/\sqrt{2}, 2/(3\sqrt{2})]$. This configuration, determined via exploratory prior predictive checks, ensures the $\mathcal{GP}$'s uncertainty covers the entire range of possible values in the Yeo-Johnson transformed data

space, providing the model with flexibility in determining the relative contribution of each length scale to the autocovariance function.

For the over-dispersion parameter ϕ , a weakly informative Inverse Gamma distribution $\phi \sim \mathcal{IG}(\nu, \sigma)$ was set, with ν and σ defined such that the 95 % HDI covers [2, 9]. This is consistent with observed ranges of over-dispersion in the data (Supplementary Figure 11).

The model was fitted using the No-U-Turn Sampler [107] (NUTS), a variant of Hamiltonian Monte Carlo, chosen for its efficiency in handling high-dimensional parameter spaces and its ability to navigate complex posterior landscapes. Four chains were run, each with 9,000 iterations, discarding the first 8,000 iterations as warm-up.

Convergence of the model parameters to a stationary distribution was verified by inspecting trace plots and using the $\hat{R}$ statistic [77], which was below 1.05 for all parameters. Comparison between prior and posterior distributions of the LGCP model hyperparameters (Supplementary Figure 12) confirmed appropriate updating of beliefs in light of the data.

Posterior predictive checks were performed for all beaches by comparing the modeled mean and variance of each beach to its sample statistics (Supplementary Figure 13), revealing generally good agreement. This indicates that the model adequately captures the underlying data structure, validating its suitability for predicting macroplastic distributions.

8.2.3 LGPC Model Posterior Inference

The seasonal effect size (σ) quantifies the degree of difference in macroplastic densities between seasons, ranging from -1 to 1 . Values near -1 or 1 indicate substantial seasonal differences, while a value of zero suggests no difference.

For any two distinct seasons s and s' from the set {Winter, Spring, Summer, Autumn}, let Y and Y' represent the macroplastic densities with posterior predictive distributions $p(y|D)$ and $p(y'|D')$, based on survey data D and D' . The effect size between these seasons is calculated as the probability that the density in season s exceeds that in season s' :

$$\begin{aligned}\hat{\sigma}_{s,s'} &= p(y > y'|D, D') \\ &= \int_{\mu} \int_{\mu'} I(\mu > \mu') p(y, y'|D, D') ds ds'\end{aligned}\tag{8.6}$$

where $I(y > y')$ is an indicator function equal to 1 when $y > y'$ and 0 otherwise.

To normalize this measure between -1 and 1 , we compute:

$$\sigma_{s,s'} = 2\hat{\sigma}_{s,s'} - 1. \quad (8.7)$$

The overall seasonal effect size σ is then defined as the maximum absolute value among all season pairs:

$$\sigma = \max_{s \neq s'} |\sigma_{s,s'}|. \quad (8.8)$$

Statistical confidence (γ) measures the likelihood that the observed seasonal differences in mean macroplastic densities are statistically significant. For mean densities μ and μ' in seasons s and s' , with posterior distributions $p(\mu)$ and $p(\mu')$, we determine the highest density intervals (HDIs) at a confidence level $c \in [0, 1]$ for each.

The statistical confidence is then given as the maximum confidence level c such that the HDIs of μ and μ' do not overlap:

$$\gamma = \max\{c \mid \text{HDI}_\mu(c) \cap \text{HDI}_{\mu'}(c) = \emptyset\}. \quad (8.9)$$

8.2.4 Clustering of Seasonal Patterns

Principal Component Analysis (PCA) serves as a fuzzy clustering method [54] to discern beaches with analogous seasonal macroplastic patterns. Utilizing the posterior predictive distributions (PPD) from the seasonal LGCP models, we systematically analyze the data as follows:

1. **Matrix representation:** Define $Y \in \mathbb{N}^{(n_b \times n_s)}$ as a matrix of realizations for $n_b = 168$ beaches across $n_s = 4$ seasons, drawn from the PPD. Here, y_{bs} denotes the modeled macroplastic density for beach b in season s .
2. **Normalization:** To compare seasonal distributions equitably, we normalize each beach's macroplastic counts by its annual total, resulting in relative seasonal shares:

$$\hat{y}_{bs} = \frac{y_{bs}}{\sum_s y_{bs}} \quad (8.10)$$

3. **Weighted PCA:** Implementing weighed PCA on the normalized matrix $\hat{Y}$ allows for the accentuation of seasonal covariance, using beach-specific statistical confidence γ_b

as weights. PCA yields eigenvalues λ_i , eigenvectors v_i and corresponding principal component (PC) scores, segregating the data into clusters based on sign. Each eigenvector reflects a distinct cluster, separated into positive and negative parts:

$$v_i^+ = \sum_{v_{bi}>0} v_{bi}^2, \quad v_i^- = \sum_{v_{bi}<0} v_{bi}^2 \quad (8.11)$$

where v_i^+/v_i^- quantifies the explained variance by positive/negative clusters, respectively, offering insight into their respective contributions to the overall seasonal variance in macroplastic distribution.

We perform these steps for $N = 4000$ realizations drawn from the PPD so that we obtain distributions for all quantities. The degree of membership of beach b to cluster i is then given by the mean of the distribution of v_{ib} . The associated statistical confidence is given by the fraction of the distribution being above/below 0, renormalized to be between 0 and 1 to ensure a proper probability. For instance, the statistical confidence of the positive clusters read

$$\gamma_{ib} = \frac{2}{N} \sum_j^N I(v_{ib} > 0) - 1, \quad (8.12)$$

where $I(\cdot)$ is the indicator function.

Supplementary Figure 14 shows that the first two clusters, cluster $C1^+$ and $C2^+$, are statistically significant and explain over 50% of the variance.

8.2.5 Sources of Macroplastics and Seasonal Potential

We analyzed riverine emissions (A), fisheries (B), and mariculture activities (C) to understand their potential impact (D) on seasonal beach litter patterns.

(A) High-resolution, 6-hourly river discharge data from 2010 to 2020 were averaged into monthly means to create climatologies. Macroplastic emissions from rivers were inferred using model data on mismanaged solid waste exports to the sea [246]. We scaled the data from annual to monthly emissions based on river discharge under the assumption of proportionality. While this simplification does not capture the full complexity of riverine macroplastic emissions [264], it provides a reasonable approximation for understanding seasonal emissions patterns [37, 45, 146, 263], excluding extreme events [265].

(B) We utilized fishing intensity data from the Global Fishing Watch dataset [135], spanning from 2013 to 2020, transformed into a gridded format on a regular 0.25° longitude-latitude

grid. To reduce dimensionality, we aggregated the high-resolution spatial data into predefined regions based on ocean basins from NaturalEarth (1:50 million scale) and extended these regions based on oceanographic settings (Figure ??). The regions include the Bay of Biscay (0), North Sea (1), Irish Sea (2), English Channel (3), Bristol Channel (4), Inner Seas (5), Skagerrak (6), West Iberian Sea (7), Gulf of Cadiz (8), West Irish Sea (9), and South Irish Sea (10). Monthly climatologies of fishing intensity were calculated for each of these 11 regions.

(C) To estimate plastic emissions from mariculture, we assumed local emission rates are proportional to production intensity. We used a global dataset of mariculture production sites [41], which partially relies on data from the European Marine Observation and Data Network (EMODnet) [59, 60]. The EMODnet dataset distinguishes between individual farms and production areas, unlike the global dataset.

To mitigate potential biases due to differing definitions, we consolidated all locations into production areas by applying a threshold distance below which farms are considered part of a single production area. We estimated this threshold distance by analyzing the distribution of distances between mariculture sites in France, where sites are exclusively production areas in the EMODnet dataset. Histograms of great circle distances revealed that distances below 3 km are uncommon in France but prevalent elsewhere, suggesting 3 km as the typical minimal separation between production areas. Using this threshold (approximately 0.027° at the equator), we created a regular 0.027° longitude-latitude grid, marking cells with at least one location. For visualization, we aggregated production areas into $0.54^\circ \times 0.54^\circ$ cells, weighting each grid cell by the cosine of its latitude to account for decreasing cell areas toward the poles. This process refined the mariculture dataset, reducing biases from differing location definitions.

To estimate seasonal variations in plastic emissions from mariculture, we considered how plastics enter the marine environment. Studies have identified pathways including routine operations, inadequate management, deliberate discharge, and extreme weather [82, 239, 257], with extreme weather being a significant source in the Northeast Atlantic (NEA) region. While routine emissions likely occur year-round, we hypothesize increased activities – and thus potential emissions – during harvest periods.

We proposed two scenarios:

1. Adverse weather scenario: Emissions are driven by adverse weather conditions. Monthly plastic emissions rates were assumed proportional to monthly mean ocean wave energy. We obtained monthly climatologies of significant wave height from the Global Waves Reanalysis [55] (2013-2020) and calculated wave energy as the square of the significant wave height. Each production area's monthly climatology was assigned based on the spatial average of wave energies within a $0.5^\circ \times 0.5^\circ$ rectangle centered on it. The cli-

matology was normalized to an annual average of one and scaled by production area density.

2. Seasonal activity scenario: Emissions increase during harvest periods. We analyzed approximately 4 million export transaction from 2009 to 2023 obtained from the European Market Observatory for Fishery and Aquaculture Products [64] (EUMOFA) to approximate harvest seasons. Although the export data do not distinguish between fisheries and aquaculture products, the main aquaculture species in the NEA region—salmon (65 %), oysters/mussels (21 %), and trout (9 %) [67]—are predominantly farmed (over 98 %).

Our methodology involved several steps. First, we summed export transaction to calculate total monthly exports per species and country. We then removed long-term trends by applying a convolution filter to detrend the data. Next, we determined the seasonal component by averaging the detrended data over all years, weighted by annual export volumes. To compute species-weighted averages, we matched species group in the mariculture dataset [41] and used the min-max-normalized seasonal components weighted by the overall mean annual aquaculture production of individual species. To calculate the mean annual aquaculture production for each species, we used a species-year-weighted average over the years 2001-2020, weighting each year by the number of OSPAR surveys per year and country to account for sampling bias and production increases over time. This process resulted in a seasonal component for each species group and country. Finally, we assigned the seasonal component to each production area based on its proximity to the nearest country and scaled it by production area density.

A third scenario considered river discharge variations, relevant for countries like Portugal where mariculture sites are situated within river mouths [219]. Seasonal variations were computed similarly to the adverse weather scenario but using river discharge climatologies.

(D) To assess how well river discharges, and fisheries and mariculture explain seasonal beach litter patterns, we calculated the coefficient of variation (c_v) to measure seasonal amplitude and the Pearson correlation coefficient (c_r) to assess pattern alignment. We adjusted seasons to match the timing of beach litter surveys, shifting them by one month due to the OSPAR sampling protocol prioritizing mid-season surveys—over 70 % occurred in January, April, July and October. Therefore, we considered November - January, February - April, May - July and August - October for comparisons. The seasonal potential, defined as $\Phi = \max(0, c_r \cdot c_v)$, identifies the strongest seasonal signals in potential sources, prioritizing those with considerable variability and temporal alignment with the observed litter patterns.

8.3 Results and Discussion

8.3.1 Seasonal Variations of Anthropogenic Beach Litter in the North-East Atlantic

Our analysis, utilizing 3866 surveys from the OSPAR dataset over two decades (2001 - 2020) and encompassing 168 beaches along the western European coastlines, employs a spatial LGCP model (Methods ??) to evaluate macroplastic pollution across the four seasons: winter, spring, summer and autumn. Macroplastic pollution was defined as the number of all artificial polymer materials larger than 2.5 cm per 100m of beach length (Methods 8.2.1). We found that median macroplastic levels peak in spring, reaching 232 items per 100 meters of beachline (items/100 m), with a 95% Bayesian credible interval (CI) ranging from 79 to 440 items/100 m. Pollution levels are slightly lower in winter (184 items/100 m), autumn (174 items/100 m) and summer (154 items/100 m), although the broad CIs suggest these differences are not statistically significant (Supplementary Table 1). These results are consistent with previously estimated annual medians for this region [93, 138].

A more detailed per-beach analysis reveals distinct interseasonal variations in macroplastic pollution, with 15% of beaches exhibiting such differences at a 95% confidence level (Figure ??, Methods 8.2.3) increasing to 39% when the confidence threshold is adjusted to 80%.

Further examination of seasonal variation patterns (Figure ??A) shows that the most common scenario, observed in 36% of confidence-weighted beaches, involves increased plastic litter in spring compared to summer. Conversely, increased summer litter over spring is the least common scenario, observed in only 9% of cases. These variations highlight the complex and diverse seasonal patterns of beach litter with a general trend for higher litter accumulation in winter/spring compared to summer/autumn. Among beaches exhibiting seasonal variations, a dominant pattern emerges where winter and spring seasons experience 95% to 118% more plastic pollution than the summer and autumn seasons (Figure ?? B). In contrast, beaches following a summer/autumn > winter/spring pattern show a smaller increase, ranging from 60% to 78% more macroplastics. The substantial range of confidence intervals for these estimates suggests only weak statistical support for the idea that relative differences are stronger in one season compared to another (Supplementary Table 2).

These findings are consistent with a growing body of scientific evidence showing pronounced seasonal variations of beach litter across various beaches worldwide [37, 189, 198, 238], particularly in the NEA region [4, 48, 73, 218, 224, 233, 274], although a clear large-scale pattern remains elusive. For example, two regional but weak patterns were reported [233] exhibiting an increase in autumn/winter pollution in the wider Atlantic region and a spring peak specifically in the North Sea. In contrast, a comprehensive study focusing on UK beaches found no pronounced seasonal patterns [181]. The discrepancy may stem from the insufficient statistical power of both studies. The neighborhood-aware LGCP model approach in our study allows better estimations of litter distributions for each beach by taking into account pollution levels from beaches nearby. Our model suggests that 60% to 77% of beaches, depending

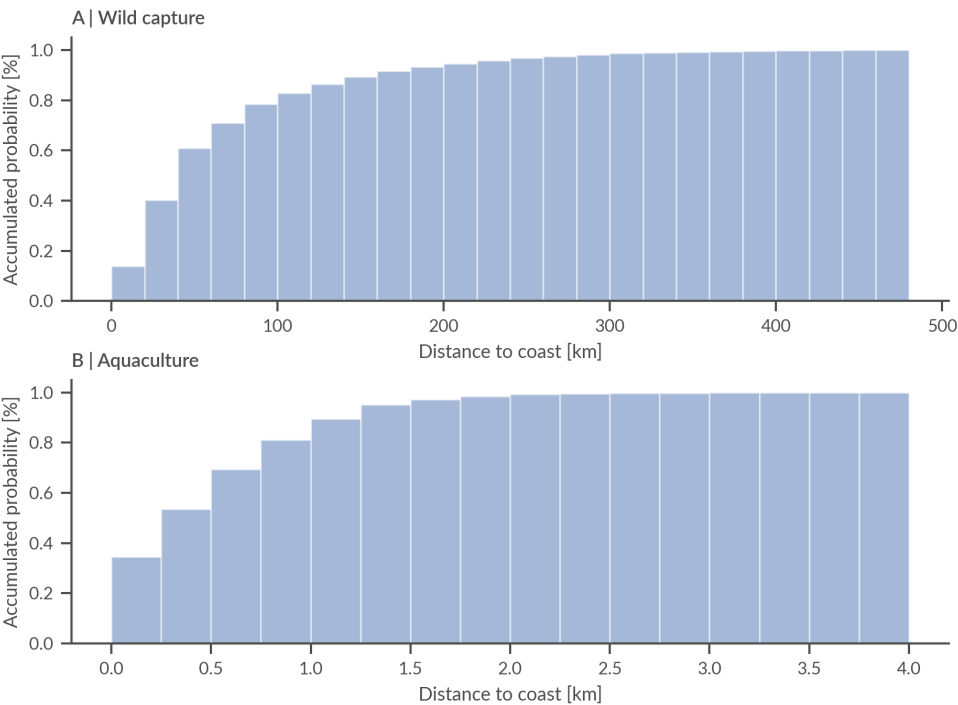


Figure 8.1 — TBD. Reproduced with permission from Springer Nature.

on the season, in the OSPAR dataset provide useful information for neighboring beaches (Supplementary Notes 2). This is supported by the fact that a non-Bayesian analysis, isolating each beach for individual analysis, finds interseasonal variations for only 12% of beaches at a 95% confidence level and 24% at an 80% confidence level (Supplementary Figure 1). This discrepancy underscores the LGCP model’s capability to more accurately estimate beach litter distributions by incorporating pollution levels from surrounding areas.

Moreover, our results are corroborated by recent studies based on numerical simulations [286] and satellite measurements [45], which show pronounced seasonal cycles in floating marine debris on large spatial scales, suggesting that beach litter should follow similar patterns.

8.3.2 Identification of Seasonal Hotspots

Our analysis reveals that seasonal variations in plastic pollution are not uniformly distributed but are spatially clustered, exhibiting stark differences across various European coastal regions (Figure ??). Notably, the western Iberian Peninsula, the western French coastline, the Irish Sea, and the Skagerrak exhibit strong seasonal patterns. Conversely, the North Sea region displays relatively low seasonal beach litter variations. By comparing standardized variance of macroplastic with the seasonal effect size on a per-beach level, we identified a strong

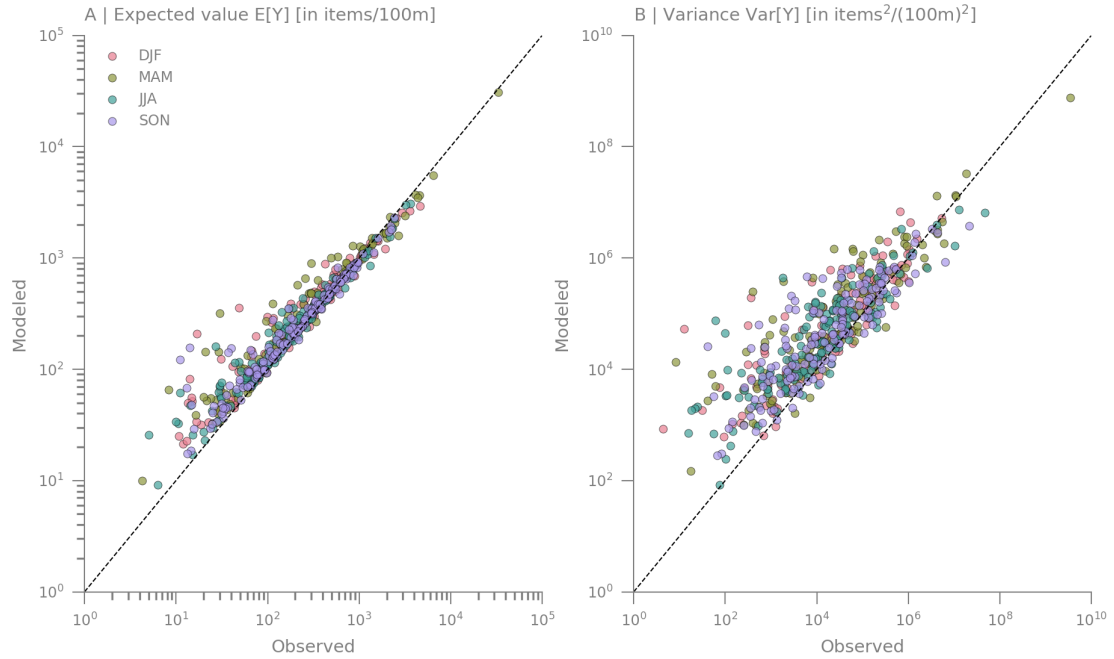


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Table 8.1 — Inference of parameters of the log-Gaussian Cox process model: variances η_i^2 , length scales ρ_i measured in km, with $i \in \{\text{short}, \text{long}\}$, spatial mean μ_μ and over-dispersion ϕ . Shown are the 50th percentile (median) together with the 95 % highest density interval (in parentheses) given by the minimum width Bayesian credible interval (BCI).

	η_{short}^2	ρ_{short}	η_{long}^2	ρ_{long}	μ_μ	ϕ
Winter	1.08 (0.92; 1.23)	16 (10; 23)	0.64 (0.33; 0.93)	777 (236; 1965)	184 (53; 352)	1.51 (1.37; 1.66)
Spring	1.19 (1.02; 1.38)	16 (10; 22)	0.72 (0.42; 1.04)	508 (215; 1097)	232 (79; 440)	1.36 (1.25; 1.48)
Summer	1.08 (0.94; 1.24)	12 (7; 17)	0.65 (0.36; 0.97)	593 (206; 1452)	154 (56; 290)	1.54 (1.41; 1.68)
Autumn	1.06 (0.93; 1.21)	9 (6; 13)	0.60 (0.33; 0.90)	690 (227; 1934)	174 (73; 332)	1.53 (1.41; 1.66)

1 correlation (+0.60), suggesting that the observed variability in OSPAR beach pollution levels
 2 can be partially attributed to seasonal factors (Supplementary Figure 2).

3 To further explore these observations, we employed Principal Component Analysis (PCA)-
 4 driven clustering to discern underlying seasonal variation patterns (see Methods 8.2.4). This
 5 analysis yielded two principal clusters indicative of distinct seasonal behaviors (Figure ??):

- 6 • **Cluster C1⁺** is characterized by a marked increase in plastic pollution during **spring**,
 7 predominantly observed in the Skagerrak and, to a lesser extent, along the French and
 8 Iberian western coastlines.

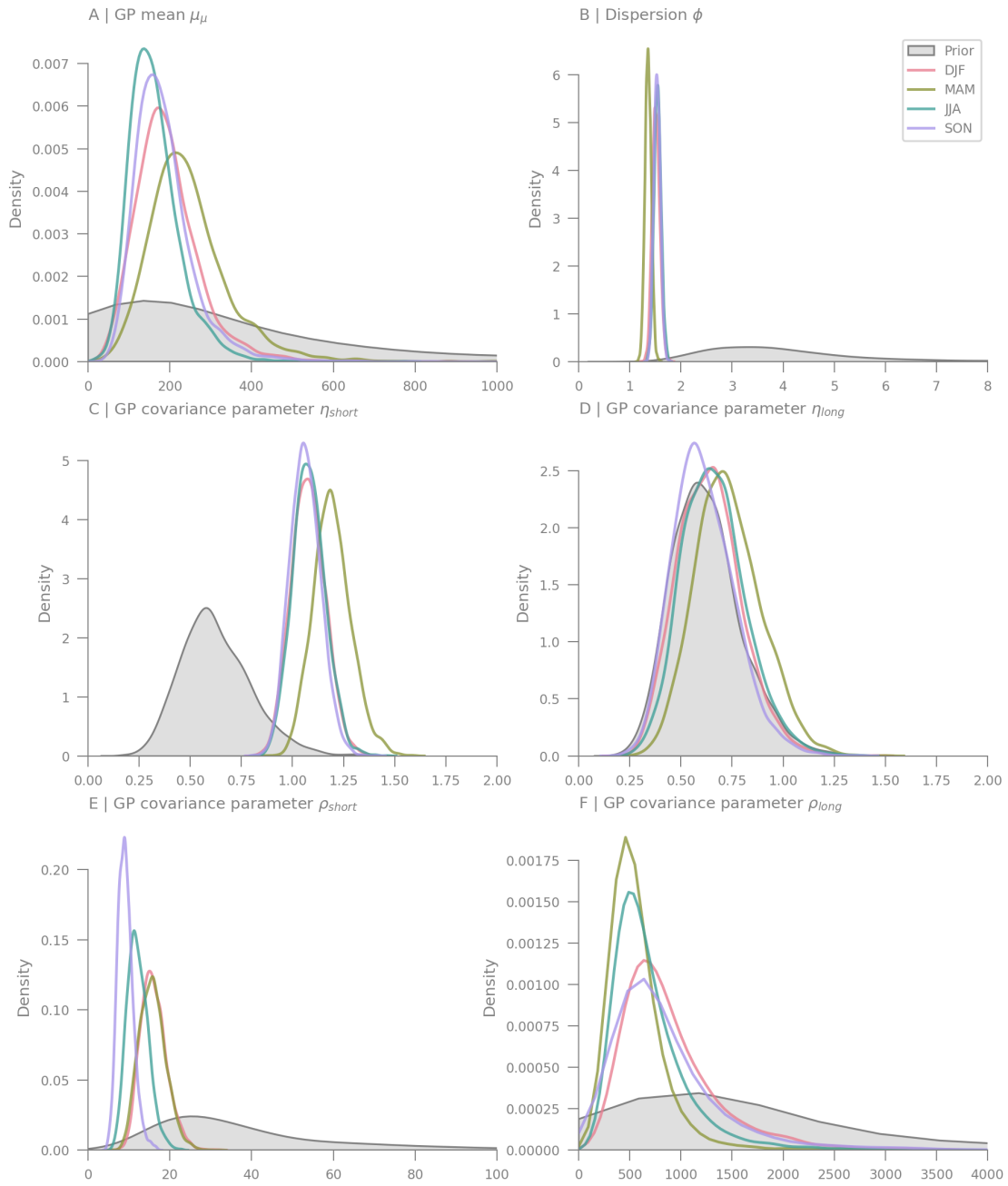


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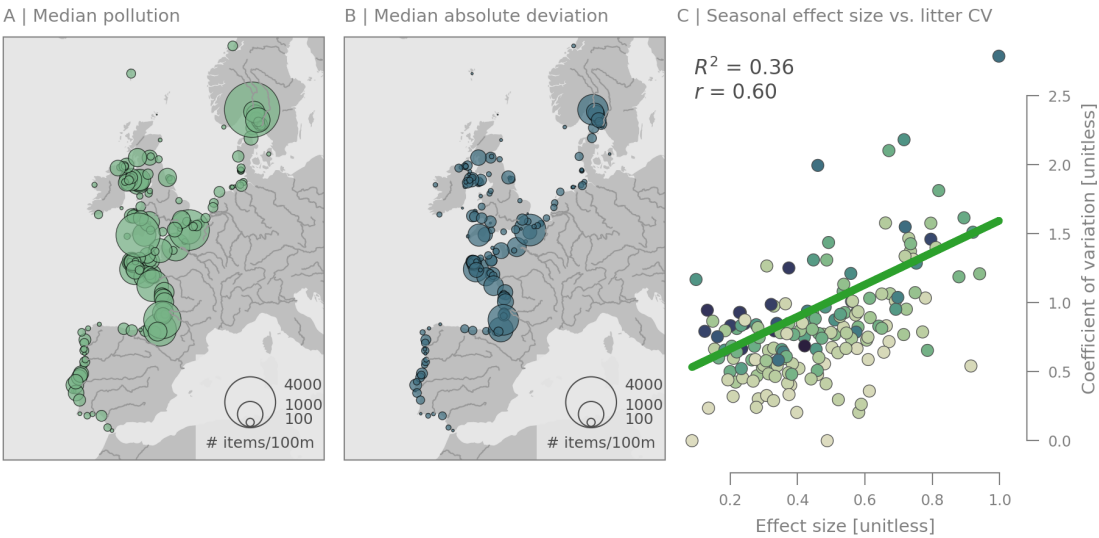


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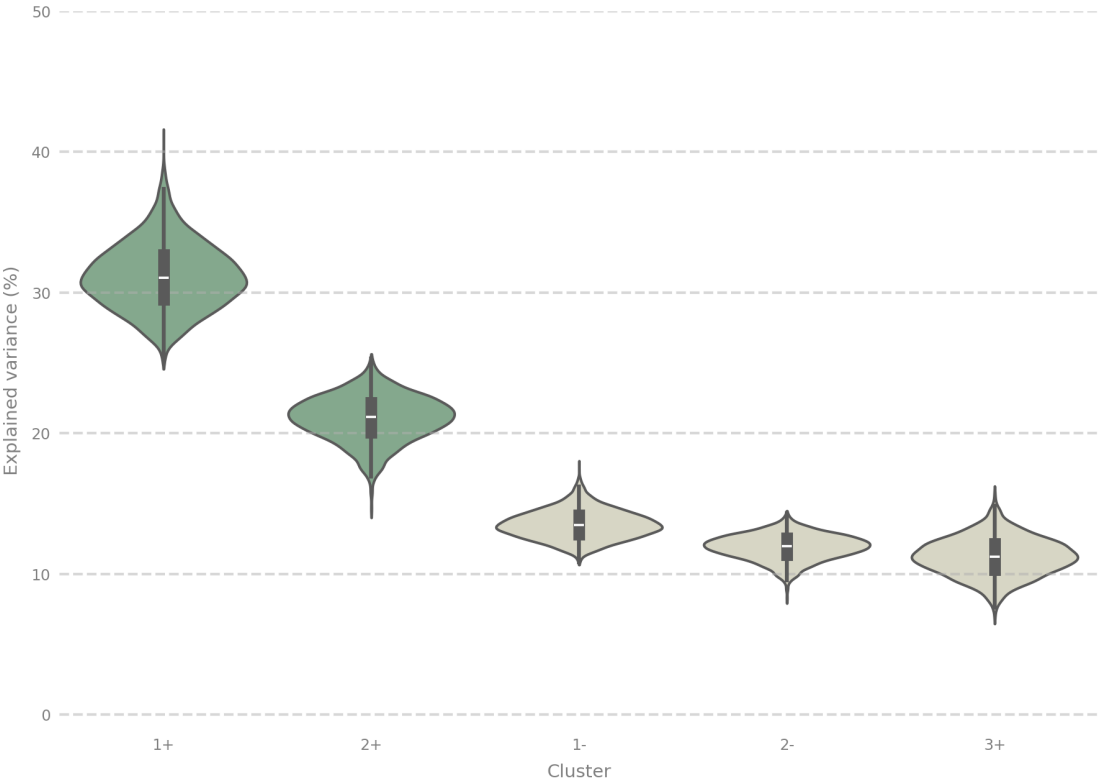


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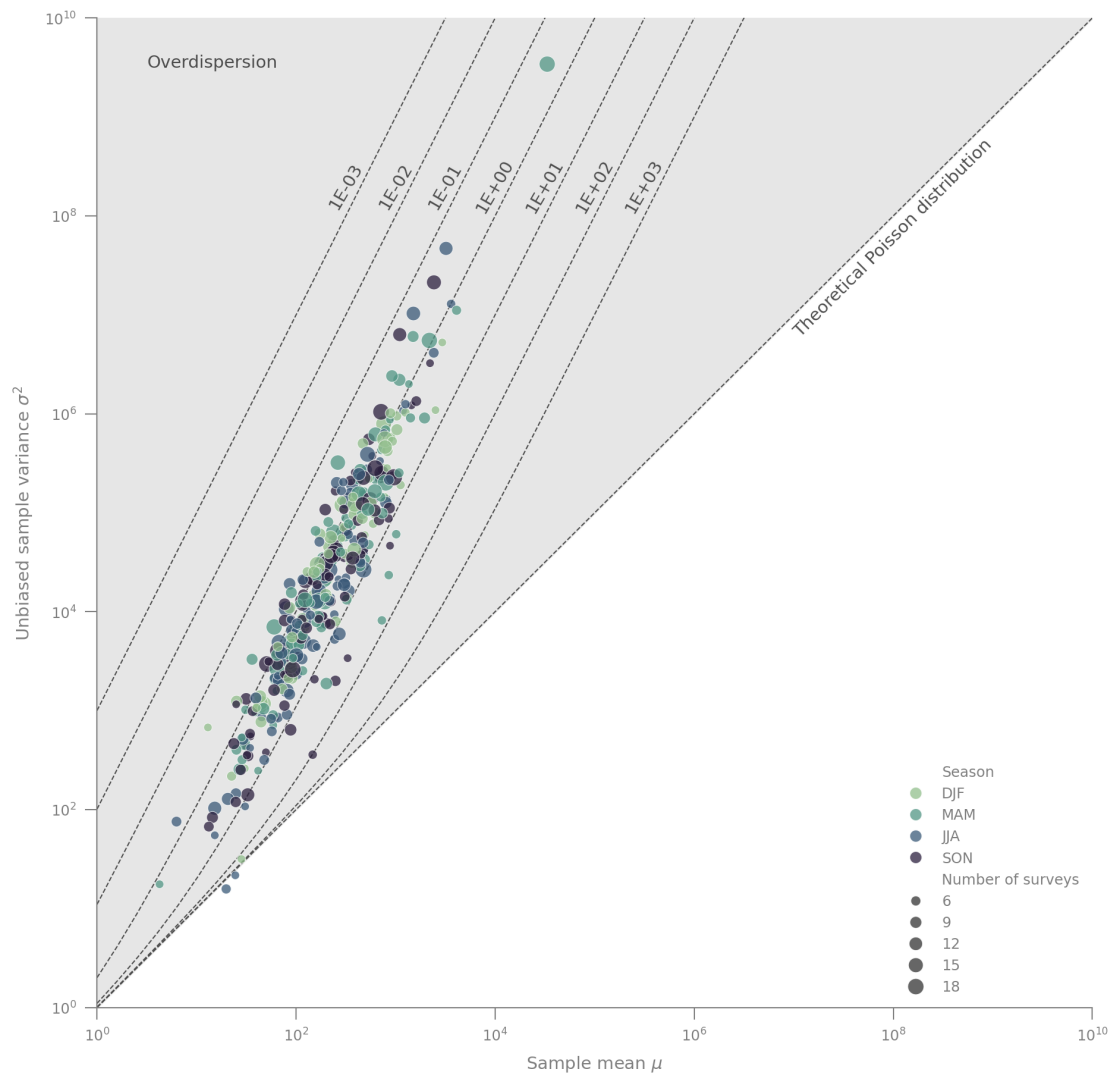


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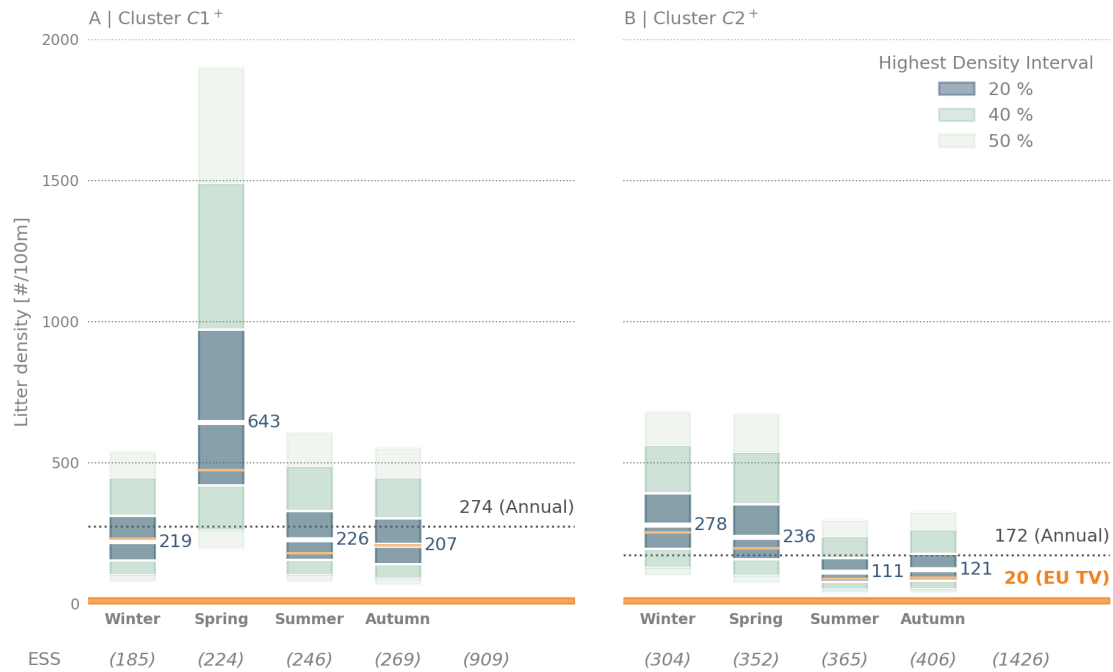


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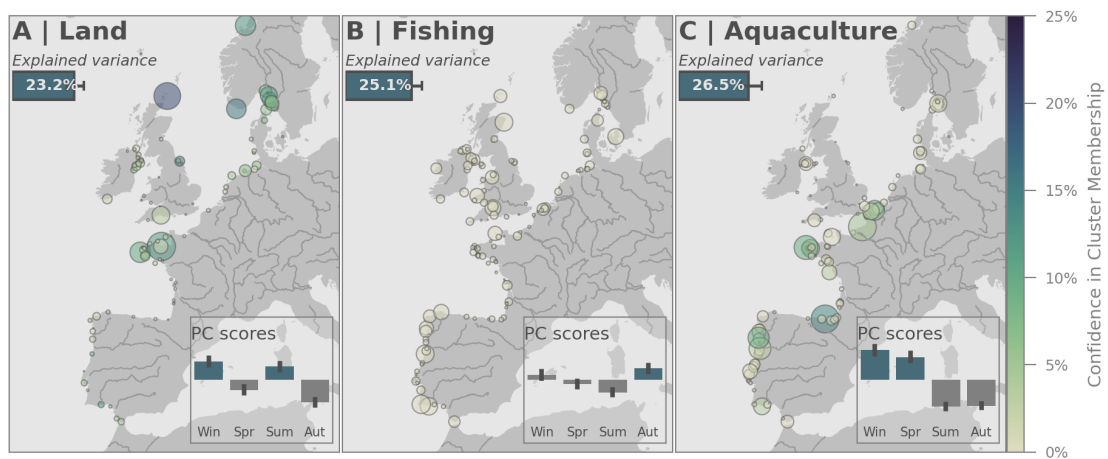


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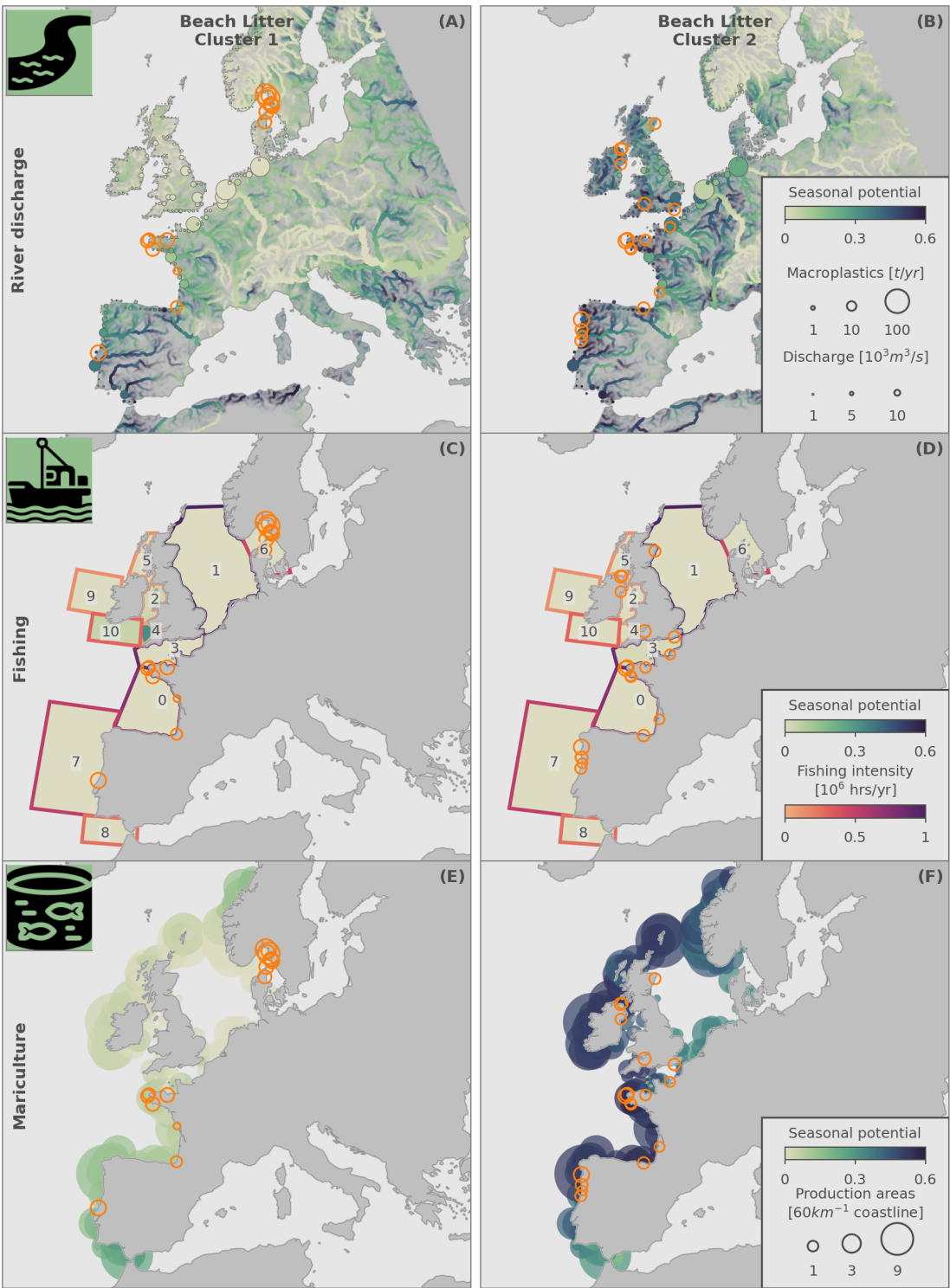


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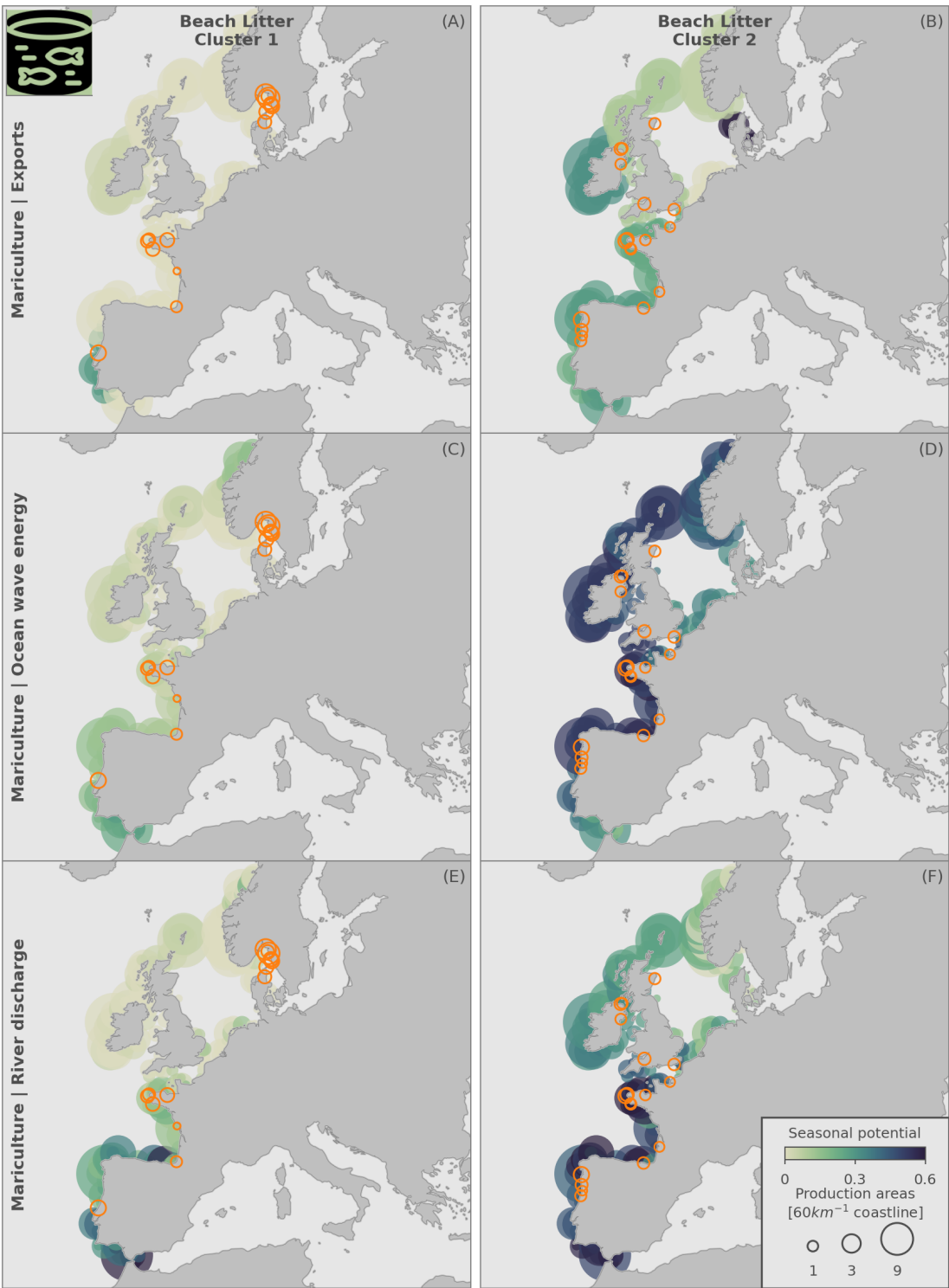


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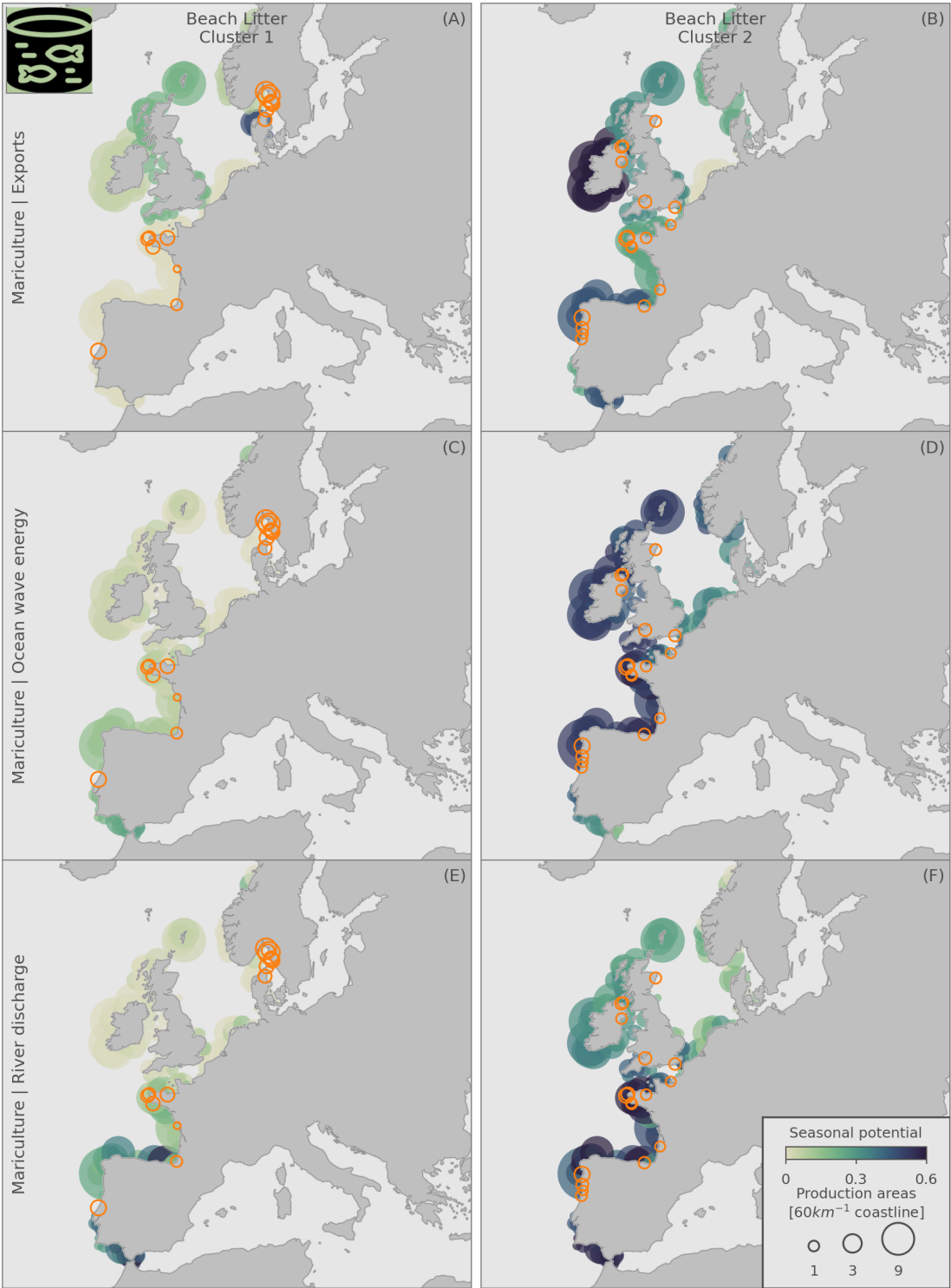


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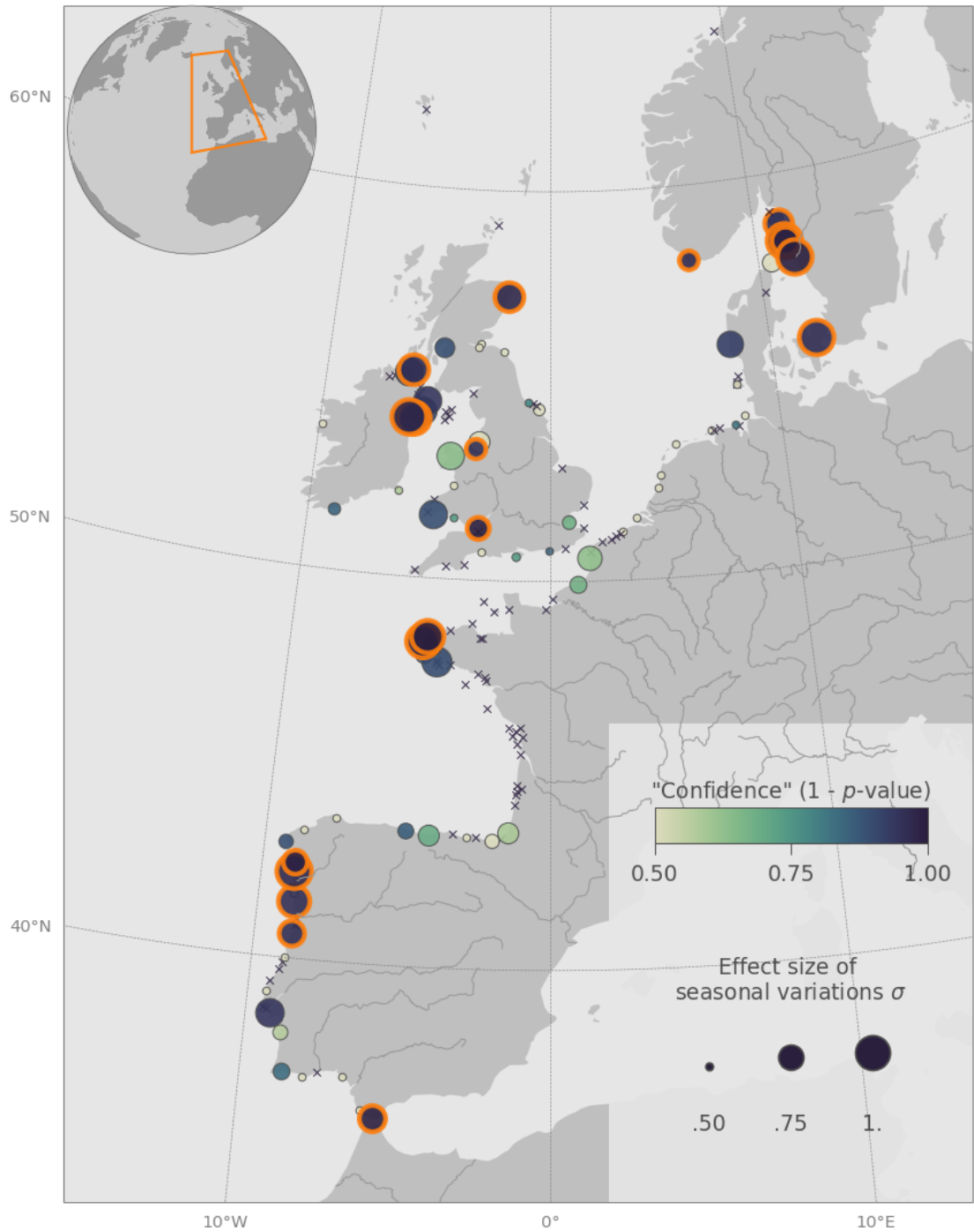


Figure 8.12 — TBD. Reproduced with permission from Springer Nature.

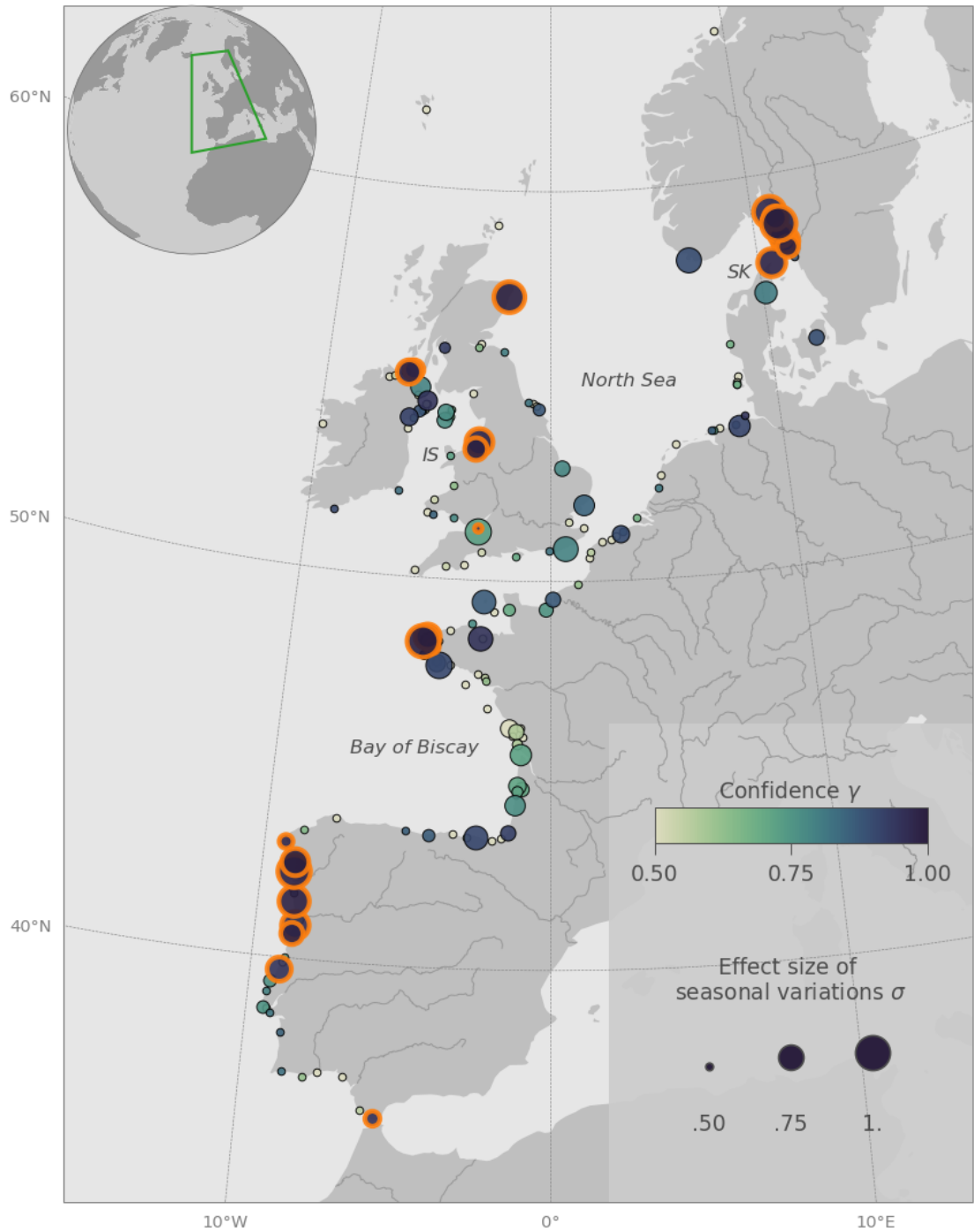


Figure 8.13 — TBD. Reproduced with permission from Springer Nature.

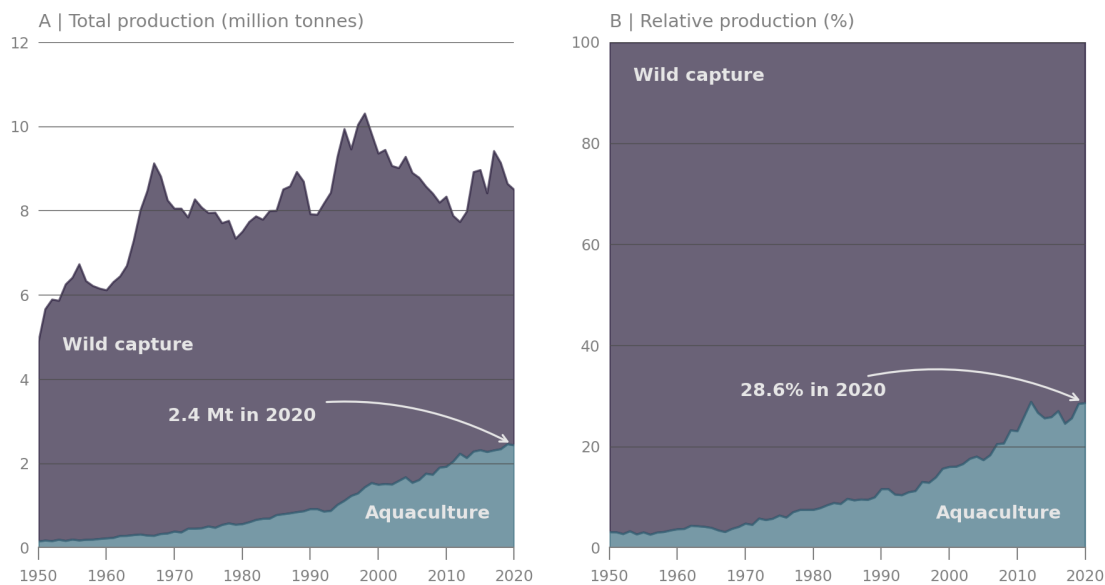


Figure 8.14 — TBD

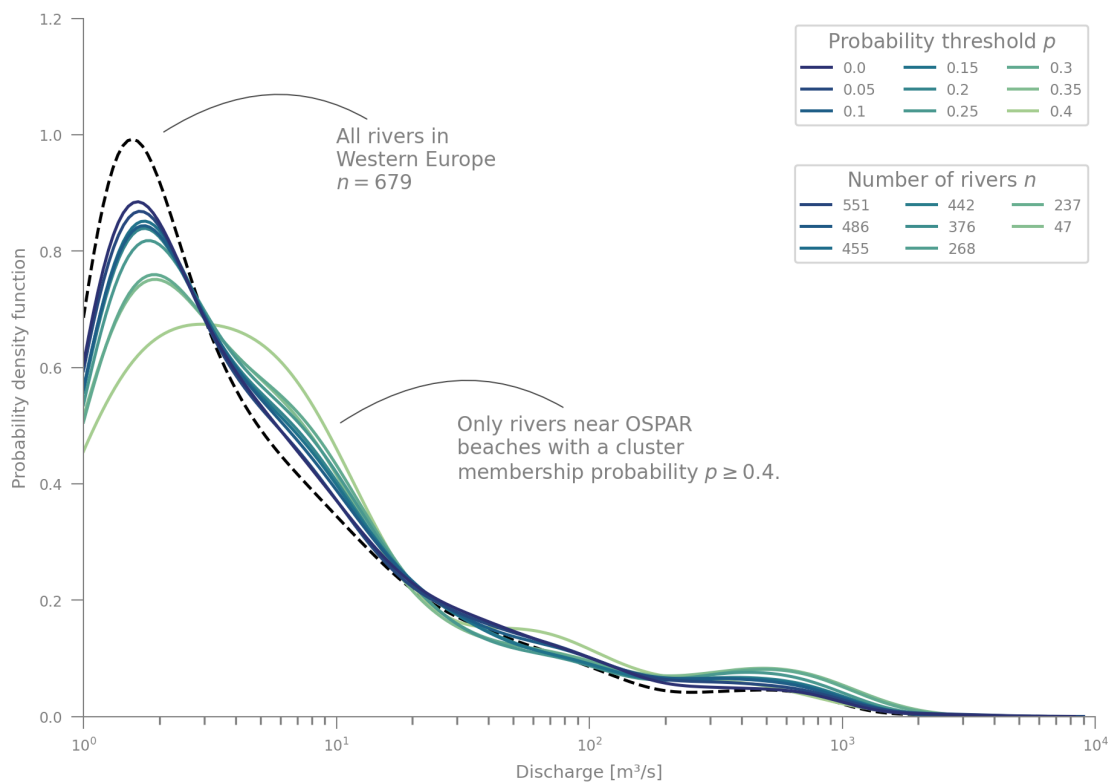


Figure 8.15 — TBD

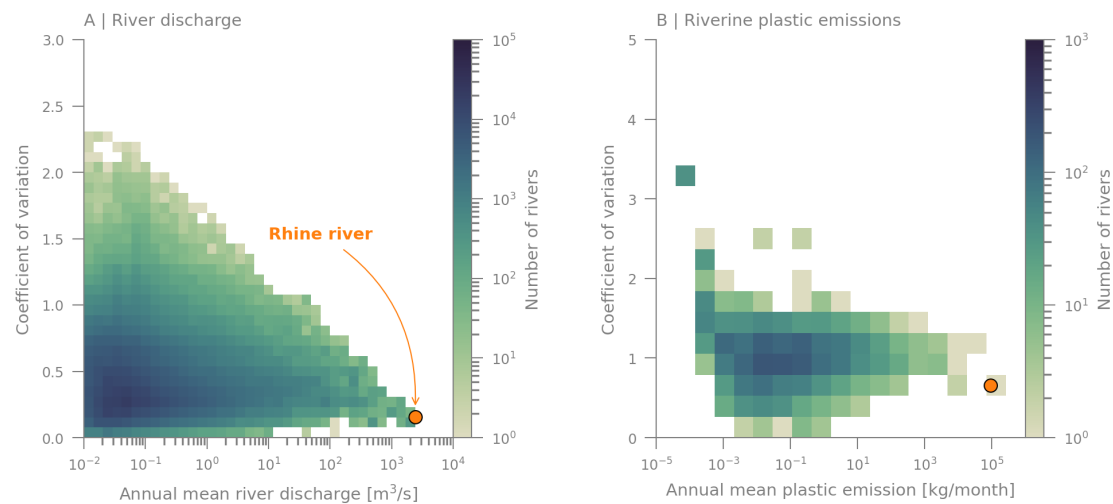


Figure 8.16 — TBD

Table 8.2 — Median relative difference of macroplastic X between season i and j computed as $\text{med}((X_j - X_i)/X)$ where $X = X_i$ if $\text{med}(X_j - X_i) \geq 0$, else $X = X_j$. Here, $\text{med}(\cdot)$ refers to the median. We then computed estimates for the NEA region as weighted median over all beaches with the individual confidence estimates as weights. For simplicity, all values are shown in percentage, together with their 25th and 75th weighted percentile.

$\leq$	Winter	Spring	Summer	Autumn
Winter	-	88(34; 238)	78(38; 160)	60(32; 184)
Spring	90(49; 190)	-	75(38; 185)	72(38; 223)
Summer	108(66; 204)	108(53; 202)	-	80(45; 121)
Autumn	95(48; 182)	118(58; 243)	61(26; 123)	-

- **Cluster C2⁺** exhibits heightened plastic abundance in the **winter/spring** seasons, spatially similar to cluster C1⁺, strongly influenced by beaches along the western Iberian and French coastlines and around the Irish Sea.

Both clusters highlight a comparative increase in pollution during the winter/spring months over summer/autumn, a pattern reported in previous studies for these specific regions [48, 73, 128, 142, 224, 233, 247]. However, our clustering supports the notion that even within these regions, beaches with different seasonal patterns than those reported as dominant may occur, indicating high spatial variability on small scales. This variability may explain the divergent findings in some studies [4, 73, 218, 224, 233, 274].

Utilizing our PCA-derived cluster weights alongside LGCP-modeled beach litter distributions, we calculated weighted percentiles for each cluster, revealing substantial seasonal differences (Figure ??). For instance, the cluster C1⁺ shows median pollution levels escalating from 207 items/100 min autumn to 643 items/100 min spring. This springtime peak signifies a more than

200% increase in median plastic pollution levels compared to autumn, identifying a strong seasonal hotspot.

Conversely, winter/spring cluster $C2^+$ presents a more uniform distribution between seasons, with the most considerable relative increase observed as a 150% rise in winter compared to summer. This pattern suggests that cluster $C2^+$ more closely represents the general seasonal trend across European beaches, whereas $C1^+$ captures regions with more extreme seasonal fluctuations.

These seasonal pollution levels are well above the European Union threshold value (EU TV) of 20 items/100 m for beach litter [267], underscoring the urgent need for targeted litter management strategies. Specifically, our results for cluster $C1^+$ indicate that relying on annual medians as a measure of beach litter pollution, which suggests a pollution level of 274 items/100 m in agreement with previous research [93, 138, 234, 267], can be misleading as it fails to account for seasonal pollution peaks.

8.3.3 Drivers of Seasonal Hotspots in Anthropogenic Beach Litter

To illuminate the origins of the identified seasonal hotspots for beach litter in the NEA, particularly during spring ($C1^+$) and winter/spring ($C2^+$), we examined the relationship between these hotspots and the global sources of marine litter: riverine macroplastic emissions [86, 126, 146, 170, 246, 290], fishing [126, 135, 195] and aquaculture activities [136, 195, 239, 257].

We analyzed the latest data on simulated riverine emissions of macrolitter [246]. Given that these data represent average annual emissions, we employed high-resolution reanalysis data of river discharge to infer seasonal variation [169], approximating the amount of macroplastics entering the ocean from rivers at different times of the year (Methods 8.2.5).

To assess the impact of fishing-related activities, we utilized high-resolution daily data on fishing intensity [135]. For estimating seasonal variations in local litter emissions from mariculture, we employed a recently developed dataset of mariculture production locations [41], combining it with seasonal information derived from changes in mean monthly ocean wave energies as a proxy for adverse weather conditions [82, 239] (Methods 8.2.5, Supplementary Figure 3). Additionally, to enhance our analysis, we proposed two alternative scenarios for the seasonality of mariculture production: one using monthly national export statistics and the other using river discharge data for farms situated in rivers [219].

To quantify each source's potential to create the observed seasonal beach litter patterns, we calculated both the coefficient of variation (c_v) and the Pearson correlation coefficient (c_r) of monthly climatologies. The former measures the magnitude of seasonal variations, while the latter indicates the degree of similarity between the seasonal variations of beach litter and litter sources. We then introduced a seasonal potential metric (Φ), calculated as $\Phi =$

$\max(c_v c_r, 0)$, which identifies the strongest seasonal signals in the potential sources of beach litter, prioritizing those with strong variability and temporal alignment with the observed litter patterns (Methods 8.2.5).

Our analysis revealed that both riverine discharge and mariculture exhibit a strong seasonal potential Φ to explain the seasonal variations in beach litter during the winter/spring pattern (Figure ??). In contrast, fisheries display a low seasonal potential, close to zero (Figure ??C-D), due to relatively minor seasonal variations that typically peak during summer in most regions. This indicates that fisheries are less likely to substantially contribute to seasonal beach litter variations. Furthermore, since less than 5% of wild capture occurs within the first 4 km of the coastline—with the majority happening tens of kilometers offshore (Supplementary Figure 4)—macroplastic emissions from fishing vessels are likely to diffuse over space and time [235].

Interestingly, our findings suggest that numerous small-scale rivers (annual mean discharge $\sim 10 \text{ m}^3 \text{ s}^{-1}$), rather than larger ones, are more likely responsible for the observed seasonal variations (Figure ??A-B, Supplementary Figure 5). Smaller rivers tend to exhibit more pronounced seasonal fluctuations than larger rivers (Supplementary Figure 6), underscoring their importance as pathways for macroplastics from continental to marine environments [86]. Our analysis supports previous studies indicating that major rivers are minor contributors to floating litter variability in the North Sea [254], where the potential influence of large river systems like the Rhine is comparatively low (Figure ??A-B, Supplementary Figure 6).

To further investigate the drivers of seasonal variations, we computed the relative fractions of plastic items likely linked to land, fishing or mariculture activities (Methods 8.2.1). This analysis aimed to determine if the seasonal patterns observed in absolute item counts could be explained by an increase in the relative fraction of these three sources. We modeled the fraction for each source and season separately using Gaussian Processes in the logit-transformed space. Subsequently, we performed PCA-clustering for each source, similar to the approach used in Figure ??, resulting in one significant cluster per source.

By examining these clusters, we found that only the mariculture-related items displayed a winter/spring pattern similar to cluster C2⁺ of the absolute item counts (Figure ??, Supplementary Figure 7). Additionally, the spatial pattern of mariculture-related items largely aligns with the identified hotspots along the western Iberian and French coastlines. This suggests that an increase in mariculture-related items could partially explain the rise in absolute item counts during winter/spring. In contrast, the temporal and spatial patterns of clusters derived from land and fisheries fractions did not match any cluster patterns found for absolute item counts. While there is a growing number of studies noting the importance of plastic emissions from mariculture systems [136, 239, 257], not much is known about their seasonality. Previous results from mariculture-exposed sandy beaches along the coast of Chile

1 suggest considerable seasonality of fishing/mariculture-related items [255]. More recently, pro-
2 nounced seasonal variations in the abundance of microplastics were found in a predominantly
3 bivalve-mariculture-influenced bay in Thailand [223].

4 To test whether bivalve aquaculture in the NEA provide a potential source of beach litter vari-
5 ability, we computed the seasonal potential specifically for bivalves/invertebrates (mainly mus-
6 sels and oysters farming in the NEA). The results were very similar to those for all mariculture
7 production (Supplementary Figure 8, 9). The primary difference is that bivalves/invertebrates
8 production is less concentrated along the Norwegian coastline, which is predominantly salmon
9 mariculture [66]. This difference better matches the observed seasonal patterns of beach litter;
10 however, it is not entirely conclusive due to the relatively low number of surveyed beaches in
11 that area, complicating validation.

12 It is important to note that not all items can be confidently linked to these specific sources, as
13 60.1% of items found in the NEA have degraded to the point where they are unrecognizable
14 and are categorized generically (e.g. "large plastics" or "others"; see Methods 8.2.1). These
15 low numbers of identifiable items likely reduce the confidence in the model's ability to detect
16 certain patterns (Supplementary Figure 7).

17 Additionally, our study makes a few key simplifications. Firstly, our approach assumes that
18 seasonal variations in macroplastics is mainly due to local sources, while the impact of more
19 distant sources is negligible. Although simulation studies show that plastics can travel vast
20 oceanic distances and in some cases remote contributions can outweigh local ones [28, 33, 187],
21 recent studies indicated that coastal areas mainly accumulate locally sourced macroplastics
22 [179]. Furthermore, our own analysis suggests that there are hardly coherent spatial structures
23 on length scales beyond 100 km (Supplementary Notes 2, Supplementary Table 1). This leads
24 us to believe the impact of distant plastics on our findings is minimal.

25 Secondly, our analysis overlooks ocean dynamics and their role in distributing plastics [235],
26 which may cause seasonal variations in beach litter independent of the original source's
27 constancy. We posit that, given the predominance of local sources for beach macroplastics in
28 Europe, ocean transport effects are secondary, though exceptions may exist [182].

29 Lastly, while methods are available to trace beach litter back to its sources [201, 262], they
30 rely on knowing each source's relative contribution, challenging to determine for fish capture
31 versus mariculture [239]. Global estimates on source shares exist [126, 144], but regional data
32 may vary greatly [18, 179, 246]. The tracking of fishing activity is feasible through automatic
33 identification system positions [135], yet our understanding of the timing and scale of plastic
34 emissions from mariculture remains limited [195, 257].

8.4 Conclusion

Our study provides evidence that seasonal variations in macroplastics pollution on beaches in the NEA are more important than previously recognized [233]. Emissions from numerous small-scale rivers and mariculture-related activities, likely driven by seasonally dependent adverse weather conditions – conditions projected to increase in the future [131] – may be key contributors. With aquaculture production in the NEA having surged by approximately 117% over the last two decades [66, 191] (Supplementary Figure 10), the role of this sector in contributing to beach litter is expected to grow [18, 136, 239, 257]. While Regulation (EU) 2021/1139 marks progress towards sustainable mariculture within the EU, our study underscores the urgent need for more specific and effective strategies to address mariculture-related plastic emissions.

Addressing the complexities of beach litter, particularly from mariculture, requires a comprehensive enhancement of the existing monitoring network. The Marine Strategy Framework Directive (MSFD) Technical Group on Marine Litter recommends that beach litter assessments be conducted seasonally [74], i.e., four times a year, to capture the fluctuating dynamics of litter accumulation. Our findings support this recommendation, highlighting the importance of sufficient temporal resolution in monitoring to effectively identify and address seasonal litter hotspots. Furthermore, a more refined categorization of litter items is necessary to confidently identify litter sources [191]. Recent studies have shown that using OSPAR categorization can underestimate actual mariculture-related items by a factor of four [239]. Accurate categorizations would also provide a robust basis for informing probabilistic Lagrangian backward simulations [201, 262] needed to identify sources considering ocean currents.

Our research also identifies strategic opportunities for more efficient removal of macroplastics from beaches by focusing efforts on regions and periods identified as seasonal hotspots. Additionally, the collection of large items from beaches serves as a useful stopgap measure to limit the formation of microplastics while effective steps to prevent plastic leakage into the environment are formulated [226]. For example, targeted cleanup operations in the Skagerrak during spring and along the French and Iberian western coastlines throughout winter and spring could enhance the efficiency of litter collection efforts. Such focused initiatives could potentially double the median quantity of items collected, providing valuable insights for NGOs and citizen science projects aimed at maximizing environmental impact, enhancing community engagement, and raising public awareness about marine litter issues. Furthermore, involving citizen science in monitoring efforts presents a promising approach to supplement traditional data collection methods [42, 104].

In addition to better quantifying plastic emissions from mariculture [257], addressing the root causes of macroplastic leakage into the marine environment is paramount. Our study emphasizes the importance of adhering to policy recommendations made by the Aqua-Lit project

1 [106], which include using high-quality materials in mariculture installations, expanding waste
2 collection, and implementing protocols for monitoring plastic waste. These efforts should be
3 complemented by incentives such as quality seals and bonuses for adherence to sustainable
4 practices [211, 239]. The advancement of extended producer responsibility schemes [143]
5 further supports these initiatives.

Conclusion & Future Work

outlook

use bayesian log gaussian cox process framework to combine datasets from different sources, including scarce yet accurate government data and dense yet noisy citizen science data.

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